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(54) **DIGITAL ASSISTANT INTEGRATION WITH TELEPHONY**

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H04M 3/42 (2006.01)

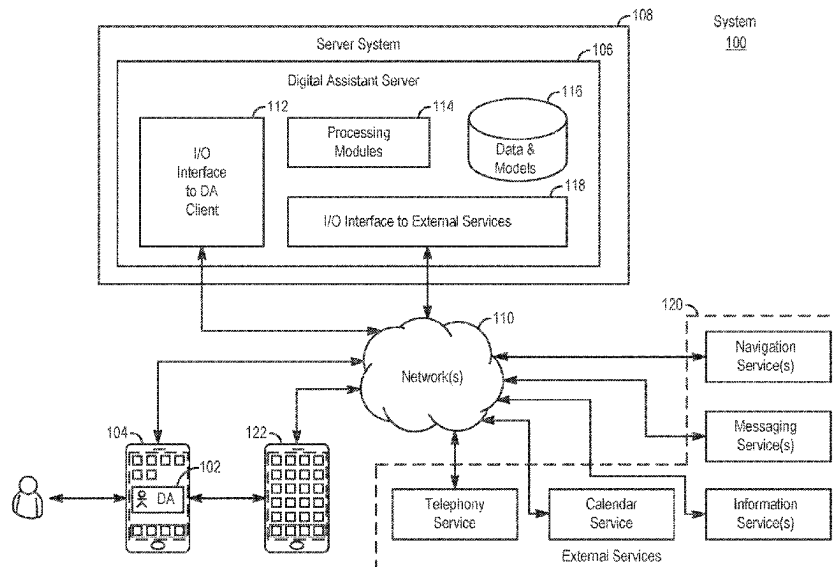
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(57) **ABSTRACT**

Systems and processes for integrating a digital assistant with telephony are provided. For example, an incoming call may be received, from a caller, at an electronic device. A communication session may be established between the caller and the digital assistant of the electronic device. In accordance with a determination that the identification of the caller is unknown, determination is made whether the caller corresponds to an automated calling system. In accordance with a determination that the identification of the caller is known, a response is provided by the digital assistant to the caller. An output including information corresponding to the communication is provided at the electronic device.

33 Claims, 23 Drawing Sheets



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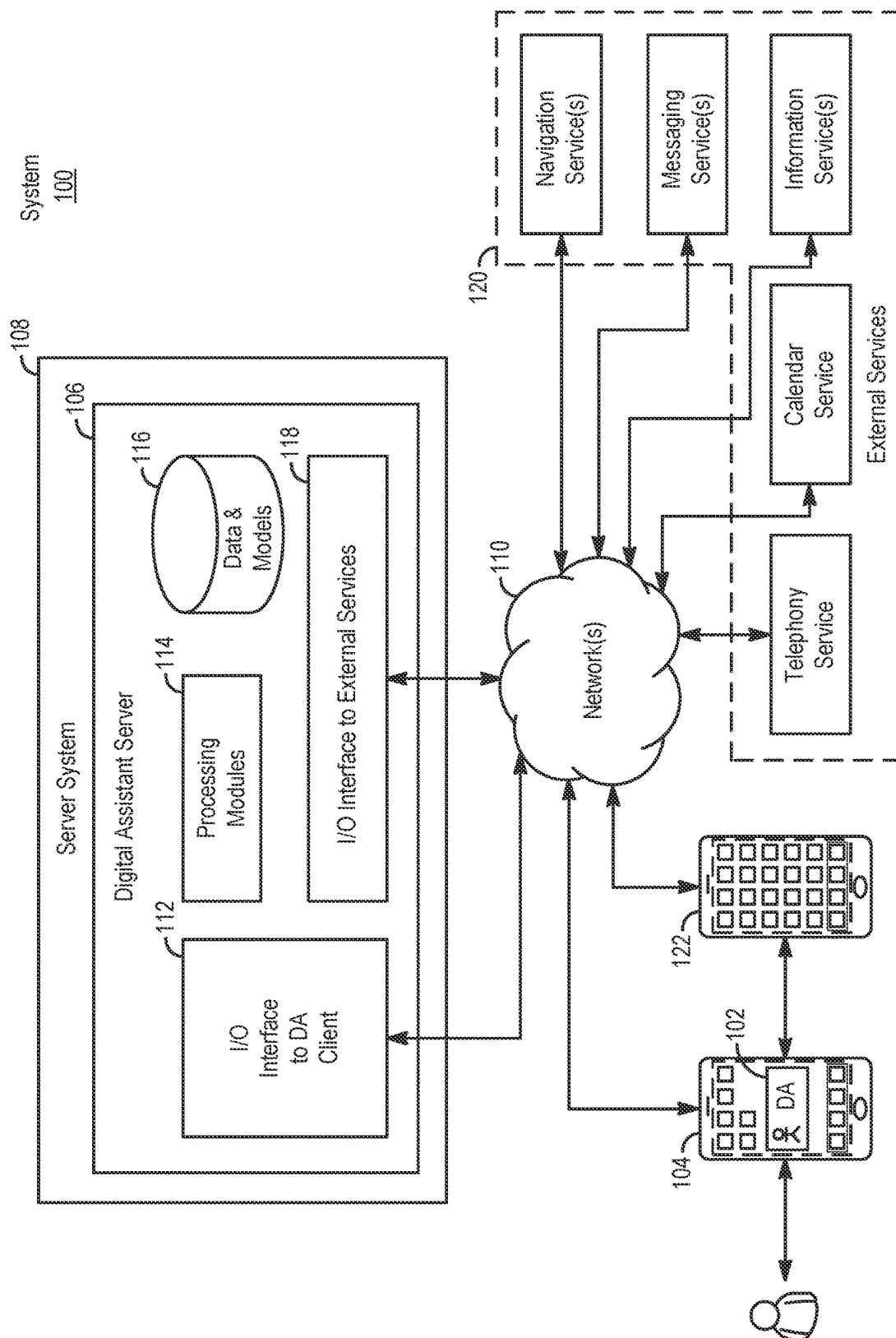


FIG. 1

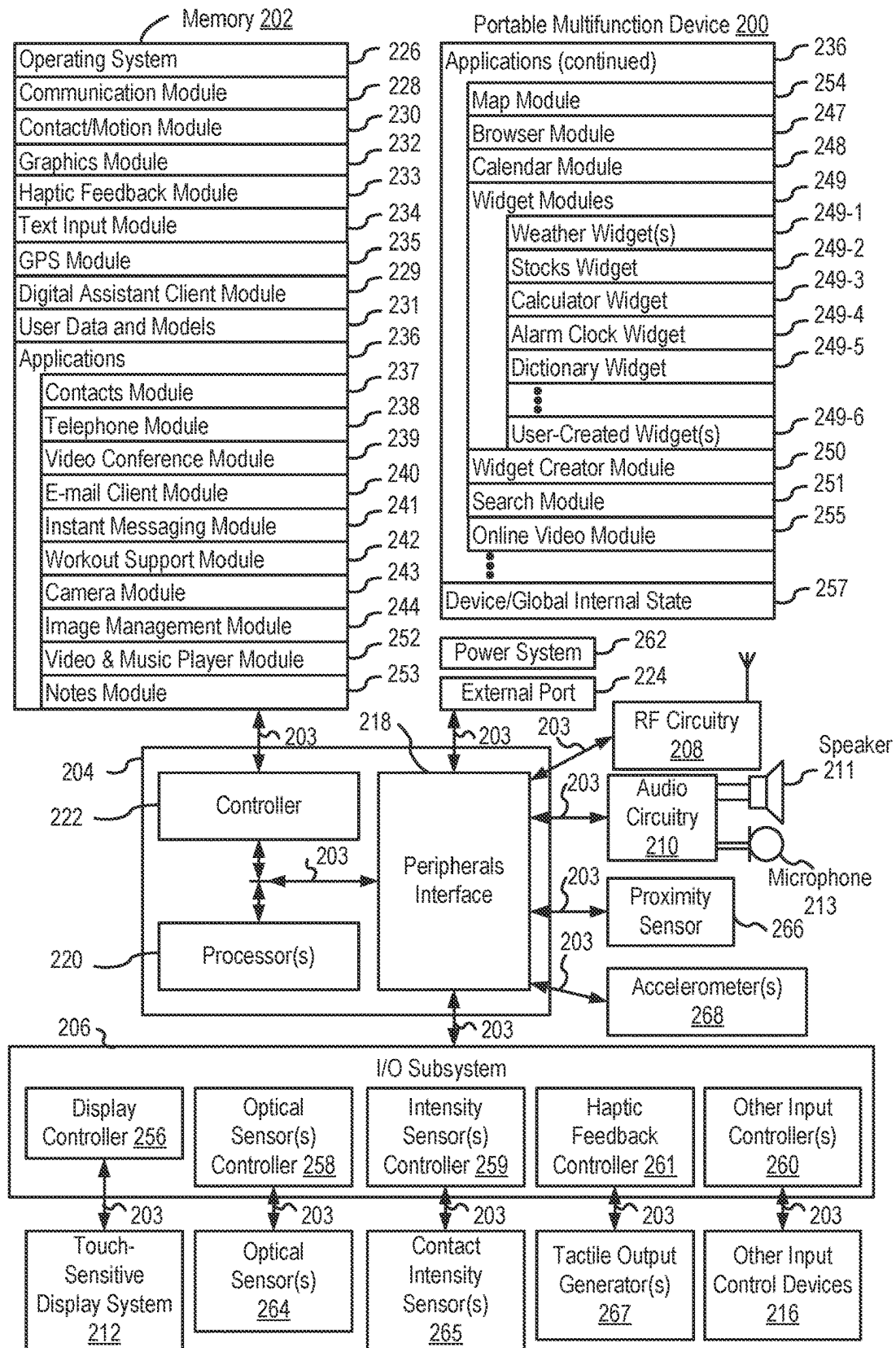
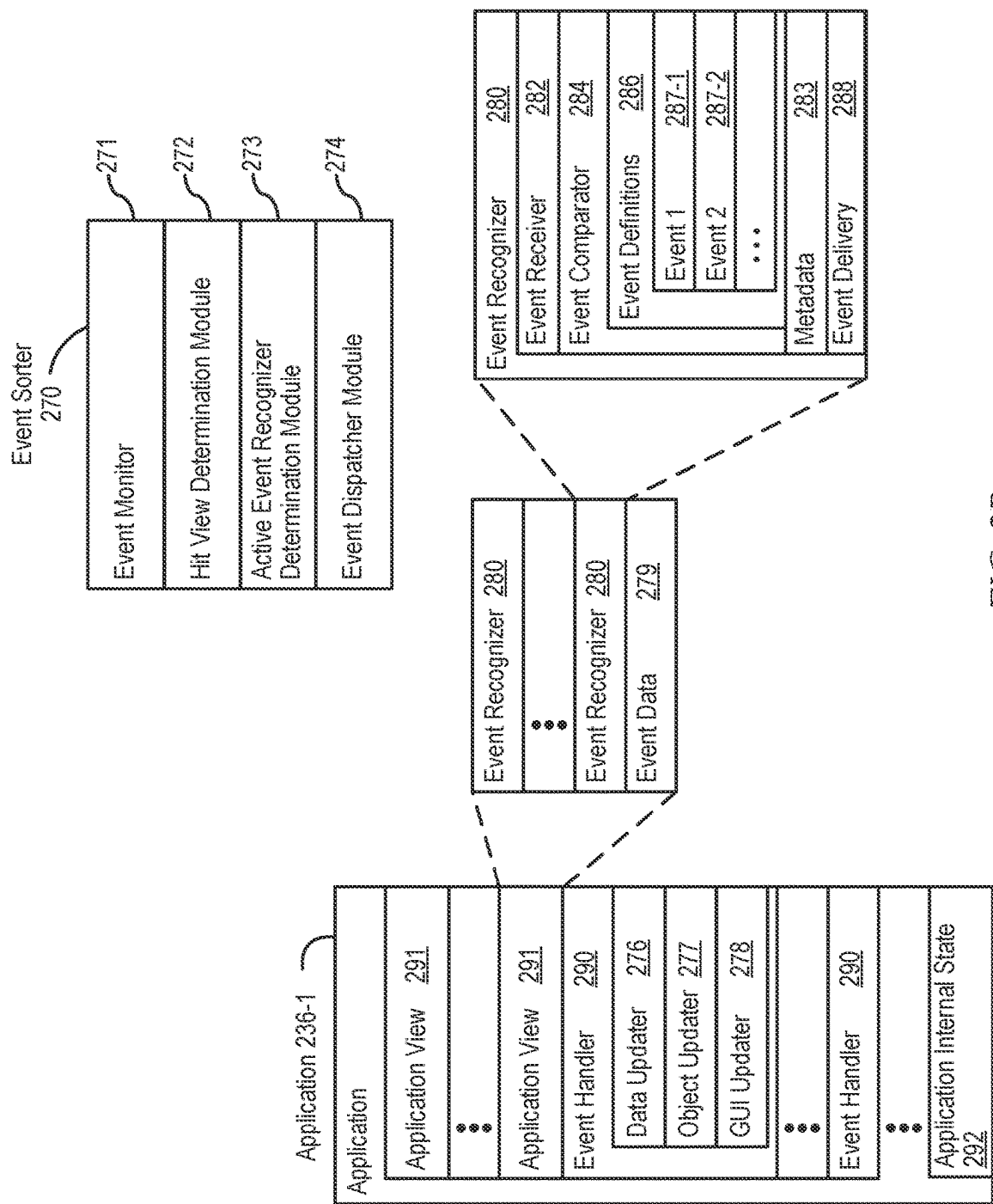


FIG. 2A



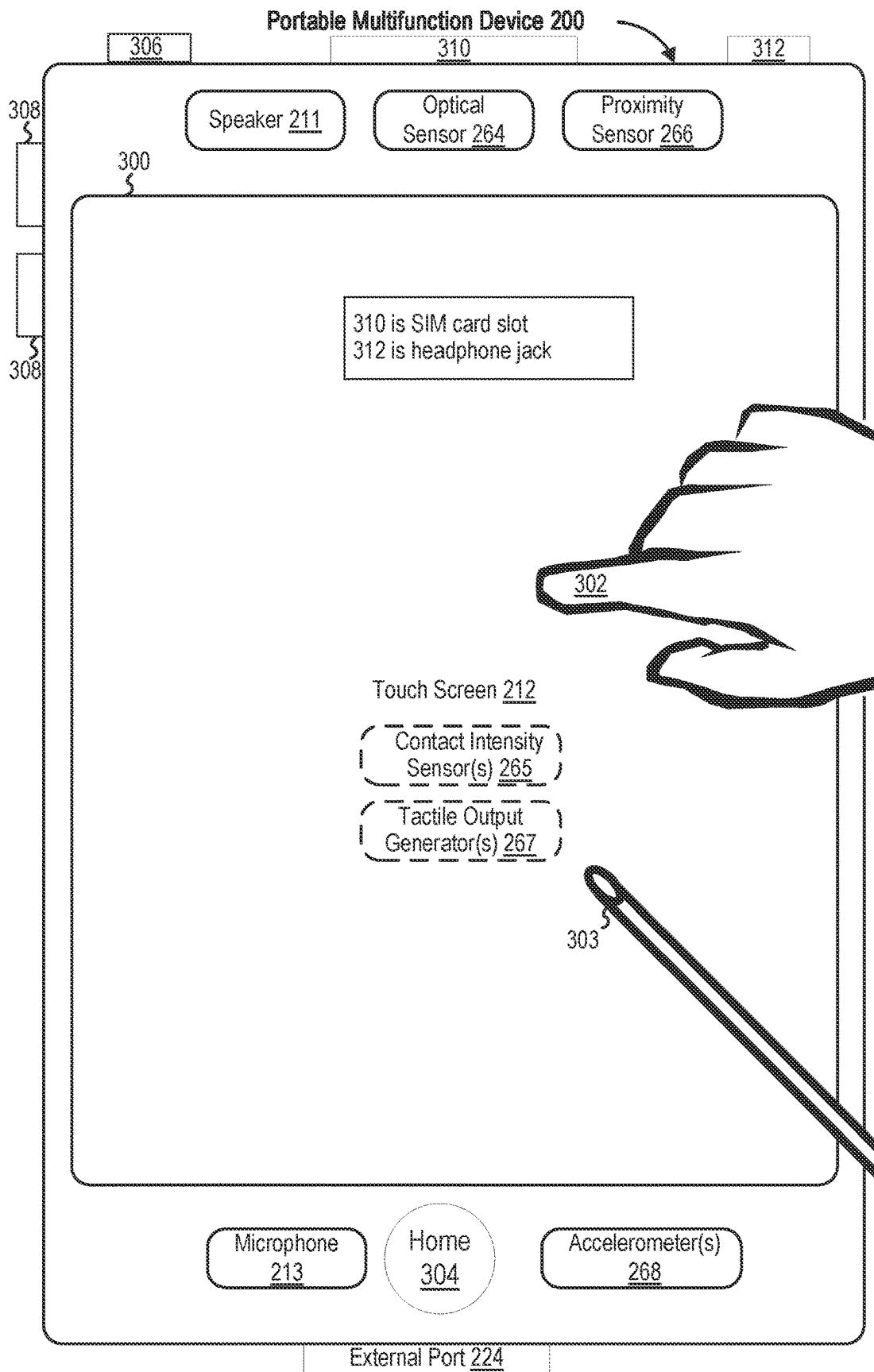


FIG. 3

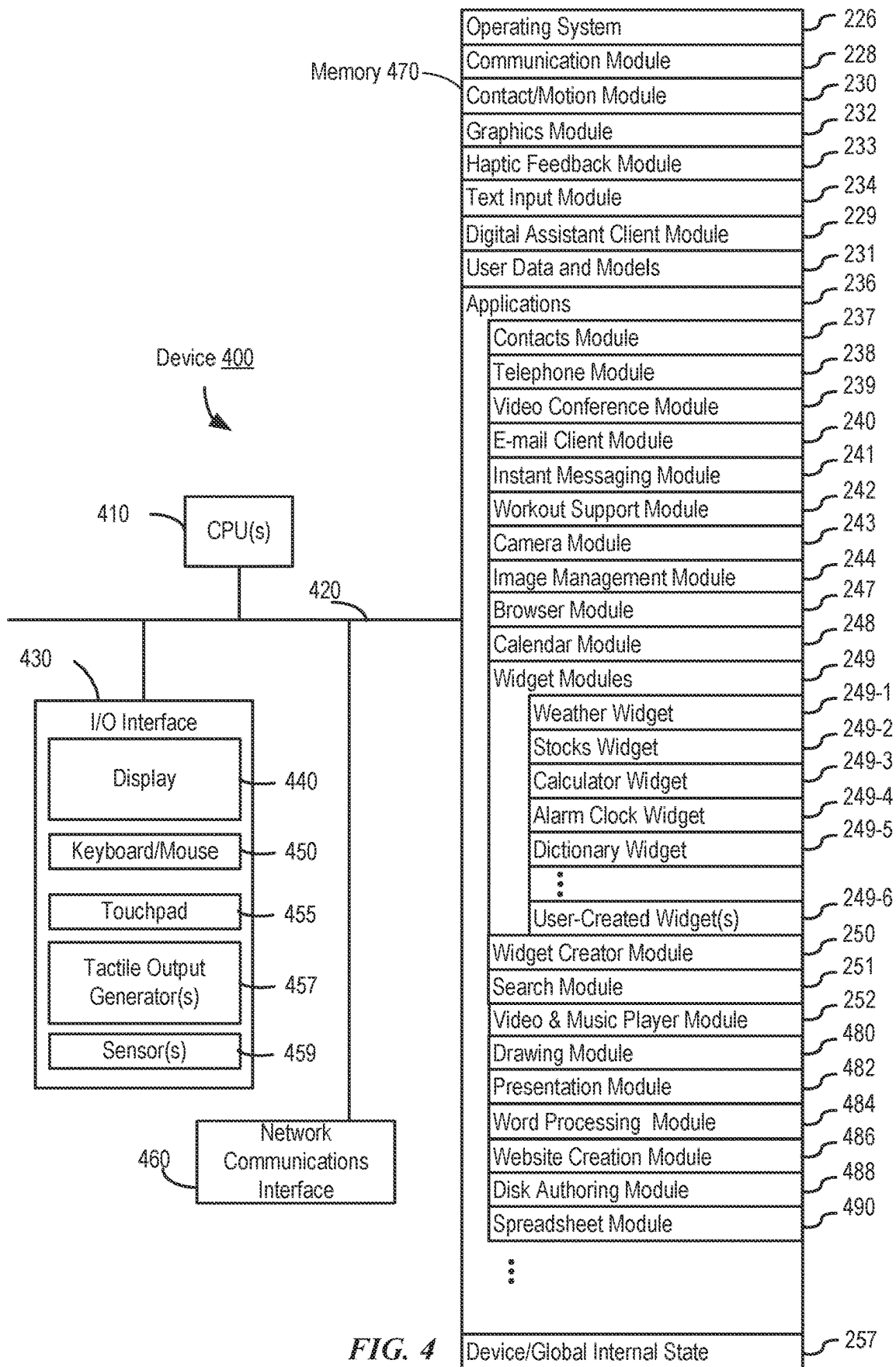


FIG. 4

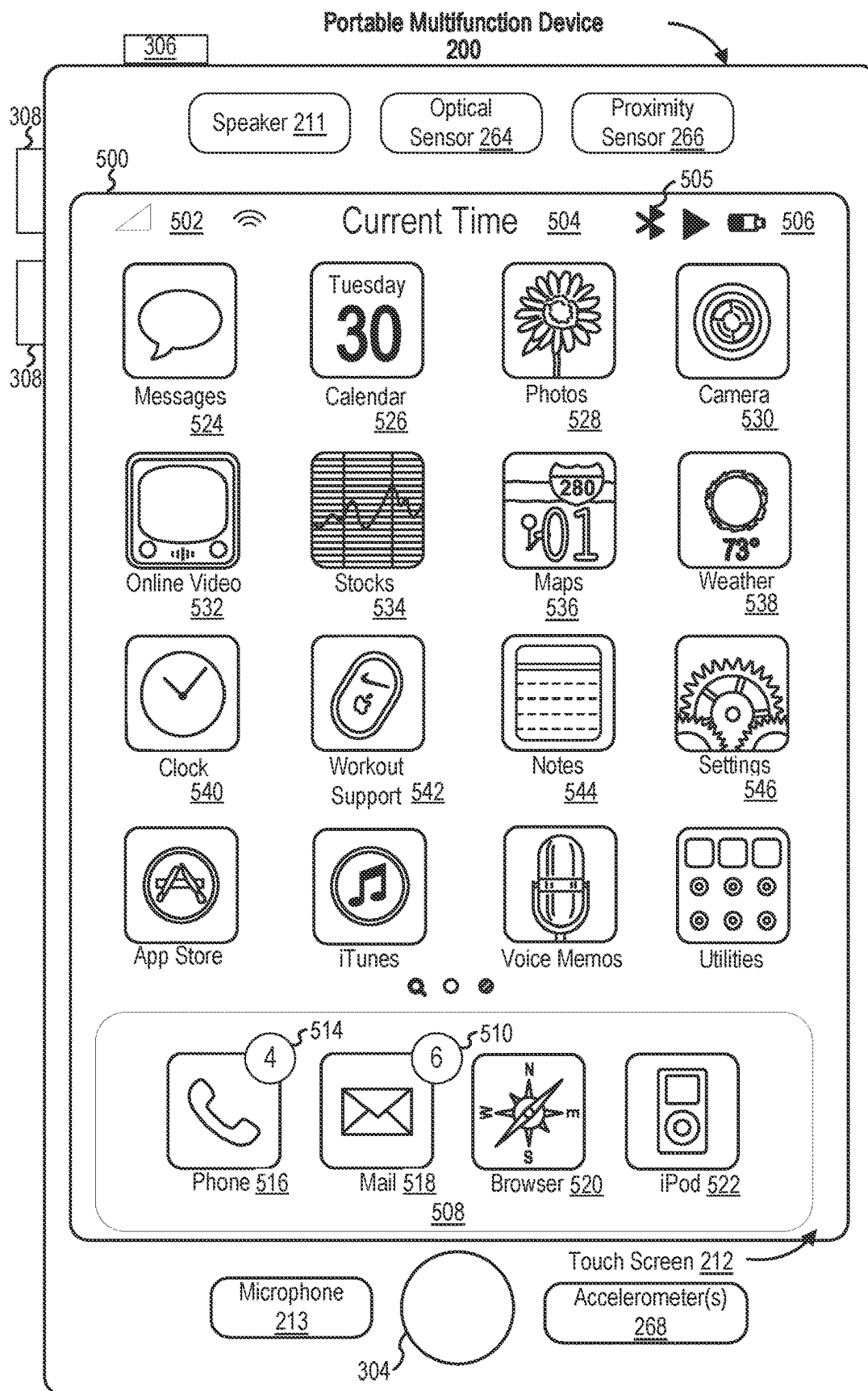


FIG. 5A

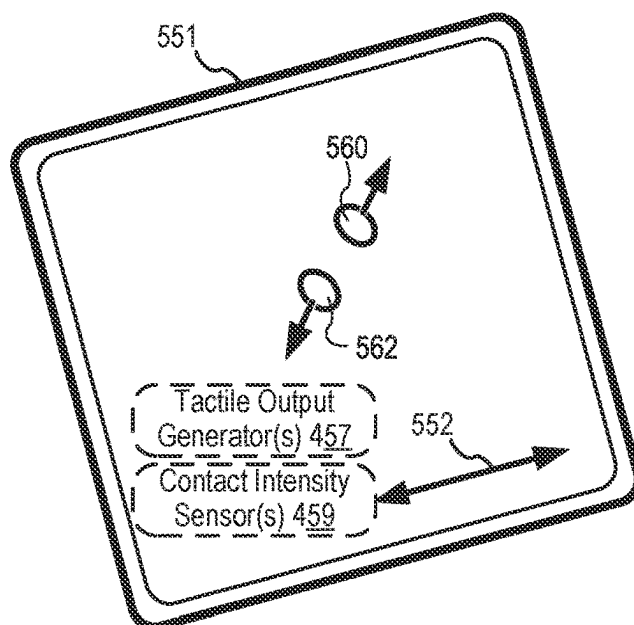
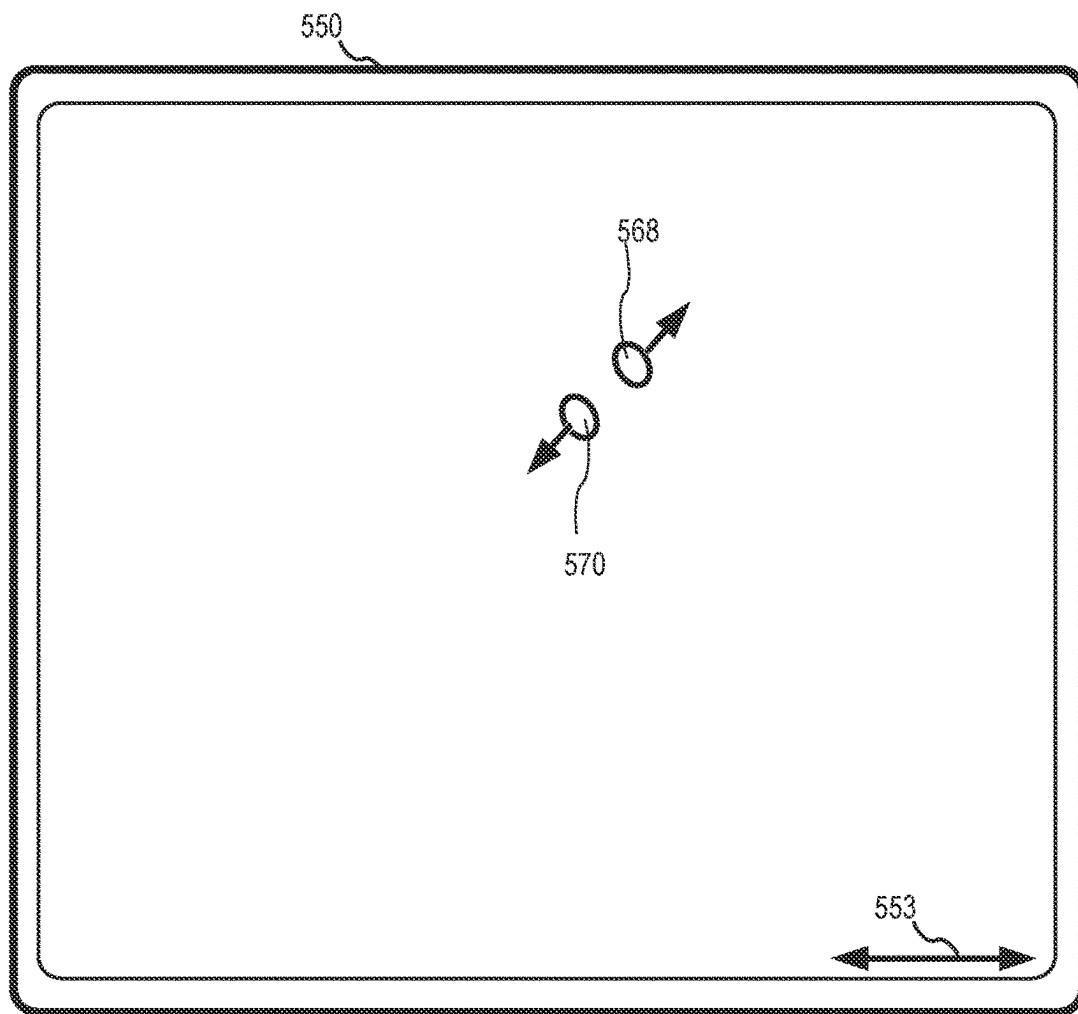


FIG. 5B

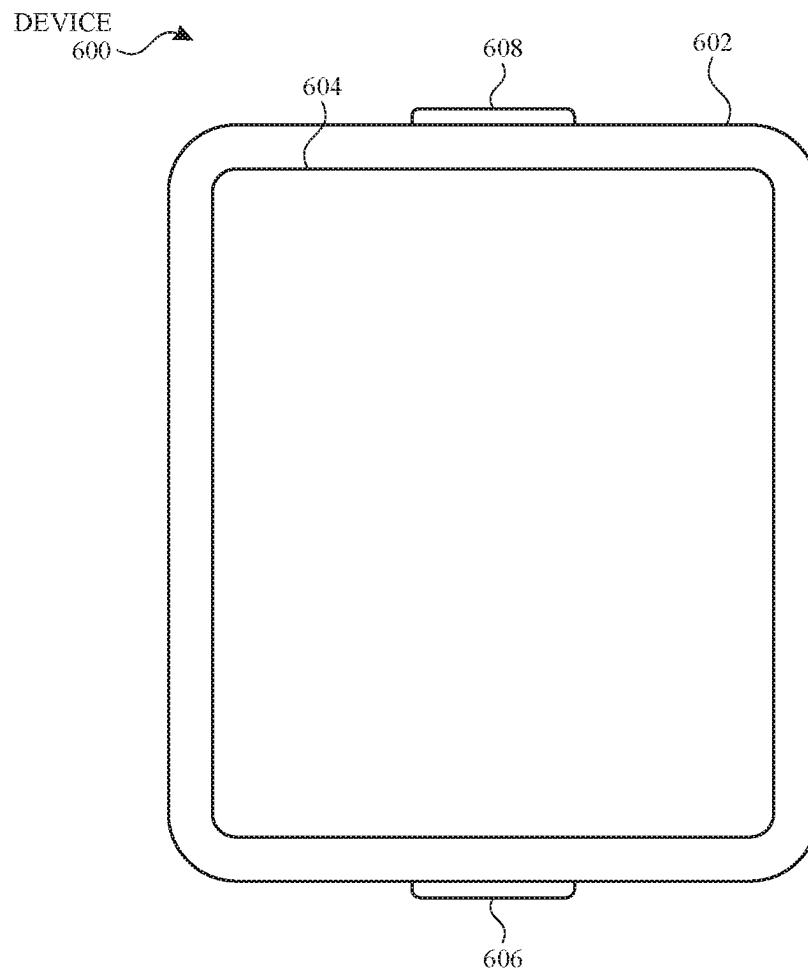


FIG. 6A

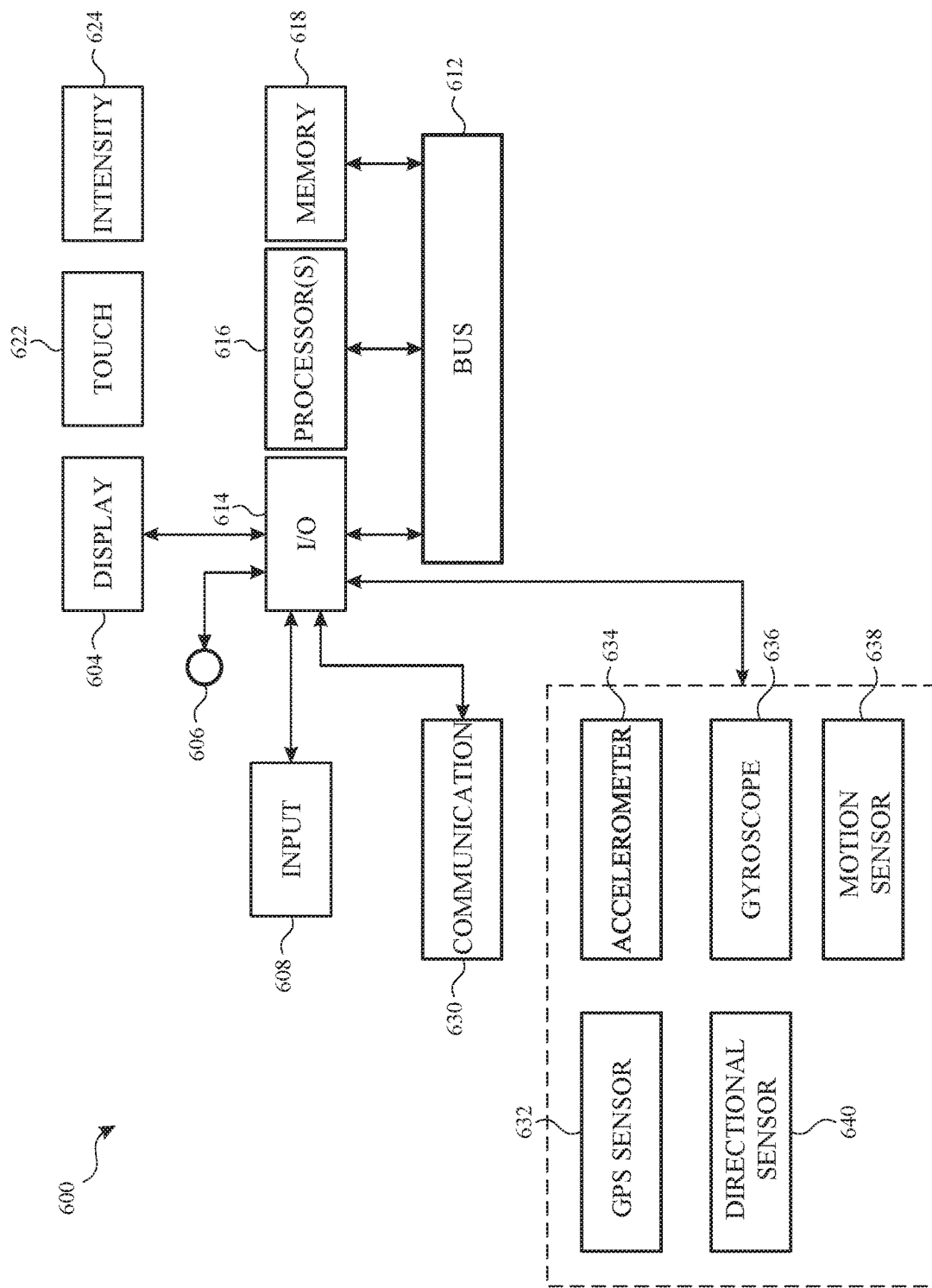


FIG. 6B

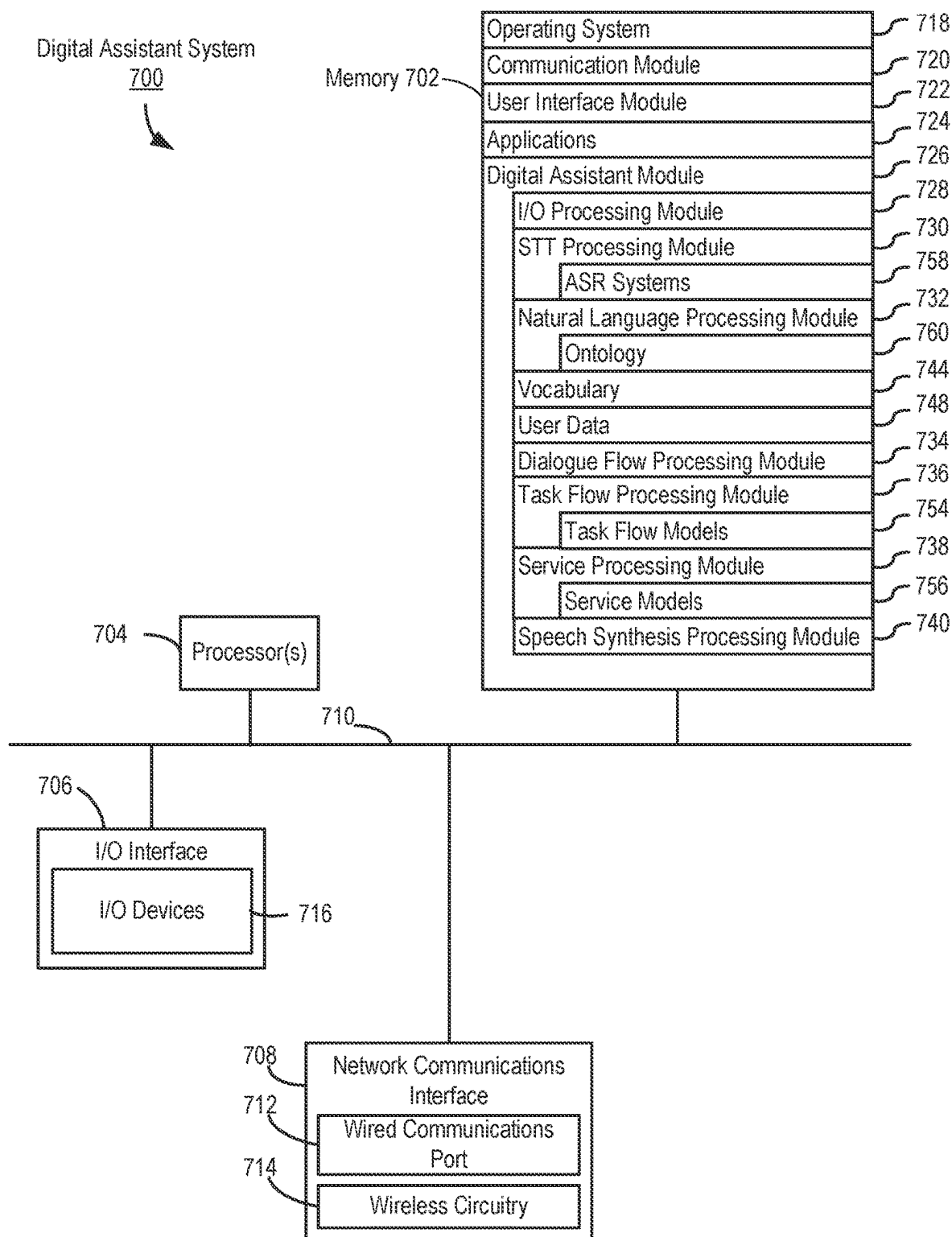


FIG. 7A

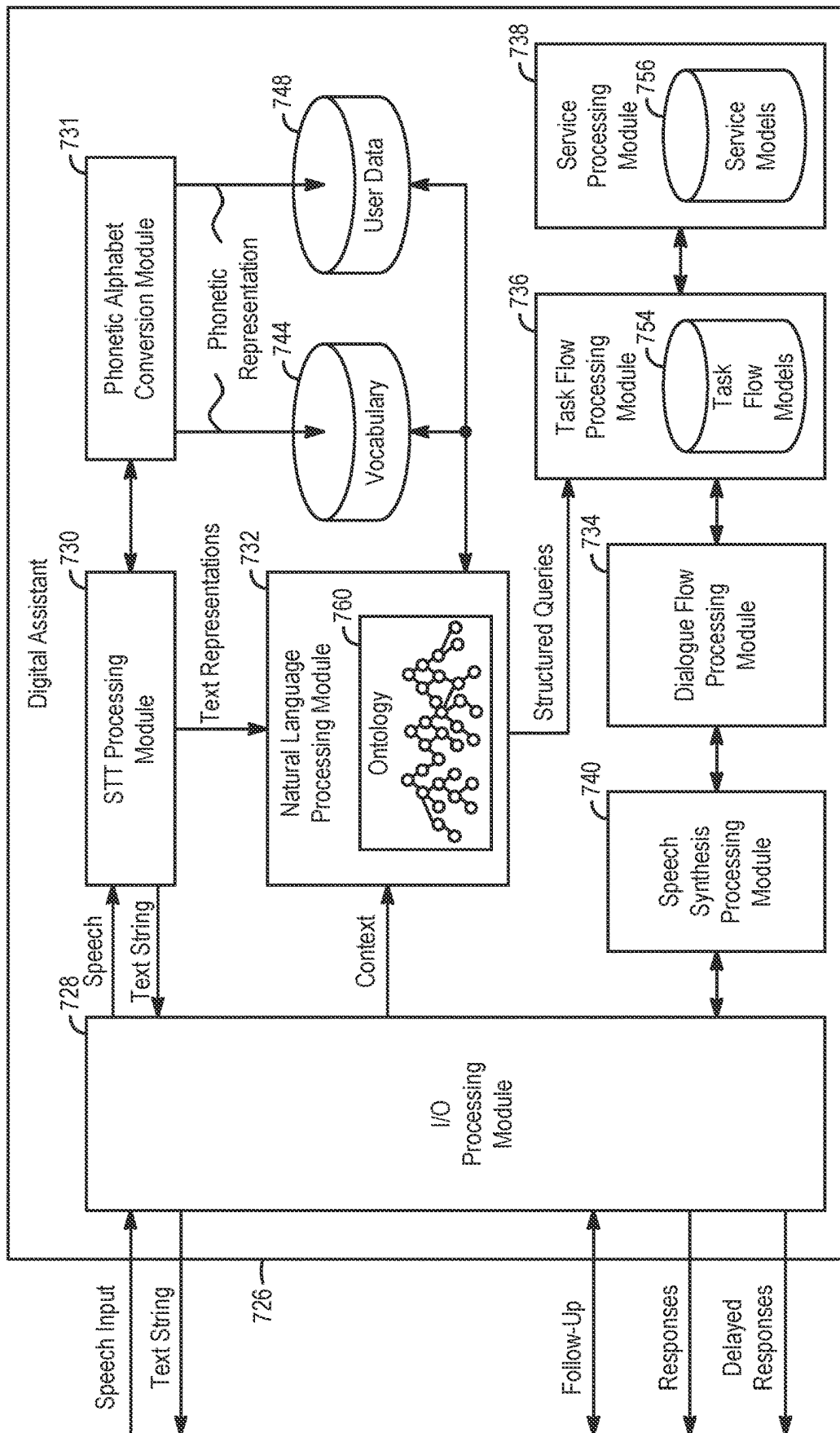


FIG. 7B

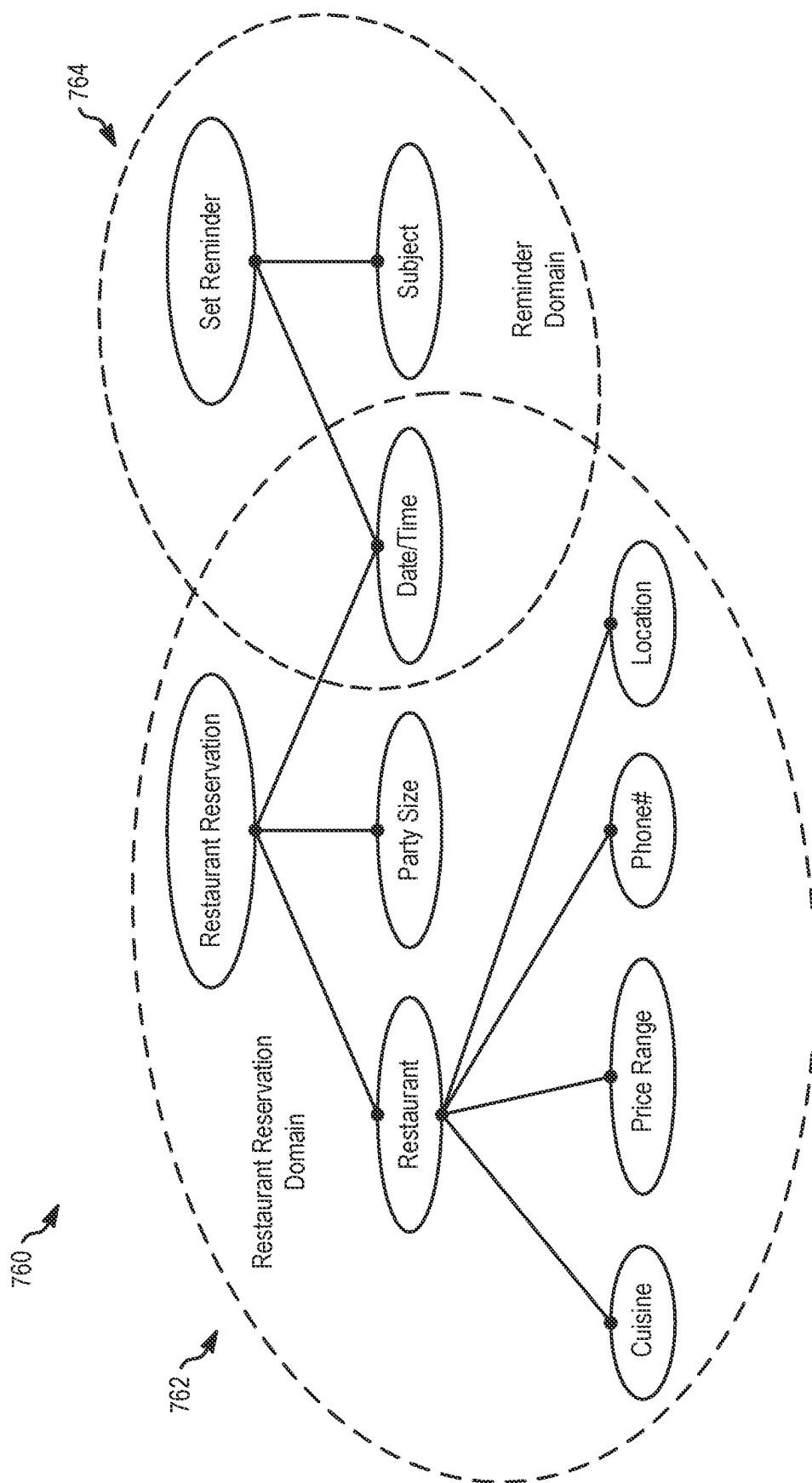


FIG. 7C

800

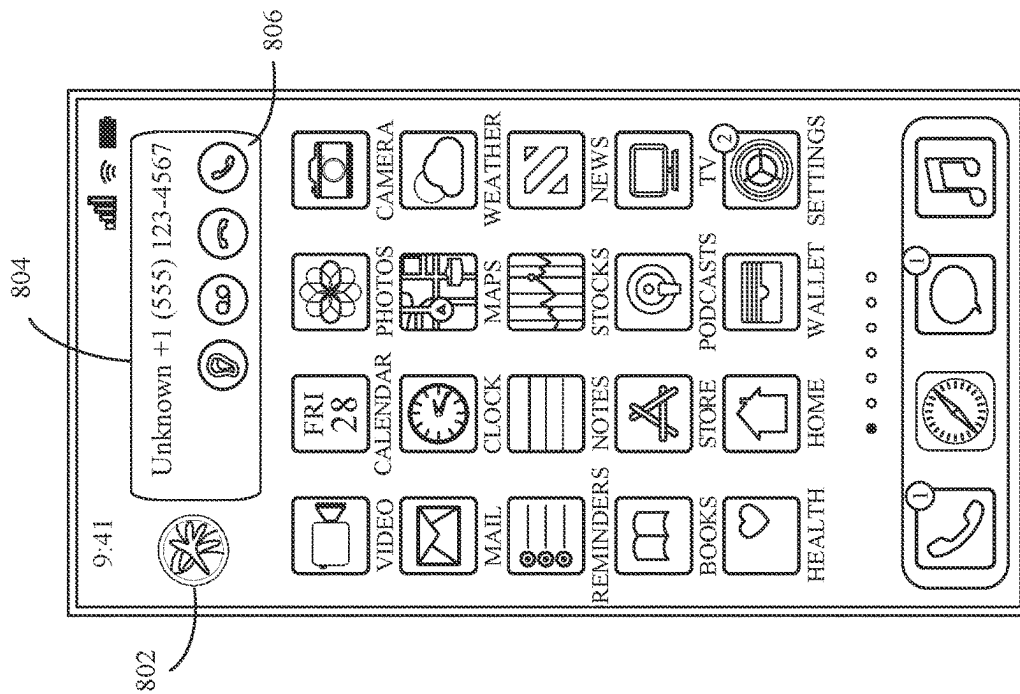


FIG. 8A

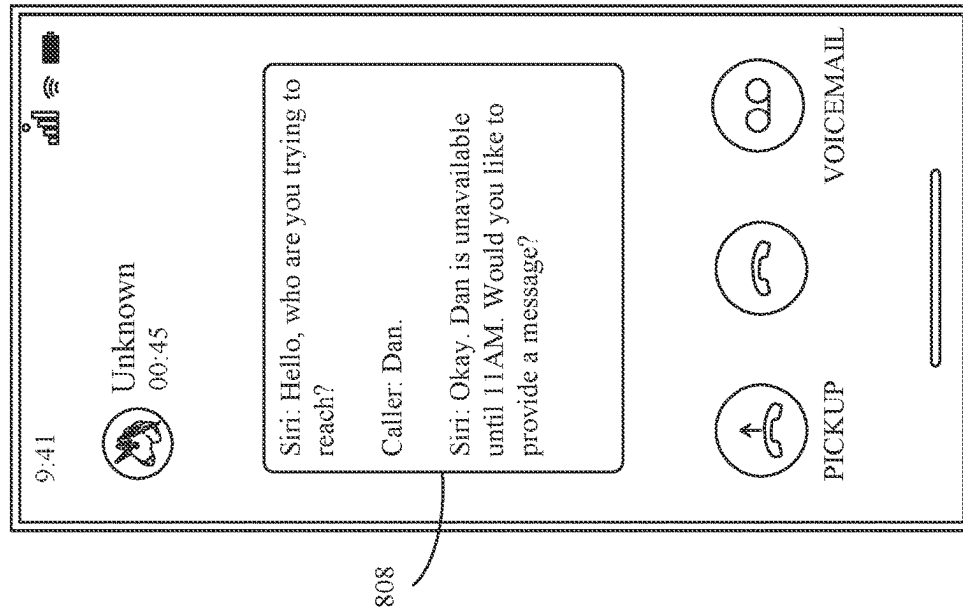


FIG. 8B

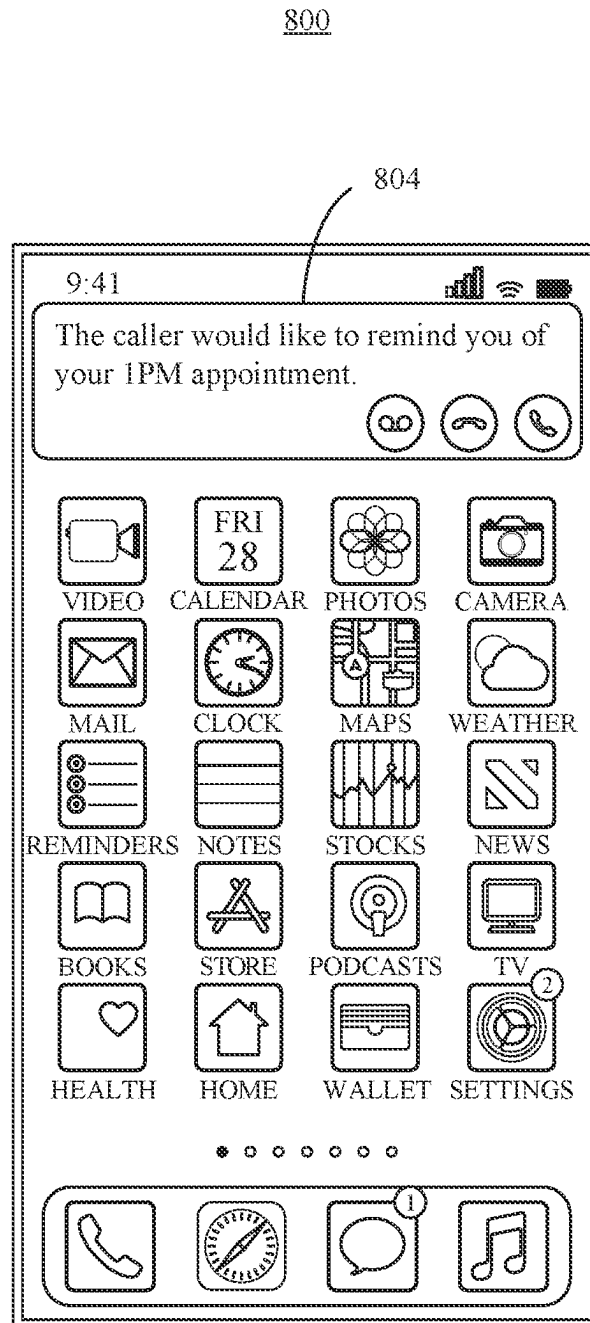


FIG. 8C

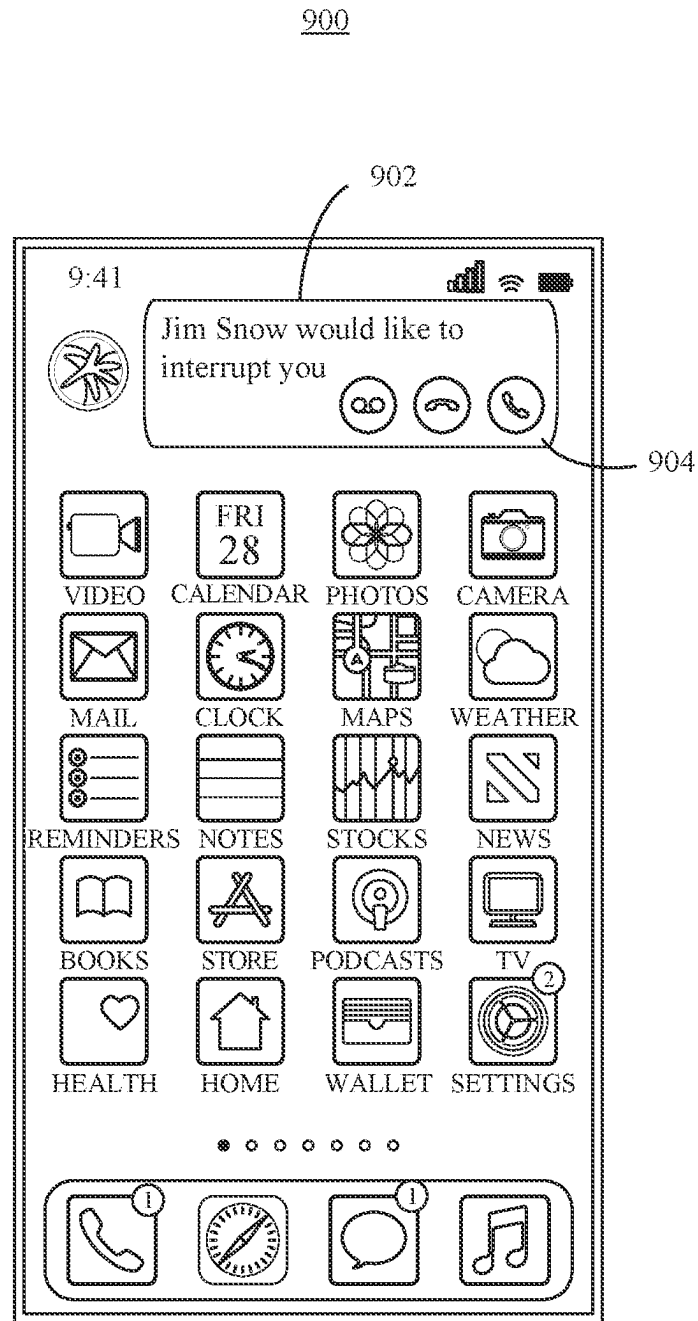


FIG. 9

1000

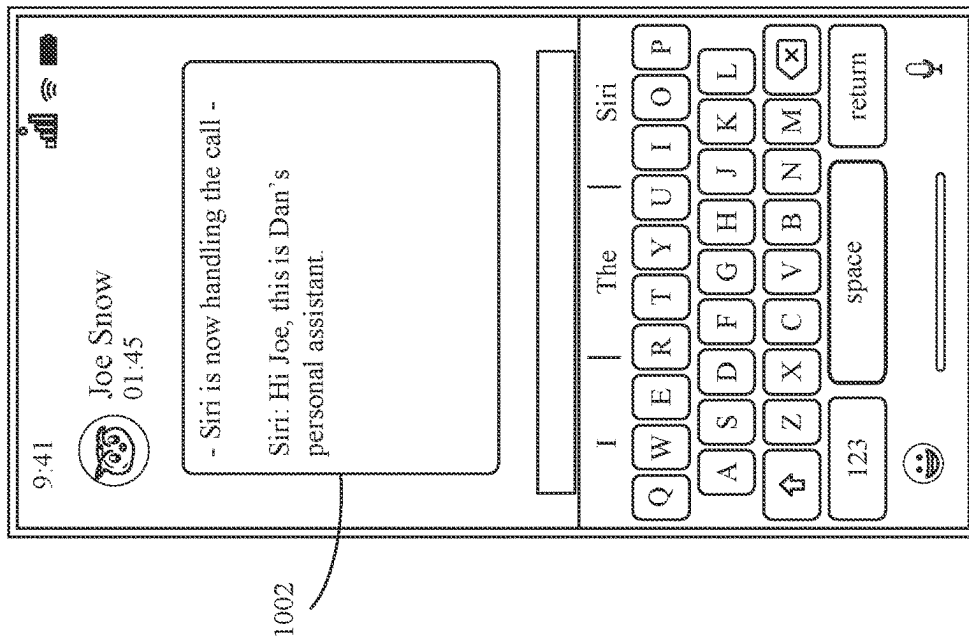


FIG. 10A

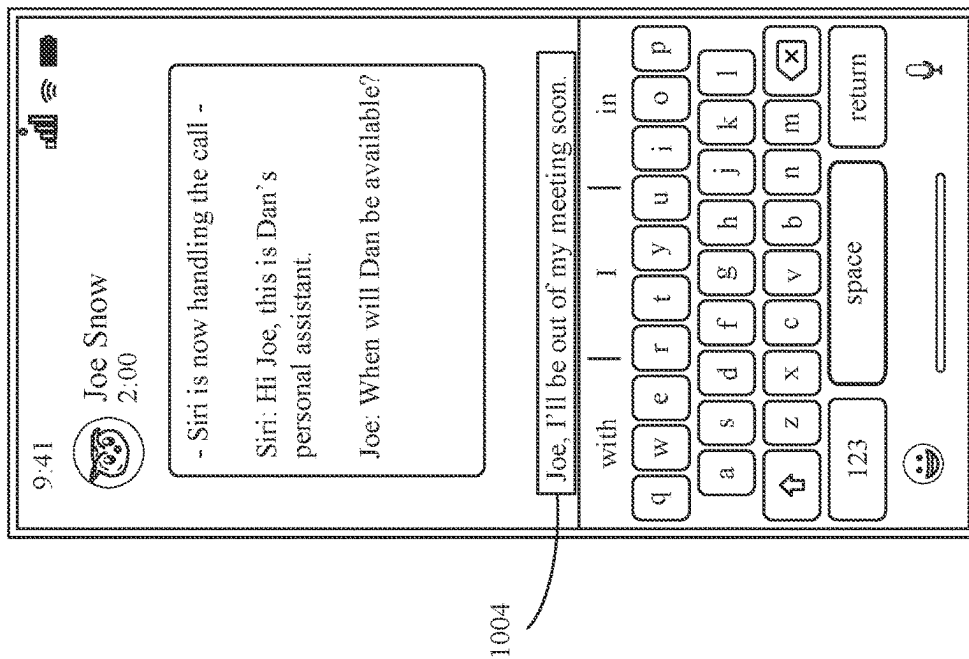


FIG. 10B

1100

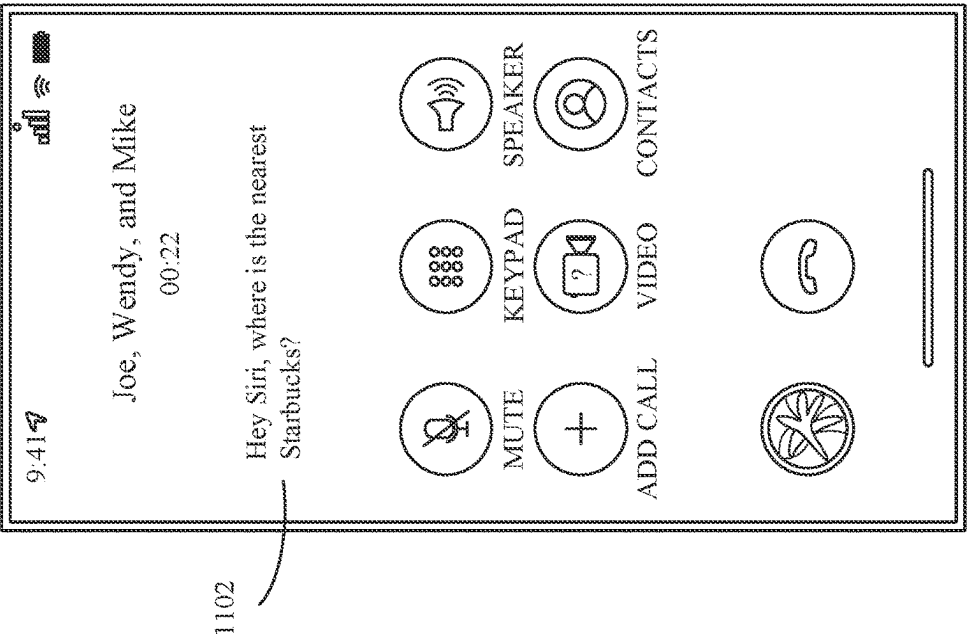


FIG. 11A

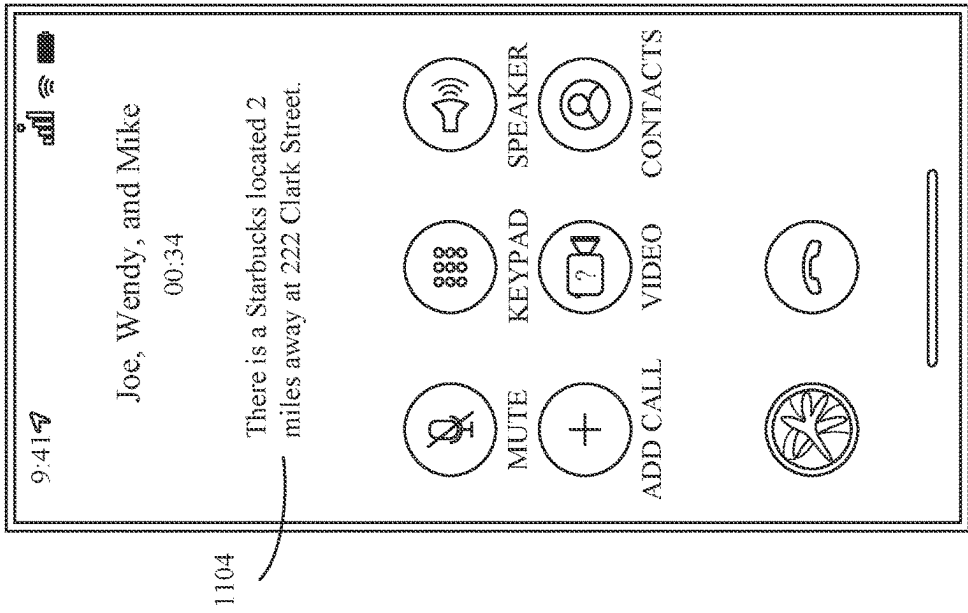


FIG. 11B

1200

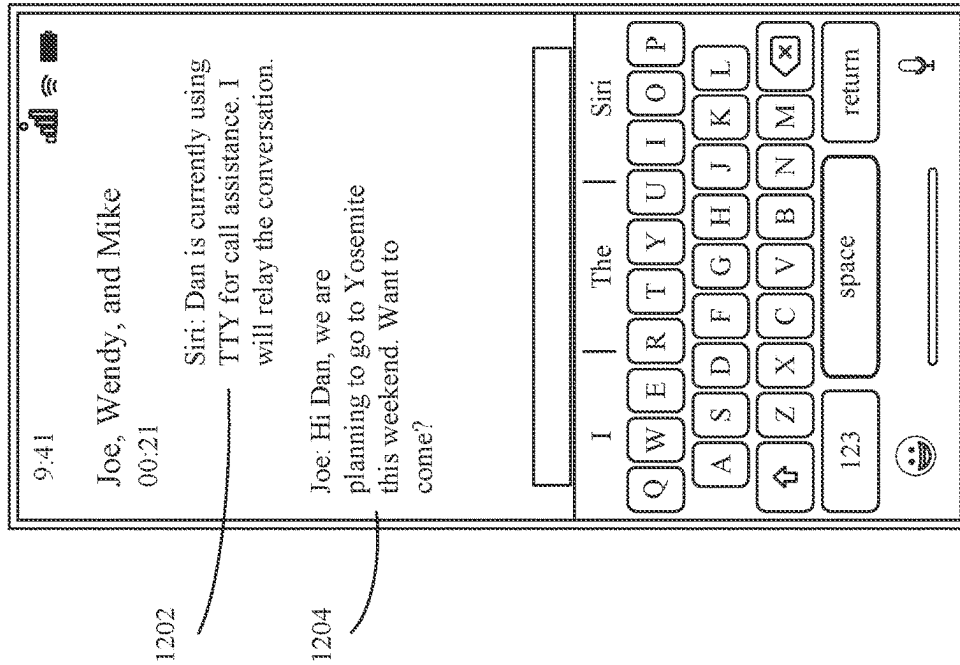


FIG. 12A

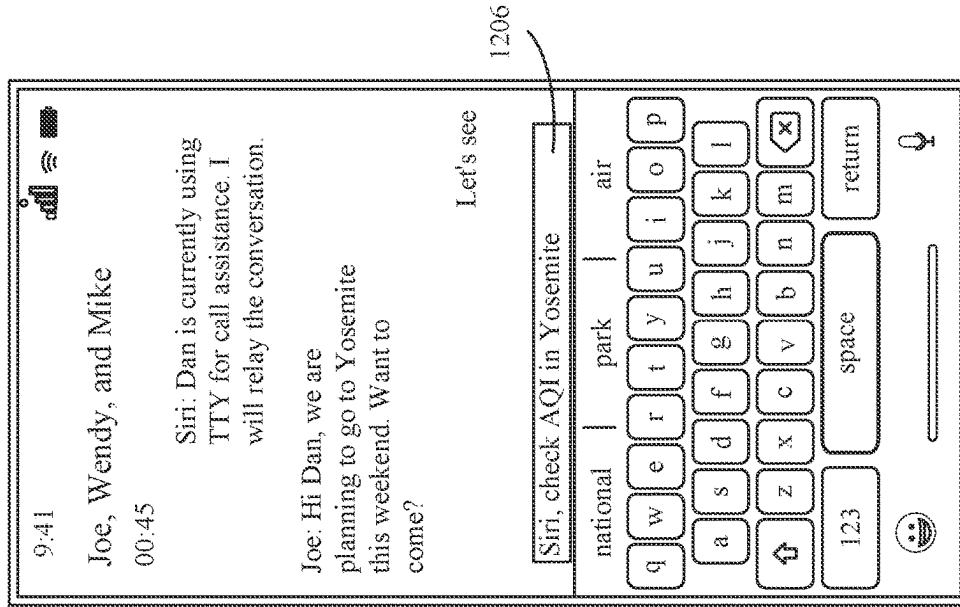


FIG. 12B

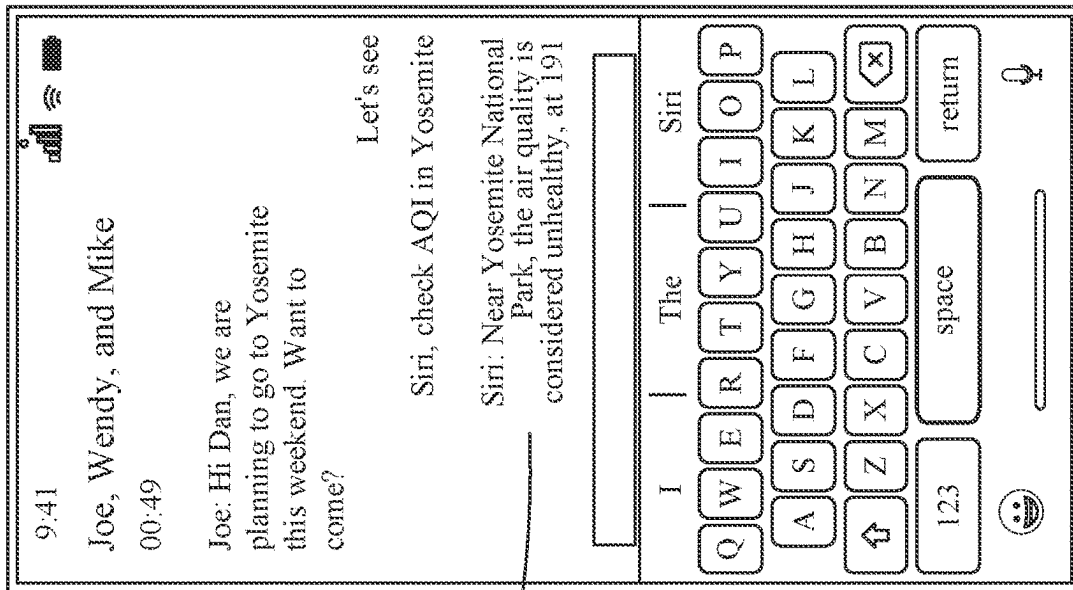


FIG. 12C

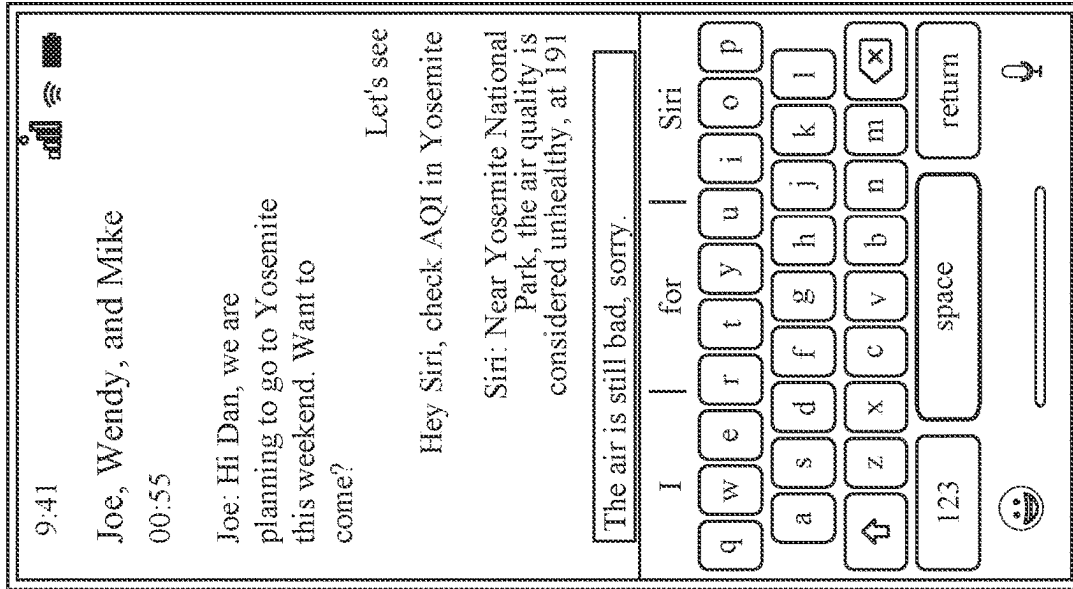
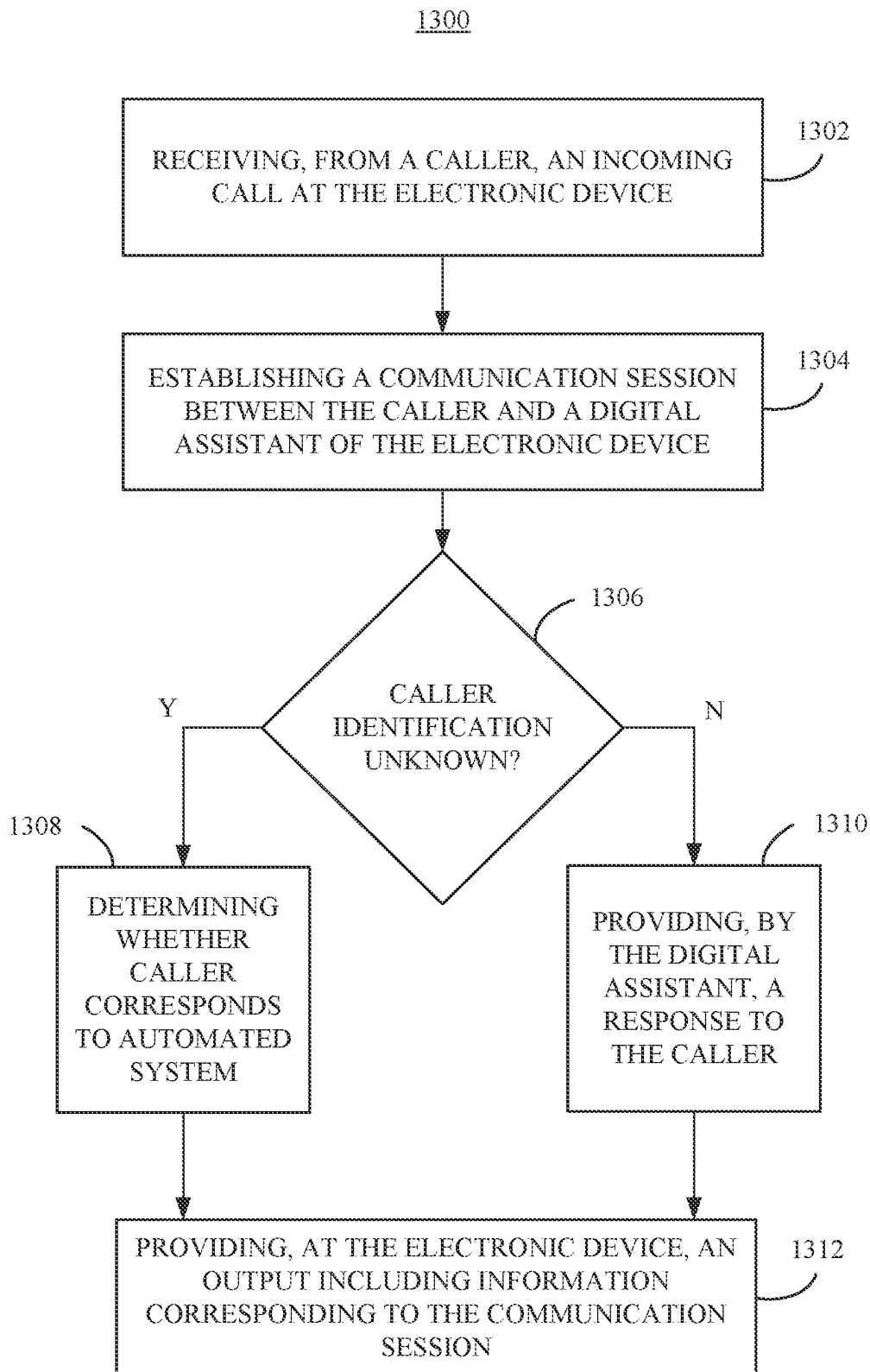
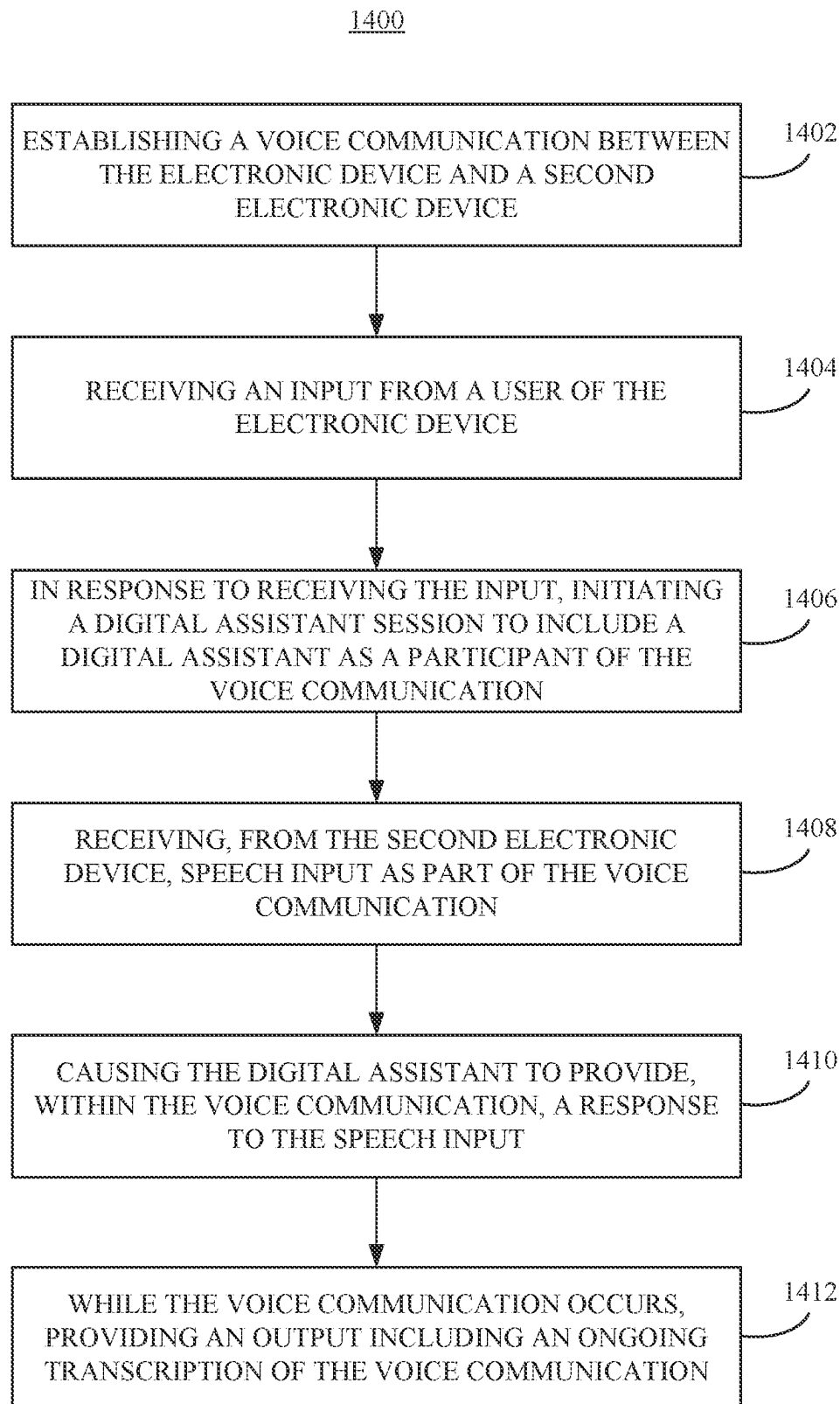
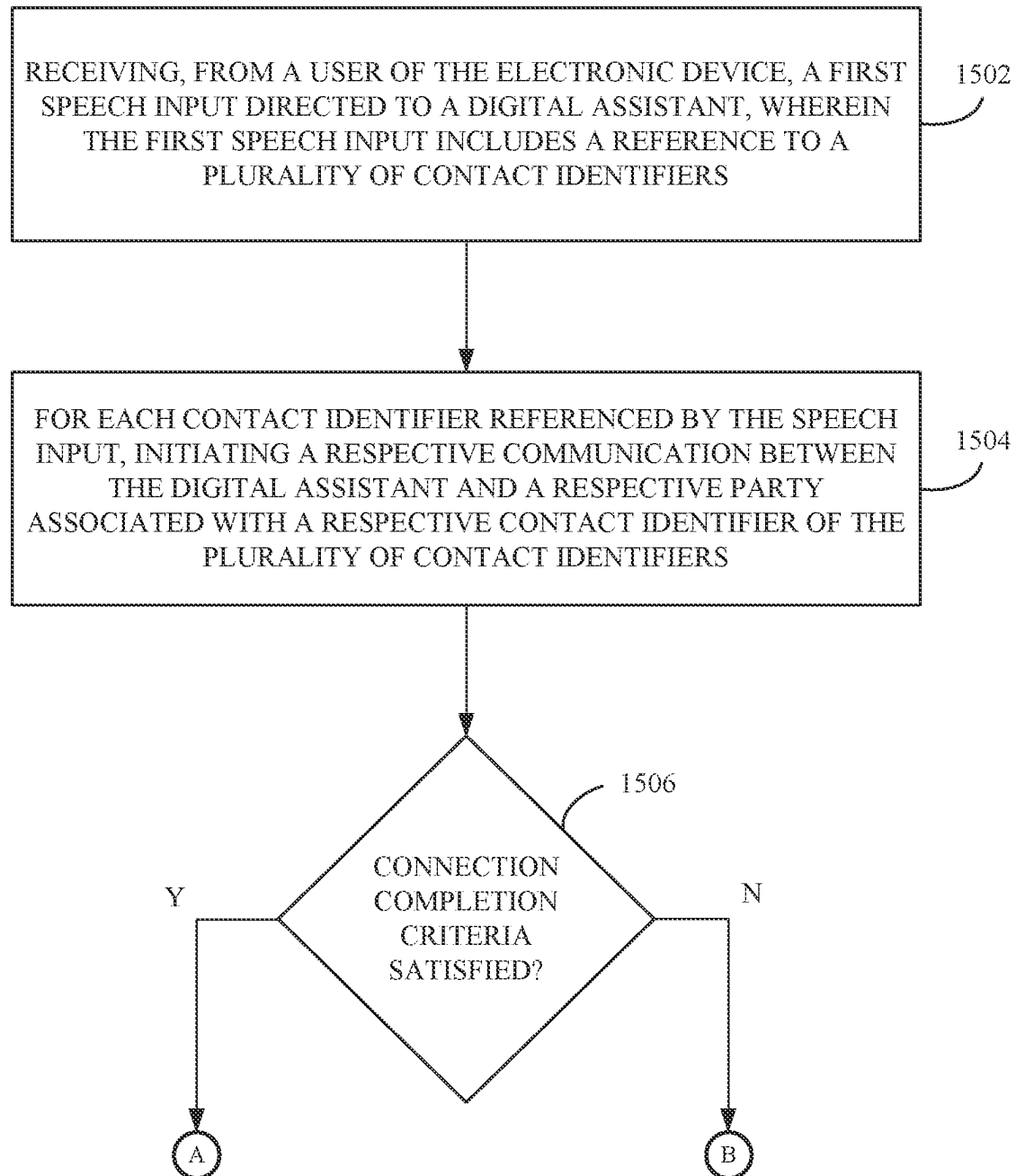
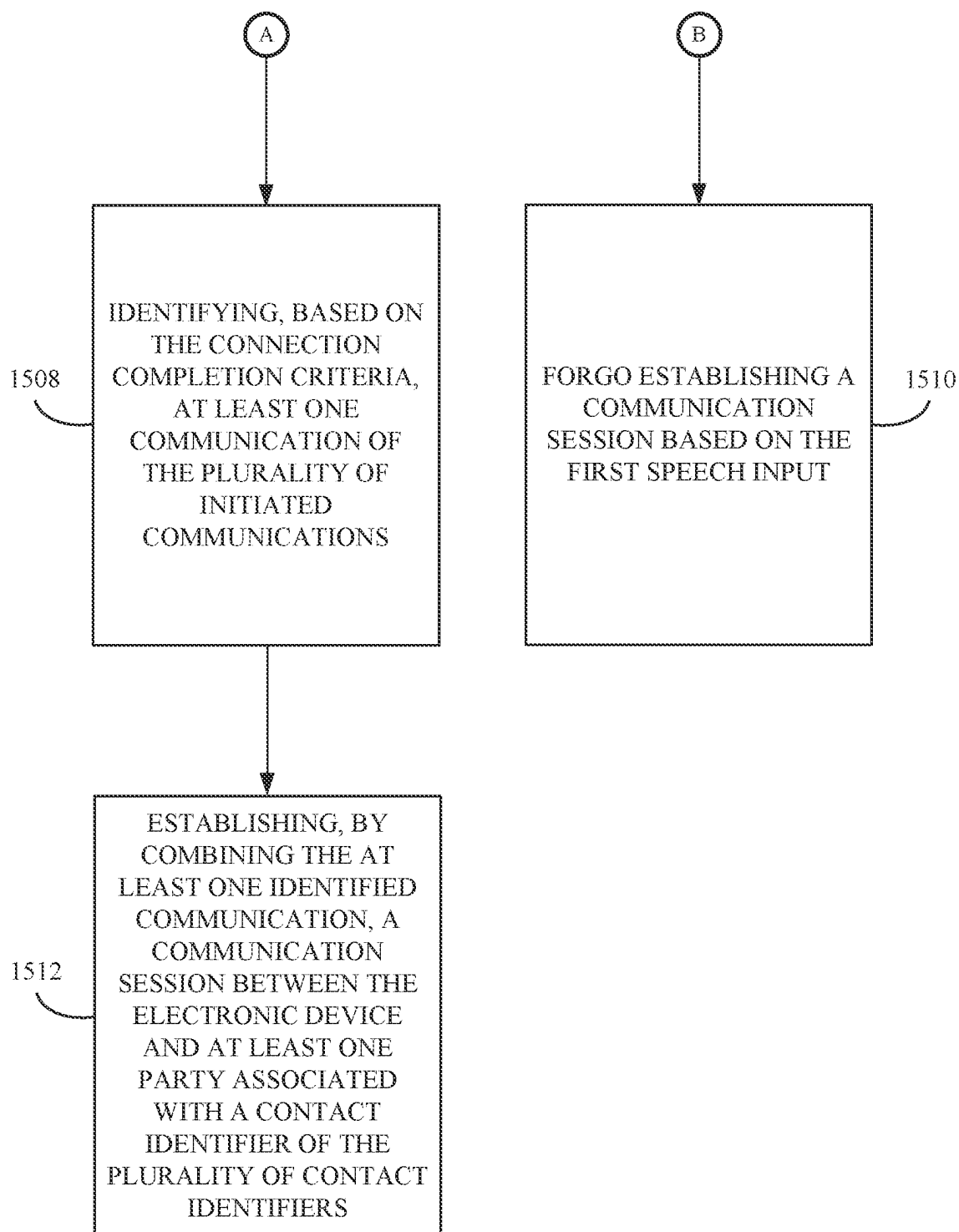


FIG. 12D

**FIG. 13**

**FIG. 14**

1500**FIG. 15A**

**FIG. 15B**

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**DIGITAL ASSISTANT INTEGRATION WITH
TELEPHONY****CROSS-REFERENCE TO RELATED
APPLICATION**

This application is a continuation of U.S. patent application Ser. No. 17/395,322, entitled “DIGITAL ASSISTANT INTEGRATION WITH TELEPHONY,” filed on Aug. 5, 2021, which claims priority to U.S. Provisional Patent Application No. 63/174,413, entitled “DIGITAL ASSISTANT INTEGRATION WITH TELEPHONY,” filed on Apr. 13, 2021, the contents of which is hereby incorporated by reference in its entirety.

FIELD

This application relates generally to intelligent automated assistants and, more specifically, to integrating a digital assistant with telephony.

BACKGROUND

Intelligent automated assistants (or digital assistants) can provide a beneficial interface between human users and electronic devices. Such assistants can allow users to interact with devices or systems using natural language in spoken and/or text forms. For example, a user can provide a speech input containing a user request to a digital assistant operating on an electronic device. The digital assistant can interpret the user's intent from the speech input and operationalize the user's intent into tasks. The tasks can then be performed by executing one or more services of the electronic device, and a relevant output responsive to the user request can be returned to the user.

Telephony has long been a fundamental technological feature for facilitating communication. Recent advancements in digital assistant technology has also improved interactions between humans and devices. However, the full potential of integrating telephony and digital assistant technology has not been fully realized. For example, conventional digital assistants are not capable of managing incoming and outgoing calls on behalf of users, much less facilitating ongoing calls. Traditional devices are also not well equipped to handle digital assistant requests from the user while the user is also engaged in a phone call. Given the speech-based nature of both digital assistant technology and telephony, an improved system for integrating digital assistants with telephony is desired.

SUMMARY

Systems and processes for integrating a digital assistant with telephony are provided. For example, an incoming call may be received, from a caller, at an electronic device. A communication session may be established between the caller and the digital assistant of the electronic device. In accordance with a determination that the identification of the caller is unknown, determination is made whether the caller corresponds to an automated calling system. In accordance with a determination that the identification of the caller is known, a response is provided by the digital assistant to the caller. An output including information corresponding to the communication is provided at the electronic device.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating a system and environment for implementing a digital assistant, according to various examples.

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FIG. 2A is a block diagram illustrating a portable multifunction device implementing the client-side portion of a digital assistant, according to various examples.

FIG. 2B is a block diagram illustrating exemplary components for event handling, according to various examples.

FIG. 3 illustrates a portable multifunction device implementing the client-side portion of a digital assistant, according to various examples.

FIG. 4 is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface, according to various examples.

FIG. 5A illustrates an exemplary user interface for a menu of applications on a portable multifunction device, according to various examples.

FIG. 5B illustrates an exemplary user interface for a multifunction device with a touch-sensitive surface that is separate from the display, according to various examples.

FIG. 6A illustrates a personal electronic device, according to various examples.

FIG. 6B is a block diagram illustrating a personal electronic device, according to various examples.

FIG. 7A is a block diagram illustrating a digital assistant system or a server portion thereof, according to various examples.

FIG. 7B illustrates the functions of the digital assistant shown in FIG. 7A, according to various examples.

FIG. 7C illustrates a portion of an ontology, according to various examples.

FIGS. 8A-8C illustrate a process for integrating a digital assistant with telephony, according to various examples.

FIG. 9 illustrates a process for integrating a digital assistant with telephony, according to various examples.

FIGS. 10A-10B illustrate a process for integrating a digital assistant with telephony, according to various examples.

FIGS. 11A-11B illustrate a process for integrating a digital assistant with telephony, according to various examples.

FIGS. 12A-12D illustrate a process for integrating a digital assistant with telephony, according to various examples.

FIG. 13 illustrates a process for integrating a digital assistant with telephony, according to various examples.

FIG. 14 illustrates a process for integrating a digital assistant with telephony, according to various examples.

FIGS. 15A-15B illustrate a process for integrating a digital assistant with telephony, according to various examples.

DETAILED DESCRIPTION

In the following description of examples, reference is made to the accompanying drawings in which are shown by way of illustration specific examples that can be practiced. It is to be understood that other examples can be used and structural changes can be made without departing from the scope of the various examples.

Conventional techniques for integrating digital assistants with telephony are generally lacking. In particular, traditional telephonic systems typically do not include functionality for effectively recognizing both speech input directed to the digital assistant and speech input intended for participants of a communication session (e.g., a phone call). Such systems are also not configured to efficiently allow a user to “hand off” a phone call to a digital assistant, such that the digital assistant may then interact with other parties within the session and otherwise act on behalf of the user

within the communication session. By virtue of the above, these conventional devices also do not permit a user to monitor a communication session in which a digital assistant is acting on their behalf. As a result, an improved system for integrating digital assistants with telephony is desired.

Although the following description uses terms “first,” “second,” etc. to describe various elements, these elements should not be limited by the terms. These terms are only used to distinguish one element from another. For example, a first input could be termed a second input, and, similarly, a second input could be termed a first input, without departing from the scope of the various described examples. The first input and the second input are both inputs and, in some cases, are separate and different inputs.

The terminology used in the description of the various described examples herein is for the purpose of describing particular examples only and is not intended to be limiting. As used in the description of the various described examples and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

The term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” may be construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

1. System and Environment

FIG. 1 illustrates a block diagram of system 100 according to various examples. In some examples, system 100 implements a digital assistant. The terms “digital assistant,” “virtual assistant,” “intelligent automated assistant,” or “automatic digital assistant” refer to any information processing system that interprets natural language input in spoken and/or textual form to infer user intent, and performs actions based on the inferred user intent. For example, to act on an inferred user intent, the system performs one or more of the following: identifying a task flow with steps and parameters designed to accomplish the inferred user intent, inputting specific requirements from the inferred user intent into the task flow; executing the task flow by invoking programs, methods, services, APIs, or the like; and generating output responses to the user in an audible (e.g., speech) and/or visual form.

Specifically, a digital assistant is capable of accepting a user request at least partially in the form of a natural language command, request, statement, narrative, and/or inquiry. Typically, the user request seeks either an informational answer or performance of a task by the digital assistant. A satisfactory response to the user request includes a provision of the requested informational answer, a performance of the requested task, or a combination of the two. For example, a user asks the digital assistant a question, such as “Where am I right now?” Based on the user’s current location, the digital assistant answers, “You are in Central

Park near the west gate.” The user also requests the performance of a task, for example, “Please invite my friends to my girlfriend’s birthday party next week.” In response, the digital assistant can acknowledge the request by saying “Yes, right away,” and then send a suitable calendar invite on behalf of the user to each of the user’s friends listed in the user’s electronic address book. During performance of a requested task, the digital assistant sometimes interacts with the user in a continuous dialogue involving multiple exchanges of information over an extended period of time. There are numerous other ways of interacting with a digital assistant to request information or performance of various tasks. In addition to providing verbal responses and taking programmed actions, the digital assistant also provides responses in other visual or audio forms, e.g., as text, alerts, music, videos, animations, etc.

As shown in FIG. 1, in some examples, a digital assistant is implemented according to a client-server model. The digital assistant includes client-side portion 102 (hereafter “DA client 102”) executed on user device 104 and server-side portion 106 (hereafter “DA server 106”) executed on server system 108. DA client 102 communicates with DA server 106 through one or more networks 110. DA client 102 provides client-side functionalities such as user-facing input and output processing and communication with DA server 106. DA server 106 provides server-side functionalities for any number of DA clients 102 each residing on a respective user device 104.

In some examples, DA server 106 includes client-facing I/O interface 112, one or more processing modules 114, data and models 116, and I/O interface to external services 118. The client-facing I/O interface 112 facilitates the client-facing input and output processing for DA server 106. One or more processing modules 114 utilize data and models 116 to process speech input and determine the user’s intent based on natural language input. Further, one or more processing modules 114 perform task execution based on inferred user intent. In some examples, DA server 106 communicates with external services 120 through network(s) 110 for task completion or information acquisition. I/O interface to external services 118 facilitates such communications.

User device 104 can be any suitable electronic device. In some examples, user device 104 is a portable multifunctional device (e.g., device 200, described below with reference to FIG. 2A), a multifunctional device (e.g., device 400, described below with reference to FIG. 4), or a personal electronic device (e.g., device 600, described below with reference to FIGS. 6A-6B.) A portable multifunctional device is, for example, a mobile telephone that also contains other functions, such as PDA and/or music player functions. Specific examples of portable multifunction devices include the Apple Watch®, iPhone®, iPod Touch®, and iPad® devices from Apple Inc. of Cupertino, California. Other examples of portable multifunction devices include, without limitation, earphones/headphones, speakers, and laptop or tablet computers. Further, in some examples, user device 104 is a non-portable multifunctional device. In particular, user device 104 is a desktop computer, a game console, a speaker, a television, or a television set-top box. In some examples, user device 104 includes a touch-sensitive surface (e.g., touch screen displays and/or touchpads). Further, user device 104 optionally includes one or more other physical user-interface devices, such as a physical keyboard, a mouse, and/or a joystick. Various examples of electronic devices, such as multifunctional devices, are described below in greater detail.

Examples of communication network(s) **110** include local area networks (LAN) and wide area networks (WAN), e.g., the Internet. Communication network(s) **110** is implemented using any known network protocol, including various wired or wireless protocols, such as, for example, Ethernet, Universal Serial Bus (USB), FIREWIRE, Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Wi-Fi, voice over Internet Protocol (VoIP), Wi-MAX, or any other suitable communication protocol.

Server system **108** is implemented on one or more standalone data processing apparatus or a distributed network of computers. In some examples, server system **108** also employs various virtual devices and/or services of third-party service providers (e.g., third-party cloud service providers) to provide the underlying computing resources and/or infrastructure resources of server system **108**.

In some examples, user device **104** communicates with DA server **106** via second user device **122**. Second user device **122** is similar or identical to user device **104**. For example, second user device **122** is similar to devices **200**, **400**, or **600** described below with reference to FIGS. 2A, 4, and 6A-6B. User device **104** is configured to communicatively couple to second user device **122** via a direct communication connection, such as Bluetooth, NFC, BTLE, or the like, or via a wired or wireless network, such as a local Wi-Fi network. In some examples, second user device **122** is configured to act as a proxy between user device **104** and DA server **106**. For example, DA client **102** of user device **104** is configured to transmit information (e.g., a user request received at user device **104**) to DA server **106** via second user device **122**. DA server **106** processes the information and returns relevant data (e.g., data content responsive to the user request) to user device **104** via second user device **122**.

In some examples, user device **104** is configured to communicate abbreviated requests for data to second user device **122** to reduce the amount of information transmitted from user device **104**. Second user device **122** is configured to determine supplemental information to add to the abbreviated request to generate a complete request to transmit to DA server **106**. This system architecture can advantageously allow user device **104** having limited communication capabilities and/or limited battery power (e.g., a watch or a similar compact electronic device) to access services provided by DA server **106** by using second user device **122**, having greater communication capabilities and/or battery power (e.g., a mobile phone, laptop computer, tablet computer, or the like), as a proxy to DA server **106**. While only two user devices **104** and **122** are shown in FIG. 1, it should be appreciated that system **100**, in some examples, includes any number and type of user devices configured in this proxy configuration to communicate with DA server system **106**.

Although the digital assistant shown in FIG. 1 includes both a client-side portion (e.g., DA client **102**) and a server-side portion (e.g., DA server **106**), in some examples, the functions of a digital assistant are implemented as a standalone application installed on a user device. In addition, the divisions of functionalities between the client and server portions of the digital assistant can vary in different implementations. For instance, in some examples, the DA client is a thin-client that provides only user-facing input and output processing functions, and delegates all other functionalities of the digital assistant to a backend server.

2. Electronic Devices

Attention is now directed toward embodiments of electronic devices for implementing the client-side portion of a digital assistant. FIG. 2A is a block diagram illustrating portable multifunction device **200** with touch-sensitive display system **212** in accordance with some embodiments. Touch-sensitive display **212** is sometimes called a “touch screen” for convenience and is sometimes known as or called a “touch-sensitive display system.” Device **200** includes memory **202** (which optionally includes one or more computer-readable storage mediums), memory controller **222**, one or more processing units (CPUs) **220**, peripherals interface **218**, RF circuitry **208**, audio circuitry **210**, speaker **211**, microphone **213**, input/output (I/O) subsystem **206**, other input control devices **216**, and external port **224**. Device **200** optionally includes one or more optical sensors **264**. Device **200** optionally includes one or more contact intensity sensors **265** for detecting intensity of contacts on device **200** (e.g., a touch-sensitive surface such as touch-sensitive display system **212** of device **200**). Device **200** optionally includes one or more tactile output generators **267** for generating tactile outputs on device **200** (e.g., generating tactile outputs on a touch-sensitive surface such as touch-sensitive display system **212** of device **200** or touchpad **455** of device **400**). These components optionally communicate over one or more communication buses or signal lines **203**.

As used in the specification and claims, the term “intensity” of a contact on a touch-sensitive surface refers to the force or pressure (force per unit area) of a contact (e.g., a finger contact) on the touch-sensitive surface, or to a substitute (proxy) for the force or pressure of a contact on the touch-sensitive surface. The intensity of a contact has a range of values that includes at least four distinct values and more typically includes hundreds of distinct values (e.g., at least 256). Intensity of a contact is, optionally, determined (or measured) using various approaches and various sensors or combinations of sensors. For example, one or more force sensors underneath or adjacent to the touch-sensitive surface are, optionally, used to measure force at various points on the touch-sensitive surface. In some implementations, force measurements from multiple force sensors are combined (e.g., a weighted average) to determine an estimated force of a contact. Similarly, a pressure-sensitive tip of a stylus is, optionally, used to determine a pressure of the stylus on the touch-sensitive surface. Alternatively, the size of the contact area detected on the touch-sensitive surface and/or changes thereto, the capacitance of the touch-sensitive surface proximate to the contact and/or changes thereto, and/or the resistance of the touch-sensitive surface proximate to the contact and/or changes thereto are, optionally, used as a substitute for the force or pressure of the contact on the touch-sensitive surface. In some implementations, the substitute measurements for contact force or pressure are used directly to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is described in units corresponding to the substitute measurements). In some implementations, the substitute measurements for contact force or pressure are converted to an estimated force or pressure, and the estimated force or pressure is used to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is a pressure threshold measured in units of pressure). Using the intensity of a contact as an attribute of a user input allows for user access to additional device functionality that may otherwise not be accessible by the user on a reduced-size device with limited real estate for displaying affordances (e.g., on a touch-sensitive display) and/or receiving user input (e.g., via a

touch-sensitive display, a touch-sensitive surface, or a physical/mechanical control such as a knob or a button).

As used in the specification and claims, the term “tactile output” refers to physical displacement of a device relative to a previous position of the device, physical displacement of a component (e.g., a touch-sensitive surface) of a device relative to another component (e.g., housing) of the device, or displacement of the component relative to a center of mass of the device that will be detected by a user with the user’s sense of touch. For example, in situations where the device or the component of the device is in contact with a surface of a user that is sensitive to touch (e.g., a finger, palm, or other part of a user’s hand), the tactile output generated by the physical displacement will be interpreted by the user as a tactile sensation corresponding to a perceived change in physical characteristics of the device or the component of the device. For example, movement of a touch-sensitive surface (e.g., a touch-sensitive display or trackpad) is, optionally, interpreted by the user as a “down click” or “up click” of a physical actuator button. In some cases, a user will feel a tactile sensation such as an “down click” or “up click” even when there is no movement of a physical actuator button associated with the touch-sensitive surface that is physically pressed (e.g., displaced) by the user’s movements. As another example, movement of the touch-sensitive surface is, optionally, interpreted or sensed by the user as “roughness” of the touch-sensitive surface, even when there is no change in smoothness of the touch-sensitive surface. While such interpretations of touch by a user will be subject to the individualized sensory perceptions of the user, there are many sensory perceptions of touch that are common to a large majority of users. Thus, when a tactile output is described as corresponding to a particular sensory perception of a user (e.g., an “up click,” a “down click,” “roughness”), unless otherwise stated, the generated tactile output corresponds to physical displacement of the device or a component thereof that will generate the described sensory perception for a typical (or average) user.

It should be appreciated that device **200** is only one example of a portable multifunction device, and that device **200** optionally has more or fewer components than shown, optionally combines two or more components, or optionally has a different configuration or arrangement of the components. The various components shown in FIG. **2A** are implemented in hardware, software, or a combination of both hardware and software, including one or more signal processing and/or application-specific integrated circuits.

Memory **202** includes one or more computer-readable storage mediums. The computer-readable storage mediums are, for example, tangible and non-transitory. Memory **202** includes high-speed random access memory and also includes non-volatile memory, such as one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Memory controller **222** controls access to memory **202** by other components of device **200**.

In some examples, a non-transitory computer-readable storage medium of memory **202** is used to store instructions (e.g., for performing aspects of processes described below) for use by or in connection with an instruction execution system, apparatus, or device, such as a computer-based system, processor-containing system, or other system that can fetch the instructions from the instruction execution system, apparatus, or device and execute the instructions. In other examples, the instructions (e.g., for performing aspects of the processes described below) are stored on a non-transitory computer-readable storage medium (not shown)

of the server system **108** or are divided between the non-transitory computer-readable storage medium of memory **202** and the non-transitory computer-readable storage medium of server system **108**.

Peripherals interface **218** is used to couple input and output peripherals of the device to CPU **220** and memory **202**. The one or more processors **220** run or execute various software programs and/or sets of instructions stored in memory **202** to perform various functions for device **200** and to process data. In some embodiments, peripherals interface **218**, CPU **220**, and memory controller **222** are implemented on a single chip, such as chip **204**. In some other embodiments, they are implemented on separate chips.

RF (radio frequency) circuitry **208** receives and sends RF signals, also called electromagnetic signals. RF circuitry **208** converts electrical signals to/from electromagnetic signals and communicates with communications networks and other communications devices via the electromagnetic signals. RF circuitry **208** optionally includes well-known circuitry for performing these functions, including but not limited to an antenna system, an RF transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a CODEC chipset, a subscriber identity module (SIM) card, memory, and so forth. RF circuitry **208** optionally communicates with networks, such as the Internet, also referred to as the World Wide Web (WWW), an intranet and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN) and/or a metropolitan area network (MAN), and other devices by wireless communication. The RF circuitry **208** optionally includes well-known circuitry for detecting near field communication (NFC) fields, such as by a short-range communication radio. The wireless communication optionally uses any of a plurality of communications standards, protocols, and technologies, including but not limited to Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), high-speed downlink packet access (HSDPA), high-speed uplink packet access (HSUPA), Evolution, Data-Only (EV-DO), HSPA, HSPA+, Dual-Cell HSPA (DC-HSPA), long term evolution (LTE), near field communication (NFC), wideband code division multiple access (W-CDMA), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Bluetooth Low Energy (BTLE), Wireless Fidelity (Wi-Fi) (e.g., IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, IEEE 802.11n, and/or IEEE 802.11ac), voice over Internet Protocol (VoIP), Wi-MAX, a protocol for e mail (e.g., Internet message access protocol (IMAP) and/or post office protocol (POP)), instant messaging (e.g., extensible messaging and presence protocol (XMPP), Session Initiation Protocol for Instant Messaging and Presence Leveraging Extensions (SIMPLE), Instant Messaging and Presence Service (IMPS)), and/or Short Message Service (SMS), or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

Audio circuitry **210**, speaker **211**, and microphone **213** provide an audio interface between a user and device **200**. Audio circuitry **210** receives audio data from peripherals interface **218**, converts the audio data to an electrical signal, and transmits the electrical signal to speaker **211**. Speaker **211** converts the electrical signal to human-audible sound waves. Audio circuitry **210** also receives electrical signals converted by microphone **213** from sound waves. Audio circuitry **210** converts the electrical signal to audio data and transmits the audio data to peripherals interface **218** for processing. Audio data are retrieved from and/or transmitted to memory **202** and/or RF circuitry **208** by peripherals

interface **218**. In some embodiments, audio circuitry **210** also includes a headset jack (e.g., **312**, FIG. 3). The headset jack provides an interface between audio circuitry **210** and removable audio input/output peripherals, such as output-only headphones or a headset with both output (e.g., a headphone for one or both ears) and input (e.g., a microphone).

I/O subsystem **206** couples input/output peripherals on device **200**, such as touch screen **212** and other input control devices **216**, to peripherals interface **218**. I/O subsystem **206** optionally includes display controller **256**, optical sensor controller **258**, intensity sensor controller **259**, haptic feedback controller **261**, and one or more input controllers **260** for other input or control devices. The one or more input controllers **260** receive/send electrical signals from/to other input control devices **216**. The other input control devices **216** optionally include physical buttons (e.g., push buttons, rocker buttons, etc.), dials, slider switches, joysticks, click wheels, and so forth. In some alternate embodiments, input controller(s) **260** are, optionally, coupled to any (or none) of the following: a keyboard, an infrared port, a USB port, and a pointer device such as a mouse. The one or more buttons (e.g., **308**, FIG. 3) optionally include an up/down button for volume control of speaker **211** and/or microphone **213**. The one or more buttons optionally include a push button (e.g., **306**, FIG. 3).

A quick press of the push button disengages a lock of touch screen **212** or begin a process that uses gestures on the touch screen to unlock the device, as described in U.S. patent application Ser. No. 11/322,549, "Unlocking a Device by Performing Gestures on an Unlock Image," filed Dec. 23, 2005, U.S. Pat. No. 7,657,849, which is hereby incorporated by reference in its entirety. A longer press of the push button (e.g., **306**) turns power to device **200** on or off. The user is able to customize a functionality of one or more of the buttons. Touch screen **212** is used to implement virtual or soft buttons and one or more soft keyboards.

Touch-sensitive display **212** provides an input interface and an output interface between the device and a user. Display controller **256** receives and/or sends electrical signals from/to touch screen **212**. Touch screen **212** displays visual output to the user. The visual output includes graphics, text, icons, video, and any combination thereof (collectively termed "graphics"). In some embodiments, some or all of the visual output correspond to user-interface objects.

Touch screen **212** has a touch-sensitive surface, sensor, or set of sensors that accepts input from the user based on haptic and/or tactile contact. Touch screen **212** and display controller **256** (along with any associated modules and/or sets of instructions in memory **202**) detect contact (and any movement or breaking of the contact) on touch screen **212** and convert the detected contact into interaction with user-interface objects (e.g., one or more soft keys, icons, web pages, or images) that are displayed on touch screen **212**. In an exemplary embodiment, a point of contact between touch screen **212** and the user corresponds to a finger of the user.

Touch screen **212** uses LCD (liquid crystal display) technology, LPD (light emitting polymer display) technology, or LED (light emitting diode) technology, although other display technologies may be used in other embodiments. Touch screen **212** and display controller **256** detect contact and any movement or breaking thereof using any of a plurality of touch sensing technologies now known or later developed, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with touch screen **212**.

In an exemplary embodiment, projected mutual capacitance sensing technology is used, such as that found in the iPhone® and iPod Touch® from Apple Inc. of Cupertino, California.

A touch-sensitive display in some embodiments of touch screen **212** is analogous to the multi-touch sensitive touchpads described in the following U.S. Pat. No. 6,323,846 (Westerman et al.), U.S. Pat. No. 6,570,557 (Westerman et al.), and/or U.S. Pat. No. 6,677,932 (Westerman), and/or U.S. Patent Publication 2002/0015024A1, each of which is hereby incorporated by reference in its entirety. However, touch screen **212** displays visual output from device **200**, whereas touch-sensitive touchpads do not provide visual output.

A touch-sensitive display in some embodiments of touch screen **212** is as described in the following applications: (1) U.S. patent application Ser. No. 11/381,313, "Multipoint Touch Surface Controller," filed May 2, 2006; (2) U.S. patent application Ser. No. 10/840,862, "Multipoint Touchscreen," filed May 6, 2004; (3) U.S. patent application Ser. No. 10/903,964, "Gestures For Touch Sensitive Input Devices," filed Jul. 30, 2004; (4) U.S. patent application Ser. No. 11/048,264, "Gestures For Touch Sensitive Input Devices," filed Jan. 31, 2005; (5) U.S. patent application Ser. No. 11/038,590, "Mode-Based Graphical User Interfaces For Touch Sensitive Input Devices," filed Jan. 18, 2005; (6) U.S. patent application Ser. No. 11/228,758, "Virtual Input Device Placement On A Touch Screen User Interface," filed Sep. 16, 2005; (7) U.S. patent application Ser. No. 11/228,700, "Operation Of A Computer With A Touch Screen Interface," filed Sep. 16, 2005; (8) U.S. patent application Ser. No. 11/228,737, "Activating Virtual Keys Of A Touch-Screen Virtual Keyboard," filed Sep. 16, 2005; and (9) U.S. patent application Ser. No. 11/367,749, "Multi-Functional Hand-Held Device," filed Mar. 3, 2006. All of these applications are incorporated by reference herein in their entirety.

Touch screen **212** has, for example, a video resolution in excess of 100 dpi. In some embodiments, the touch screen has a video resolution of approximately 160 dpi. The user makes contact with touch screen **212** using any suitable object or appendage, such as a stylus, a finger, and so forth. In some embodiments, the user interface is designed to work primarily with finger-based contacts and gestures, which can be less precise than stylus-based input due to the larger area of contact of a finger on the touch screen. In some embodiments, the device translates the rough finger-based input into a precise pointer/cursor position or command for performing the actions desired by the user.

In some embodiments, in addition to the touch screen, device **200** includes a touchpad (not shown) for activating or deactivating particular functions. In some embodiments, the touchpad is a touch-sensitive area of the device that, unlike the touch screen, does not display visual output. The touchpad is a touch-sensitive surface that is separate from touch screen **212** or an extension of the touch-sensitive surface formed by the touch screen.

Device **200** also includes power system **262** for powering the various components. Power system **262** includes a power management system, one or more power sources (e.g., battery, alternating current (AC)), a recharging system, a power failure detection circuit, a power converter or inverter, a power status indicator (e.g., a light-emitting diode (LED)) and any other components associated with the generation, management and distribution of power in portable devices.

Device **200** also includes one or more optical sensors **264**. FIG. 2A shows an optical sensor coupled to optical sensor

controller **258** in I/O subsystem **206**. Optical sensor **264** includes charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS) phototransistors. Optical sensor **264** receives light from the environment, projected through one or more lenses, and converts the light to data representing an image. In conjunction with imaging module **243** (also called a camera module), optical sensor **264** captures still images or video. In some embodiments, an optical sensor is located on the back of device **200**, opposite touch screen display **212** on the front of the device so that the touch screen display is used as a viewfinder for still and/or video image acquisition. In some embodiments, an optical sensor is located on the front of the device so that the user's image is obtained for video conferencing while the user views the other video conference participants on the touch screen display. In some embodiments, the position of optical sensor **264** can be changed by the user (e.g., by rotating the lens and the sensor in the device housing) so that a single optical sensor **264** is used along with the touch screen display for both video conferencing and still and/or video image acquisition.

Device **200** optionally also includes one or more contact intensity sensors **265**. FIG. 2A shows a contact intensity sensor coupled to intensity sensor controller **259** in I/O subsystem **206**. Contact intensity sensor **265** optionally includes one or more piezoresistive strain gauges, capacitive force sensors, electric force sensors, piezoelectric force sensors, optical force sensors, capacitive touch-sensitive surfaces, or other intensity sensors (e.g., sensors used to measure the force (or pressure) of a contact on a touch-sensitive surface). Contact intensity sensor **265** receives contact intensity information (e.g., pressure information or a proxy for pressure information) from the environment. In some embodiments, at least one contact intensity sensor is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system **212**). In some embodiments, at least one contact intensity sensor is located on the back of device **200**, opposite touch screen display **212**, which is located on the front of device **200**.

Device **200** also includes one or more proximity sensors **266**. FIG. 2A shows proximity sensor **266** coupled to peripherals interface **218**. Alternately, proximity sensor **266** is coupled to input controller **260** in I/O subsystem **206**. Proximity sensor **266** is performed as described in U.S. patent application Ser. No. 11/241,839, "Proximity Detector In Handheld Device"; Ser. No. 11/240,788, "Proximity Detector In Handheld Device"; Ser. No. 11/620,702, "Using Ambient Light Sensor To Augment Proximity Sensor Output"; Ser. No. 11/586,862, "Automated Response To And Sensing Of User Activity In Portable Devices"; and Ser. No. 11/638,251, "Methods And Systems For Automatic Configuration Of Peripherals," which are hereby incorporated by reference in their entirety. In some embodiments, the proximity sensor turns off and disables touch screen **212** when the multifunction device is placed near the user's ear (e.g., when the user is making a phone call).

Device **200** optionally also includes one or more tactile output generators **267**. FIG. 2A shows a tactile output generator coupled to haptic feedback controller **261** in I/O subsystem **206**. Tactile output generator **267** optionally includes one or more electroacoustic devices such as speakers or other audio components and/or electromechanical devices that convert energy into linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating component (e.g., a component that converts electrical signals into tactile outputs on the device). Contact intensity

sensor **265** receives tactile feedback generation instructions from haptic feedback module **233** and generates tactile outputs on device **200** that are capable of being sensed by a user of device **200**. In some embodiments, at least one tactile output generator is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system **212**) and, optionally, generates a tactile output by moving the touch-sensitive surface vertically (e.g., in/out of a surface of device **200**) or laterally (e.g., back and forth in the same plane as a surface of device **200**). In some embodiments, at least one tactile output generator sensor is located on the back of device **200**, opposite touch screen display **212**, which is located on the front of device **200**.

Device **200** also includes one or more accelerometers **268**. FIG. 2A shows accelerometer **268** coupled to peripherals interface **218**. Alternately, accelerometer **268** is coupled to an input controller **260** in I/O subsystem **206**. Accelerometer **268** performs, for example, as described in U.S. Patent Publication No. 20050190059, "Acceleration-based Theft Detection System for Portable Electronic Devices," and U.S. Patent Publication No. 20060017692, "Methods And Apparatuses For Operating A Portable Device Based On An Accelerometer," both of which are incorporated by reference herein in their entirety. In some embodiments, information is displayed on the touch screen display in a portrait view or a landscape view based on an analysis of data received from the one or more accelerometers. Device **200** optionally includes, in addition to accelerometer(s) **268**, a magnetometer (not shown) and a GPS (or GLONASS or other global navigation system) receiver (not shown) for obtaining information concerning the location and orientation (e.g., portrait or landscape) of device **200**.

In some embodiments, the software components stored in memory **202** include operating system **226**, communication module (or set of instructions) **228**, contact/motion module (or set of instructions) **230**, graphics module (or set of instructions) **232**, text input module (or set of instructions) **234**, Global Positioning System (GPS) module (or set of instructions) **235**, Digital Assistant Client Module **229**, and applications (or sets of instructions) **236**. Further, memory **202** stores data and models, such as user data and models **231**. Furthermore, in some embodiments, memory **202** (FIG. 2A) or **470** (FIG. 4) stores device/global internal state **257**, as shown in FIGS. 2A and 4. Device/global internal state **257** includes one or more of: active application state, indicating which applications, if any, are currently active; display state, indicating what applications, views or other information occupy various regions of touch screen display **212**; sensor state, including information obtained from the device's various sensors and input control devices **216**; and location information concerning the device's location and/or attitude.

Operating system **226** (e.g., Darwin, RTXC, LINUX, UNIX, OS X, iOS, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communication between various hardware and software components.

Communication module **228** facilitates communication with other devices over one or more external ports **224** and also includes various software components for handling data received by RF circuitry **208** and/or external port **224**. External port **224** (e.g., Universal Serial Bus (USB), FIREWIRE, etc.) is adapted for coupling directly to other devices or indirectly over a network (e.g., the Internet, wireless LAN, etc.). In some embodiments, the external port is a multi-pin (e.g., 30-pin) connector that is the same as, or

similar to and/or compatible with, the 30-pin connector used on iPod® (trademark of Apple Inc.) devices.

Contact/motion module **230** optionally detects contact with touch screen **212** (in conjunction with display controller **256**) and other touch-sensitive devices (e.g., a touchpad or physical click wheel). Contact/motion module **230** includes various software components for performing various operations related to detection of contact, such as determining if contact has occurred (e.g., detecting a finger-down event), determining an intensity of the contact (e.g., the force or pressure of the contact or a substitute for the force or pressure of the contact), determining if there is movement of the contact and tracking the movement across the touch-sensitive surface (e.g., detecting one or more finger-dragging events), and determining if the contact has ceased (e.g., detecting a finger-up event or a break in contact). Contact/motion module **230** receives contact data from the touch-sensitive surface. Determining movement of the point of contact, which is represented by a series of contact data, optionally includes determining speed (magnitude), velocity (magnitude and direction), and/or an acceleration (a change in magnitude and/or direction) of the point of contact. These operations are, optionally, applied to single contacts (e.g., one finger contacts) or to multiple simultaneous contacts (e.g., “multitouch”/multiple finger contacts). In some embodiments, contact/motion module **230** and display controller **256** detect contact on a touchpad.

In some embodiments, contact/motion module **230** uses a set of one or more intensity thresholds to determine whether an operation has been performed by a user (e.g., to determine whether a user has “clicked” on an icon). In some embodiments, at least a subset of the intensity thresholds are determined in accordance with software parameters (e.g., the intensity thresholds are not determined by the activation thresholds of particular physical actuators and can be adjusted without changing the physical hardware of device **200**). For example, a mouse “click” threshold of a trackpad or touch screen display can be set to any of a large range of predefined threshold values without changing the trackpad or touch screen display hardware. Additionally, in some implementations, a user of the device is provided with software settings for adjusting one or more of the set of intensity thresholds (e.g., by adjusting individual intensity thresholds and/or by adjusting a plurality of intensity thresholds at once with a system-level click “intensity” parameter).

Contact/motion module **230** optionally detects a gesture input by a user. Different gestures on the touch-sensitive surface have different contact patterns (e.g., different motions, timings, and/or intensities of detected contacts). Thus, a gesture is, optionally, detected by detecting a particular contact pattern. For example, detecting a finger tap gesture includes detecting a finger-down event followed by detecting a finger-up (liftoff) event at the same position (or substantially the same position) as the finger-down event (e.g., at the position of an icon). As another example, detecting a finger swipe gesture on the touch-sensitive surface includes detecting a finger-down event followed by detecting one or more finger-dragging events, and subsequently followed by detecting a finger-up (liftoff) event.

Graphics module **232** includes various known software components for rendering and displaying graphics on touch screen **212** or other display, including components for changing the visual impact (e.g., brightness, transparency, saturation, contrast, or other visual property) of graphics that are displayed. As used herein, the term “graphics” includes any object that can be displayed to a user, including, without

limitation, text, web pages, icons (such as user-interface objects including soft keys), digital images, videos, animations, and the like.

In some embodiments, graphics module **232** stores data representing graphics to be used. Each graphic is, optionally, assigned a corresponding code. Graphics module **232** receives, from applications etc., one or more codes specifying graphics to be displayed along with, if necessary, coordinate data and other graphic property data, and then generates screen image data to output to display controller **256**.

Haptic feedback module **233** includes various software components for generating instructions used by tactile output generator(s) **267** to produce tactile outputs at one or more locations on device **200** in response to user interactions with device **200**.

Text input module **234**, which is, in some examples, a component of graphics module **232**, provides soft keyboards for entering text in various applications (e.g., contacts module **237**, e-mail client module **240**, IM module **241**, browser module **247**, and any other application that needs text input).

GPS module **235** determines the location of the device and provides this information for use in various applications (e.g., to telephone module **238** for use in location-based dialing; to camera module **243** as picture/video metadata; and to applications that provide location-based services such as weather widgets, local yellow page widgets, and map/navigation widgets).

Digital assistant client module **229** includes various client-side digital assistant instructions to provide the client-side functionalities of the digital assistant. For example, digital assistant client module **229** is capable of accepting voice input (e.g., speech input), text input, touch input, and/or gestural input through various user interfaces (e.g., microphone **213**, accelerometer(s) **268**, touch-sensitive display system **212**, optical sensor(s) **264**, other input control devices **216**, etc.) of portable multifunction device **200**. Digital assistant client module **229** is also capable of providing output in audio (e.g., speech output), visual, and/or tactile forms through various output interfaces (e.g., speaker **211**, touch-sensitive display system **212**, tactile output generator(s) **267**, etc.) of portable multifunction device **200**. For example, output is provided as voice, sound, alerts, text messages, menus, graphics, videos, animations, vibrations, and/or combinations of two or more of the above. During operation, digital assistant client module **229** communicates with DA server **106** using RF circuitry **208**.

User data and models **231** include various data associated with the user (e.g., user-specific vocabulary data, user preference data, user-specified name pronunciations, data from the user’s electronic address book, to-do lists, shopping lists, etc.) to provide the client-side functionalities of the digital assistant. Further, user data and models **231** include various models (e.g., speech recognition models, statistical language models, natural language processing models, ontology, task flow models, service models, etc.) for processing user input and determining user intent.

In some examples, digital assistant client module **229** utilizes the various sensors, subsystems, and peripheral devices of portable multifunction device **200** to gather additional information from the surrounding environment of the portable multifunction device **200** to establish a context associated with a user, the current user interaction, and/or the current user input. In some examples, digital assistant client module **229** provides the contextual information or a subset thereof with the user input to DA server **106** to help infer the user’s intent. In some examples, the digital assistant

also uses the contextual information to determine how to prepare and deliver outputs to the user. Contextual information is referred to as context data.

In some examples, the contextual information that accompanies the user input includes sensor information, e.g., lighting, ambient noise, ambient temperature, images or videos of the surrounding environment, etc. In some examples, the contextual information can also include the physical state of the device, e.g., device orientation, device location, device temperature, power level, speed, acceleration, motion patterns, cellular signals strength, etc. In some examples, information related to the software state of DA server 106, e.g., running processes, installed programs, past and present network activities, background services, error logs, resources usage, etc., and of portable multifunction device 200 is provided to DA server 106 as contextual information associated with a user input.

In some examples, the digital assistant client module 229 selectively provides information (e.g., user data 231) stored on the portable multifunction device 200 in response to requests from DA server 106. In some examples, digital assistant client module 229 also elicits additional input from the user via a natural language dialogue or other user interfaces upon request by DA server 106. Digital assistant client module 229 passes the additional input to DA server 106 to help DA server 106 in intent deduction and/or fulfillment of the user's intent expressed in the user request.

A more detailed description of a digital assistant is described below with reference to FIGS. 7A-7C. It should be recognized that digital assistant client module 229 can include any number of the sub-modules of digital assistant module 726 described below.

Applications 236 include the following modules (or sets of instructions), or a subset or superset thereof:

- Contacts module 237 (sometimes called an address book or contact list);
- Telephone module 238;
- Video conference module 239;
- E-mail client module 240;
- Instant messaging (IM) module 241;
- Workout support module 242;
- Camera module 243 for still and/or video images;
- Image management module 244;
- Video player module;
- Music player module;
- Browser module 247;
- Calendar module 248;
- Widget modules 249, which includes, in some examples, one or more of: weather widget 249-1, stocks widget 249-2, calculator widget 249-3, alarm clock widget 249-4, dictionary widget 249-5, and other widgets obtained by the user, as well as user-created widgets 249-6;
- Widget creator module 250 for making user-created widgets 249-6;
- Search module 251;
- Video and music player module 252, which merges video player module and music player module;
- Notes module 253;
- Map module 254; and/or
- Online video module 255.

Examples of other applications 236 that are stored in memory 202 include other word processing applications, other image editing applications, drawing applications, presentation applications, JAVA-enabled applications, encryption, digital rights management, voice recognition, and voice replication.

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, contacts module 237 are used to manage an address book or contact list (e.g., stored in application internal state 292 of contacts module 237 in memory 202 or memory 470), including: adding name(s) to the address book; deleting name(s) from the address book; associating telephone number(s), e-mail address(es), physical address(es) or other information with a name; associating an image with a name; categorizing and sorting names; providing telephone numbers or e-mail addresses to initiate and/or facilitate communications by telephone module 238, video conference module 239, e-mail client module 240, or IM module 241; and so forth.

In conjunction with RF circuitry 208, audio circuitry 210, speaker 211, microphone 213, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, telephone module 238 are used to enter a sequence of characters corresponding to a telephone number, access one or more telephone numbers in contacts module 237, modify a telephone number that has been entered, dial a respective telephone number, conduct a conversation, and disconnect or hang up when the conversation is completed. As noted above, the wireless communication uses any of a plurality of communications standards, protocols, and technologies.

In conjunction with RF circuitry 208, audio circuitry 210, speaker 211, microphone 213, touch screen 212, display controller 256, optical sensor 264, optical sensor controller 258, contact/motion module 230, graphics module 232, text input module 234, contacts module 237, and telephone module 238, video conference module 239 includes executable instructions to initiate, conduct, and terminate a video conference between a user and one or more other participants in accordance with user instructions.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, e-mail client module 240 includes executable instructions to create, send, receive, and manage e-mail in response to user instructions. In conjunction with image management module 244, e-mail client module 240 makes it very easy to create and send e-mails with still or video images taken with camera module 243.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, the instant messaging module 241 includes executable instructions to enter a sequence of characters corresponding to an instant message, to modify previously entered characters, to transmit a respective instant message (for example, using a Short Message Service (SMS) or Multimedia Message Service (MMS) protocol for telephony-based instant messages or using XMPP, SIMPLE, or IMPS for Internet-based instant messages), to receive instant messages, and to view received instant messages. In some embodiments, transmitted and/or received instant messages include graphics, photos, audio files, video files and/or other attachments as are supported in an MMS and/or an Enhanced Messaging Service (EMS). As used herein, "instant messaging" refers to both telephony-based messages (e.g., messages sent using SMS or MMS) and Internet-based messages (e.g., messages sent using XMPP, SIMPLE, or IMPS).

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, GPS module 235, map module 254, and music player module, workout support

module **242** includes executable instructions to create work-outs (e.g., with time, distance, and/or calorie burning goals); communicate with workout sensors (sports devices); receive workout sensor data; calibrate sensors used to monitor a workout; select and play music for a workout; and display, store, and transmit workout data.

In conjunction with touch screen **212**, display controller **256**, optical sensor(s) **264**, optical sensor controller **258**, contact/motion module **230**, graphics module **232**, and image management module **244**, camera module **243** includes executable instructions to capture still images or video (including a video stream) and store them into memory **202**, modify characteristics of a still image or video, or delete a still image or video from memory **202**.

In conjunction with touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, text input module **234**, and camera module **243**, image management module **244** includes executable instructions to arrange, modify (e.g., edit), or otherwise manipulate, label, delete, present (e.g., in a digital slide show or album), and store still and/or video images.

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, and text input module **234**, browser module **247** includes executable instructions to browse the Internet in accordance with user instructions, including searching, linking to, receiving, and displaying web pages or portions thereof, as well as attachments and other files linked to web pages.

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, text input module **234**, e-mail client module **240**, and browser module **247**, calendar module **248** includes executable instructions to create, display, modify, and store calendars and data associated with calendars (e.g., calendar entries, to-do lists, etc.) in accordance with user instructions.

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, text input module **234**, and browser module **247**, widget modules **249** are mini-applications that can be downloaded and used by a user (e.g., weather widget **249-1**, stocks widget **249-2**, calculator widget **249-3**, alarm clock widget **249-4**, and dictionary widget **249-5**) or created by the user (e.g., user-created widget **249-6**). In some embodiments, a widget includes an HTML (Hypertext Markup Language) file, a CSS (Cascading Style Sheets) file, and a JavaScript file. In some embodiments, a widget includes an XML (Extensible Markup Language) file and a JavaScript file (e.g., Yahoo! Widgets).

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, text input module **234**, and browser module **247**, the widget creator module **250** are used by a user to create widgets (e.g., turning a user-specified portion of a web page into a widget).

In conjunction with touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, and text input module **234**, search module **251** includes executable instructions to search for text, music, sound, image, video, and/or other files in memory **202** that match one or more search criteria (e.g., one or more user-specified search terms) in accordance with user instructions.

In conjunction with touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, audio circuitry **210**, speaker **211**, RF circuitry **208**, and browser module **247**, video and music player module **252** includes executable instructions that allow the user to down-

load and play back recorded music and other sound files stored in one or more file formats, such as MP3 or AAC files, and executable instructions to display, present, or otherwise play back videos (e.g., on touch screen **212** or on an external, connected display via external port **224**). In some embodiments, device **200** optionally includes the functionality of an MP3 player, such as an iPod (trademark of Apple Inc.).

In conjunction with touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, and text input module **234**, notes module **253** includes executable instructions to create and manage notes, to-do lists, and the like in accordance with user instructions.

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, text input module **234**, GPS module **235**, and browser module **247**, map module **254** are used to receive, display, modify, and store maps and data associated with maps (e.g., driving directions, data on stores and other points of interest at or near a particular location, and other location-based data) in accordance with user instructions.

In conjunction with touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, audio circuitry **210**, speaker **211**, RF circuitry **208**, text input module **234**, e-mail client module **240**, and browser module **247**, online video module **255** includes instructions that allow the user to access, browse, receive (e.g., by streaming and/or download), play back (e.g., on the touch screen or on an external, connected display via external port **224**), send an e-mail with a link to a particular online video, and otherwise manage online videos in one or more file formats, such as H.264. In some embodiments, instant messaging module **241**, rather than e-mail client module **240**, is used to send a link to a particular online video. Additional description of the online video application can be found in U.S. Provisional Patent Application No. 60/936,562, "Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos," filed Jun. 20, 2007, and U.S. patent application Ser. No. 11/968,067, "Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos," filed Dec. 31, 2007, the contents of which are hereby incorporated by reference in their entirety.

Each of the above-identified modules and applications corresponds to a set of executable instructions for performing one or more functions described above and the methods described in this application (e.g., the computer-implemented methods and other information processing methods described herein). These modules (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules can be combined or otherwise rearranged in various embodiments. For example, video player module can be combined with music player module into a single module (e.g., video and music player module **252**, FIG. 2A). In some embodiments, memory **202** stores a subset of the modules and data structures identified above. Furthermore, memory **202** stores additional modules and data structures not described above.

In some embodiments, device **200** is a device where operation of a predefined set of functions on the device is performed exclusively through a touch screen and/or a touchpad. By using a touch screen and/or a touchpad as the primary input control device for operation of device **200**, the number of physical input control devices (such as push buttons, dials, and the like) on device **200** is reduced.

The predefined set of functions that are performed exclusively through a touch screen and/or a touchpad optionally include navigation between user interfaces. In some embodi-

ments, the touchpad, when touched by the user, navigates device **200** to a main, home, or root menu from any user interface that is displayed on device **200**. In such embodiments, a “menu button” is implemented using a touchpad. In some other embodiments, the menu button is a physical push button or other physical input control device instead of a touchpad.

FIG. 2B is a block diagram illustrating exemplary components for event handling in accordance with some embodiments. In some embodiments, memory **202** (FIG. 2A) or **470** (FIG. 4) includes event sorter **270** (e.g., in operating system **226**) and a respective application **236-1** (e.g., any of the aforementioned applications **237-251**, **255**, **480-490**).

Event sorter **270** receives event information and determines the application **236-1** and application view **291** of application **236-1** to which to deliver the event information. Event sorter **270** includes event monitor **271** and event dispatcher module **274**. In some embodiments, application **236-1** includes application internal state **292**, which indicates the current application view(s) displayed on touch-sensitive display **212** when the application is active or executing. In some embodiments, device/global internal state **257** is used by event sorter **270** to determine which application(s) is (are) currently active, and application internal state **292** is used by event sorter **270** to determine application views **291** to which to deliver event information.

In some embodiments, application internal state **292** includes additional information, such as one or more of: resume information to be used when application **236-1** resumes execution, user interface state information that indicates information being displayed or that is ready for display by application **236-1**, a state queue for enabling the user to go back to a prior state or view of application **236-1**, and a redo/undo queue of previous actions taken by the user.

Event monitor **271** receives event information from peripherals interface **218**. Event information includes information about a sub-event (e.g., a user touch on touch-sensitive display **212**, as part of a multi-touch gesture). Peripherals interface **218** transmits information it receives from I/O subsystem **206** or a sensor, such as proximity sensor **266**, accelerometer(s) **268**, and/or microphone **213** (through audio circuitry **210**). Information that peripherals interface **218** receives from I/O subsystem **206** includes information from touch-sensitive display **212** or a touch-sensitive surface.

In some embodiments, event monitor **271** sends requests to the peripherals interface **218** at predetermined intervals. In response, peripherals interface **218** transmits event information. In other embodiments, peripherals interface **218** transmits event information only when there is a significant event (e.g., receiving an input above a predetermined noise threshold and/or for more than a predetermined duration).

In some embodiments, event sorter **270** also includes a hit view determination module **272** and/or an active event recognizer determination module **273**.

Hit view determination module **272** provides software procedures for determining where a sub-event has taken place within one or more views when touch-sensitive display **212** displays more than one view. Views are made up of controls and other elements that a user can see on the display.

Another aspect of the user interface associated with an application is a set of views, sometimes herein called application views or user interface windows, in which information is displayed and touch-based gestures occur. The application views (of a respective application) in which

a touch is detected correspond to programmatic levels within a programmatic or view hierarchy of the application. For example, the lowest level view in which a touch is detected is called the hit view, and the set of events that are recognized as proper inputs is determined based, at least in part, on the hit view of the initial touch that begins a touch-based gesture.

Hit view determination module **272** receives information related to sub events of a touch-based gesture. When an application has multiple views organized in a hierarchy, hit view determination module **272** identifies a hit view as the lowest view in the hierarchy which should handle the sub-event. In most circumstances, the hit view is the lowest level view in which an initiating sub-event occurs (e.g., the first sub-event in the sequence of sub-events that form an event or potential event). Once the hit view is identified by the hit view determination module **272**, the hit view typically receives all sub-events related to the same touch or input source for which it was identified as the hit view.

Active event recognizer determination module **273** determines which view or views within a view hierarchy should receive a particular sequence of sub-events. In some embodiments, active event recognizer determination module **273** determines that only the hit view should receive a particular sequence of sub-events. In other embodiments, active event recognizer determination module **273** determines that all views that include the physical location of a sub-event are actively involved views, and therefore determines that all actively involved views should receive a particular sequence of sub-events. In other embodiments, even if touch sub-events were entirely confined to the area associated with one particular view, views higher in the hierarchy would still remain as actively involved views.

Event dispatcher module **274** dispatches the event information to an event recognizer (e.g., event recognizer **280**). In embodiments including active event recognizer determination module **273**, event dispatcher module **274** delivers the event information to an event recognizer determined by active event recognizer determination module **273**. In some embodiments, event dispatcher module **274** stores in an event queue the event information, which is retrieved by a respective event receiver **282**.

In some embodiments, operating system **226** includes event sorter **270**. Alternatively, application **236-1** includes event sorter **270**. In yet other embodiments, event sorter **270** is a stand-alone module, or a part of another module stored in memory **202**, such as contact/motion module **230**.

In some embodiments, application **236-1** includes a plurality of event handlers **290** and one or more application views **291**, each of which includes instructions for handling touch events that occur within a respective view of the application's user interface. Each application view **291** of the application **236-1** includes one or more event recognizers **280**. Typically, a respective application view **291** includes a plurality of event recognizers **280**. In other embodiments, one or more of event recognizers **280** are part of a separate module, such as a user interface kit (not shown) or a higher level object from which application **236-1** inherits methods and other properties. In some embodiments, a respective event handler **290** includes one or more of: data updater **276**, object updater **277**, GUI updater **278**, and/or event data **279** received from event sorter **270**. Event handler **290** utilizes or calls data updater **276**, object updater **277**, or GUI updater **278** to update the application internal state **292**. Alternatively, one or more of the application views **291** include one or more respective event handlers **290**. Also, in some embodiments, one or more of data updater

276, object updater 277, and GUI updater 278 are included in a respective application view 291.

A respective event recognizer 280 receives event information (e.g., event data 279) from event sorter 270 and identifies an event from the event information. Event recognizer 280 includes event receiver 282 and event comparator 284. In some embodiments, event recognizer 280 also includes at least a subset of: metadata 283, and event delivery instructions 288 (which include sub-event delivery instructions).

Event receiver 282 receives event information from event sorter 270. The event information includes information about a sub-event, for example, a touch or a touch movement. Depending on the sub-event, the event information also includes additional information, such as location of the sub-event. When the sub-event concerns motion of a touch, the event information also includes speed and direction of the sub-event. In some embodiments, events include rotation of the device from one orientation to another (e.g., from a portrait orientation to a landscape orientation, or vice versa), and the event information includes corresponding information about the current orientation (also called device attitude) of the device.

Event comparator 284 compares the event information to predefined event or sub-event definitions and, based on the comparison, determines an event or sub event, or determines or updates the state of an event or sub-event. In some embodiments, event comparator 284 includes event definitions 286. Event definitions 286 contain definitions of events (e.g., predefined sequences of sub-events), for example, event 1 (287-1), event 2 (287-2), and others. In some embodiments, sub-events in an event (287) include, for example, touch begin, touch end, touch movement, touch cancellation, and multiple touching. In one example, the definition for event 1 (287-1) is a double tap on a displayed object. The double tap, for example, comprises a first touch (touch begin) on the displayed object for a predetermined phase, a first liftoff (touch end) for a predetermined phase, a second touch (touch begin) on the displayed object for a predetermined phase, and a second liftoff (touch end) for a predetermined phase. In another example, the definition for event 2 (287-2) is a dragging on a displayed object. The dragging, for example, comprises a touch (or contact) on the displayed object for a predetermined phase, a movement of the touch across touch-sensitive display 212, and liftoff of the touch (touch end). In some embodiments, the event also includes information for one or more associated event handlers 290.

In some embodiments, event definition 287 includes a definition of an event for a respective user-interface object. In some embodiments, event comparator 284 performs a hit test to determine which user-interface object is associated with a sub-event. For example, in an application view in which three user-interface objects are displayed on touch-sensitive display 212, when a touch is detected on touch-sensitive display 212, event comparator 284 performs a hit test to determine which of the three user-interface objects is associated with the touch (sub-event). If each displayed object is associated with a respective event handler 290, the event comparator uses the result of the hit test to determine which event handler 290 should be activated. For example, event comparator 284 selects an event handler associated with the sub-event and the object triggering the hit test.

In some embodiments, the definition for a respective event (287) also includes delayed actions that delay delivery of the event information until after it has been determined

whether the sequence of sub-events does or does not correspond to the event recognizer's event type.

When a respective event recognizer 280 determines that the series of sub-events do not match any of the events in event definitions 286, the respective event recognizer 280 enters an event impossible, event failed, or event ended state, after which it disregards subsequent sub-events of the touch-based gesture. In this situation, other event recognizers, if any, that remain active for the hit view continue to track and process sub-events of an ongoing touch-based gesture.

In some embodiments, a respective event recognizer 280 includes metadata 283 with configurable properties, flags, and/or lists that indicate how the event delivery system should perform sub-event delivery to actively involved event recognizers. In some embodiments, metadata 283 includes configurable properties, flags, and/or lists that indicate how event recognizers interact, or are enabled to interact, with one another. In some embodiments, metadata 283 includes configurable properties, flags, and/or lists that indicate whether sub-events are delivered to varying levels in the view or programmatic hierarchy.

In some embodiments, a respective event recognizer 280 activates event handler 290 associated with an event when one or more particular sub-events of an event are recognized. In some embodiments, a respective event recognizer 280 delivers event information associated with the event to event handler 290. Activating an event handler 290 is distinct from sending (and deferred sending) sub-events to a respective hit view. In some embodiments, event recognizer 280 throws a flag associated with the recognized event, and event handler 290 associated with the flag catches the flag and performs a predefined process.

In some embodiments, event delivery instructions 288 include sub-event delivery instructions that deliver event information about a sub-event without activating an event handler. Instead, the sub-event delivery instructions deliver event information to event handlers associated with the series of sub-events or to actively involved views. Event handlers associated with the series of sub-events or with actively involved views receive the event information and perform a predetermined process.

In some embodiments, data updater 276 creates and updates data used in application 236-1. For example, data updater 276 updates the telephone number used in contacts module 237, or stores a video file used in video player module. In some embodiments, object updater 277 creates and updates objects used in application 236-1. For example, object updater 277 creates a new user-interface object or updates the position of a user-interface object. GUI updater 278 updates the GUI. For example, GUI updater 278 prepares display information and sends it to graphics module 232 for display on a touch-sensitive display.

In some embodiments, event handler(s) 290 includes or has access to data updater 276, object updater 277, and GUI updater 278. In some embodiments, data updater 276, object updater 277, and GUI updater 278 are included in a single module of a respective application 236-1 or application view 291. In other embodiments, they are included in two or more software modules.

It shall be understood that the foregoing discussion regarding event handling of user touches on touch-sensitive displays also applies to other forms of user inputs to operate multifunction devices 200 with input devices, not all of which are initiated on touch screens. For example, mouse movement and mouse button presses, optionally coordinated with single or multiple keyboard presses or holds; contact

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movements such as taps, drags, scrolls, etc. on touchpads; pen stylus inputs; movement of the device; oral instructions; detected eye movements; biometric inputs; and/or any combination thereof are optionally utilized as inputs corresponding to sub-events which define an event to be recognized.

FIG. 3 illustrates a portable multifunction device 200 having a touch screen 212 in accordance with some embodiments. The touch screen optionally displays one or more graphics within user interface (UI) 300. In this embodiment, as well as others described below, a user is enabled to select one or more of the graphics by making a gesture on the graphics, for example, with one or more fingers 302 (not drawn to scale in the figure) or one or more styluses 303 (not drawn to scale in the figure). In some embodiments, selection of one or more graphics occurs when the user breaks contact with the one or more graphics. In some embodiments, the gesture optionally includes one or more taps, one or more swipes (from left to right, right to left, upward and/or downward), and/or a rolling of a finger (from right to left, left to right, upward and/or downward) that has made contact with device 200. In some implementations or circumstances, inadvertent contact with a graphic does not select the graphic. For example, a swipe gesture that sweeps over an application icon optionally does not select the corresponding application when the gesture corresponding to selection is a tap.

Device 200 also includes one or more physical buttons, such as “home” or menu button 304. As described previously, menu button 304 is used to navigate to any application 236 in a set of applications that is executed on device 200. Alternatively, in some embodiments, the menu button is implemented as a soft key in a GUI displayed on touch screen 212.

In one embodiment, device 200 includes touch screen 212, menu button 304, push button 306 for powering the device on/off and locking the device, volume adjustment button(s) 308, subscriber identity module (SIM) card slot 310, headset jack 312, and docking/charging external port 224. Push button 306 is, optionally, used to turn the power on/off on the device by depressing the button and holding the button in the depressed state for a predefined time interval; to lock the device by depressing the button and releasing the button before the predefined time interval has elapsed; and/or to unlock the device or initiate an unlock process. In an alternative embodiment, device 200 also accepts verbal input for activation or deactivation of some functions through microphone 213. Device 200 also, optionally, includes one or more contact intensity sensors 265 for detecting intensity of contacts on touch screen 212 and/or one or more tactile output generators 267 for generating tactile outputs for a user of device 200.

FIG. 4 is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface in accordance with some embodiments. Device 400 need not be portable. In some embodiments, device 400 is a laptop computer, a desktop computer, a tablet computer, a multimedia player device, a navigation device, an educational device (such as a child’s learning toy), a gaming system, or a control device (e.g., a home or industrial controller). Device 400 typically includes one or more processing units (CPUs) 410, one or more network or other communications interfaces 460, memory 470, and one or more communication buses 420 for interconnecting these components. Communication buses 420 optionally include circuitry (sometimes called a chipset) that interconnects and controls communications between system components. Device 400 includes input/output (I/O) interface 430 comprising display

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440, which is typically a touch screen display. I/O interface 430 also optionally includes a keyboard and/or mouse (or other pointing device) 450 and touchpad 455, tactile output generator 457 for generating tactile outputs on device 400 (e.g., similar to tactile output generator(s) 267 described above with reference to FIG. 2A), sensors 459 (e.g., optical, acceleration, proximity, touch-sensitive, and/or contact intensity sensors similar to contact intensity sensor(s) 265 described above with reference to FIG. 2A). Memory 470 includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices; and optionally includes non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. Memory 470 optionally includes one or more storage devices remotely located from CPU(s) 410. In some embodiments, memory 470 stores programs, modules, and data structures analogous to the programs, modules, and data structures stored in memory 202 of portable multifunction device 200 (FIG. 2A), or a subset thereof. Furthermore, memory 470 optionally stores additional programs, modules, and data structures not present in memory 202 of portable multifunction device 200. For example, memory 470 of device 400 optionally stores drawing module 480, presentation module 482, word processing module 484, website creation module 486, disk authoring module 488, and/or spreadsheet module 490, while memory 202 of portable multifunction device 200 (FIG. 2A) optionally does not store these modules.

Each of the above-identified elements in FIG. 4 is, in some examples, stored in one or more of the previously mentioned memory devices. Each of the above-identified modules corresponds to a set of instructions for performing a function described above. The above-identified modules or programs (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules are combined or otherwise rearranged in various embodiments. In some embodiments, memory 470 stores a subset of the modules and data structures identified above. Furthermore, memory 470 stores additional modules and data structures not described above.

Attention is now directed towards embodiments of user interfaces that can be implemented on, for example, portable multifunction device 200.

FIG. 5A illustrates an exemplary user interface for a menu of applications on portable multifunction device 200 in accordance with some embodiments. Similar user interfaces are implemented on device 400. In some embodiments, user interface 500 includes the following elements, or a subset or superset thereof:

Signal strength indicator(s) 502 for wireless communication(s), such as cellular and Wi-Fi signals;

Time 504;

Bluetooth indicator 505;

Battery status indicator 506;

Tray 508 with icons for frequently used applications, such as:

Icon 516 for telephone module 238, labeled “Phone,” which optionally includes an indicator 514 of the number of missed calls or voicemail messages;

Icon 518 for e-mail client module 240, labeled “Mail,” which optionally includes an indicator 510 of the number of unread e-mails;

Icon 520 for browser module 247, labeled “Browser;” and

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Icon **522** for video and music player module **252**, also referred to as iPod (trademark of Apple Inc.) module **252**, labeled “iPod;” and

Icons for other applications, such as:

Icon **524** for IM module **241**, labeled “Messages;”

Icon **526** for calendar module **248**, labeled “Calendar;”

Icon **528** for image management module **244**, labeled “Photos;”

Icon **530** for camera module **243**, labeled “Camera;”

Icon **532** for online video module **255**, labeled “Online Video;”

Icon **534** for stocks widget **249-2**, labeled “Stocks;”

Icon **536** for map module **254**, labeled “Maps;”

Icon **538** for weather widget **249-1**, labeled “Weather;”

Icon **540** for alarm clock widget **249-4**, labeled “Clock;”

Icon **542** for workout support module **242**, labeled “Workout Support;”

Icon **544** for notes module **253**, labeled “Notes;” and

Icon **546** for a settings application or module, labeled “Settings;” which provides access to settings for device **200** and its various applications **236**.

It should be noted that the icon labels illustrated in FIG. 5A are merely exemplary. For example, icon **522** for video and music player module **252** is optionally labeled “Music” or “Music Player.” Other labels are, optionally, used for various application icons. In some embodiments, a label for a respective application icon includes a name of an application corresponding to the respective application icon. In some embodiments, a label for a particular application icon is distinct from a name of an application corresponding to the particular application icon.

FIG. 5B illustrates an exemplary user interface on a device (e.g., device **400**, FIG. 4) with a touch-sensitive surface **551** (e.g., a tablet or touchpad **455**, FIG. 4) that is separate from the display **550** (e.g., touch screen display **212**). Device **400** also, optionally, includes one or more contact intensity sensors (e.g., one or more of sensors **459**) for detecting intensity of contacts on touch-sensitive surface **551** and/or one or more tactile output generators **457** for generating tactile outputs for a user of device **400**.

Although some of the examples which follow will be given with reference to inputs on touch screen display **212** (where the touch-sensitive surface and the display are combined), in some embodiments, the device detects inputs on a touch-sensitive surface that is separate from the display, as shown in FIG. 5B. In some embodiments, the touch-sensitive surface (e.g., **551** in FIG. 5B) has a primary axis (e.g., **552** in FIG. 5B) that corresponds to a primary axis (e.g., **553** in FIG. 5B) on the display (e.g., **550**). In accordance with these embodiments, the device detects contacts (e.g., **560** and **562** in FIG. 5B) with the touch-sensitive surface **551** at locations that correspond to respective locations on the display (e.g., in FIG. 5B, **560** corresponds to **568** and **562** corresponds to **570**). In this way, user inputs (e.g., contacts **560** and **562**, and movements thereof) detected by the device on the touch-sensitive surface (e.g., **551** in FIG. 5B) are used by the device to manipulate the user interface on the display (e.g., **550** in FIG. 5B) of the multifunction device when the touch-sensitive surface is separate from the display. It should be understood that similar methods are, optionally, used for other user interfaces described herein.

Additionally, while the following examples are given primarily with reference to finger inputs (e.g., finger contacts, finger tap gestures, finger swipe gestures), it should be understood that, in some embodiments, one or more of the finger inputs are replaced with input from another input

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device (e.g., a mouse-based input or stylus input). For example, a swipe gesture is, optionally, replaced with a mouse click (e.g., instead of a contact) followed by movement of the cursor along the path of the swipe (e.g., instead of movement of the contact). As another example, a tap gesture is, optionally, replaced with a mouse click while the cursor is located over the location of the tap gesture (e.g., instead of detection of the contact followed by ceasing to detect the contact). Similarly, when multiple user inputs are simultaneously detected, it should be understood that multiple computer mice are, optionally, used simultaneously, or a mouse and finger contacts are, optionally, used simultaneously.

FIG. 6A illustrates exemplary personal electronic device **600**. Device **600** includes body **602**. In some embodiments, device **600** includes some or all of the features described with respect to devices **200** and **400** (e.g., FIGS. 2A-4). In some embodiments, device **600** has touch-sensitive display screen **604**, hereafter touch screen **604**. Alternatively, or in addition to touch screen **604**, device **600** has a display and a touch-sensitive surface. As with devices **200** and **400**, in some embodiments, touch screen **604** (or the touch-sensitive surface) has one or more intensity sensors for detecting intensity of contacts (e.g., touches) being applied. The one or more intensity sensors of touch screen **604** (or the touch-sensitive surface) provide output data that represents the intensity of touches. The user interface of device **600** responds to touches based on their intensity, meaning that touches of different intensities can invoke different user interface operations on device **600**.

Techniques for detecting and processing touch intensity are found, for example, in related applications: International Patent Application Serial No. PCT/US2013/040061, titled “Device, Method, and Graphical User Interface for Displaying User Interface Objects Corresponding to an Application,” filed May 8, 2013, and International Patent Application Serial No. PCT/US2013/069483, titled “Device, Method, and Graphical User Interface for Transitioning Between Touch Input to Display Output Relationships,” filed Nov. 11, 2013, each of which is hereby incorporated by reference in their entirety.

In some embodiments, device **600** has one or more input mechanisms **606** and **608**. Input mechanisms **606** and **608**, if included, are physical. Examples of physical input mechanisms include push buttons and rotatable mechanisms. In some embodiments, device **600** has one or more attachment mechanisms. Such attachment mechanisms, if included, can permit attachment of device **600** with, for example, hats, eyewear, earrings, necklaces, shirts, jackets, bracelets, watch straps, chains, trousers, belts, shoes, purses, backpacks, and so forth. These attachment mechanisms permit device **600** to be worn by a user.

FIG. 6B depicts exemplary personal electronic device **600**. In some embodiments, device **600** includes some or all of the components described with respect to FIGS. 2A, 2B, and 4. Device **600** has bus **612** that operatively couples I/O section **614** with one or more computer processors **616** and memory **618**. I/O section **614** is connected to display **604**, which can have touch-sensitive component **622** and, optionally, touch-intensity sensitive component **624**. In addition, I/O section **614** is connected with communication unit **630** for receiving application and operating system data, using Wi-Fi, Bluetooth, near field communication (NFC), cellular, and/or other wireless communication techniques. Device **600** includes input mechanisms **606** and/or **608**. Input mechanism **606** is a rotatable input device or a depressible

and rotatable input device, for example. Input mechanism **608** is a button, in some examples.

Input mechanism **608** is a microphone, in some examples. Personal electronic device **600** includes, for example, various sensors, such as GPS sensor **632**, accelerometer **634**, directional sensor **640** (e.g., compass), gyroscope **636**, motion sensor **638**, and/or a combination thereof, all of which are operatively connected to I/O section **614**.

Memory **618** of personal electronic device **600** is a non-transitory computer-readable storage medium, for storing computer-executable instructions, which, when executed by one or more computer processors **616**, for example, cause the computer processors to perform the techniques and processes described below. The computer-executable instructions, for example, are also stored and/or transported within any non-transitory computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device, such as a computer-based system, processor-containing system, or other system that can fetch the instructions from the instruction execution system, apparatus, or device and execute the instructions. Personal electronic device **600** is not limited to the components and configuration of FIG. **6B**, but can include other or additional components in multiple configurations.

As used here, the term “affordance” refers to a user-interactive graphical user interface object that is, for example, displayed on the display screen of devices **200**, **400**, and/or **600** (FIGS. **2A**, **4**, and **6A-6B**). For example, an image (e.g., icon), a button, and text (e.g., hyperlink) each constitutes an affordance.

As used herein, the term “focus selector” refers to an input element that indicates a current part of a user interface with which a user is interacting. In some implementations that include a cursor or other location marker, the cursor acts as a “focus selector” so that when an input (e.g., a press input) is detected on a touch-sensitive surface (e.g., touchpad **455** in FIG. **4** or touch-sensitive surface **551** in FIG. **5B**) while the cursor is over a particular user interface element (e.g., a button, window, slider or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations that include a touch screen display (e.g., touch-sensitive display system **212** in FIG. **2A** or touch screen **212** in FIG. **5A**) that enables direct interaction with user interface elements on the touch screen display, a detected contact on the touch screen acts as a “focus selector” so that when an input (e.g., a press input by the contact) is detected on the touch screen display at a location of a particular user interface element (e.g., a button, window, slider, or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations, focus is moved from one region of a user interface to another region of the user interface without corresponding movement of a cursor or movement of a contact on a touch screen display (e.g., by using a tab key or arrow keys to move focus from one button to another button); in these implementations, the focus selector moves in accordance with movement of focus between different regions of the user interface. Without regard to the specific form taken by the focus selector, the focus selector is generally the user interface element (or contact on a touch screen display) that is controlled by the user so as to communicate the user’s intended interaction with the user interface (e.g., by indicating, to the device, the element of the user interface with which the user is intending to interact). For example, the location of a focus selector (e.g., a cursor, a contact, or a selection box) over a respective button while a press input is detected on the touch-sensitive

surface (e.g., a touchpad or touch screen) will indicate that the user is intending to activate the respective button (as opposed to other user interface elements shown on a display of the device).

As used in the specification and claims, the term “characteristic intensity” of a contact refers to a characteristic of the contact based on one or more intensities of the contact. In some embodiments, the characteristic intensity is based on multiple intensity samples. The characteristic intensity is, optionally, based on a predefined number of intensity samples, or a set of intensity samples collected during a predetermined time period (e.g., 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10 seconds) relative to a predefined event (e.g., after detecting the contact, prior to detecting liftoff of the contact, before or after detecting a start of movement of the contact, prior to detecting an end of the contact, before or after detecting an increase in intensity of the contact, and/or before or after detecting a decrease in intensity of the contact). A characteristic intensity of a contact is, optionally based on one or more of: a maximum value of the intensities of the contact, a mean value of the intensities of the contact, an average value of the intensities of the contact, a top 10 percentile value of the intensities of the contact, a value at the half maximum of the intensities of the contact, a value at the 90 percent maximum of the intensities of the contact, or the like. In some embodiments, the duration of the contact is used in determining the characteristic intensity (e.g., when the characteristic intensity is an average of the intensity of the contact over time). In some embodiments, the characteristic intensity is compared to a set of one or more intensity thresholds to determine whether an operation has been performed by a user. For example, the set of one or more intensity thresholds includes a first intensity threshold and a second intensity threshold. In this example, a contact with a characteristic intensity that does not exceed the first threshold results in a first operation, a contact with a characteristic intensity that exceeds the first intensity threshold and does not exceed the second intensity threshold results in a second operation, and a contact with a characteristic intensity that exceeds the second threshold results in a third operation. In some embodiments, a comparison between the characteristic intensity and one or more thresholds is used to determine whether or not to perform one or more operations (e.g., whether to perform a respective operation or forgo performing the respective operation) rather than being used to determine whether to perform a first operation or a second operation.

In some embodiments, a portion of a gesture is identified for purposes of determining a characteristic intensity. For example, a touch-sensitive surface receives a continuous swipe contact transitioning from a start location and reaching an end location, at which point the intensity of the contact increases. In this example, the characteristic intensity of the contact at the end location is based on only a portion of the continuous swipe contact, and not the entire swipe contact (e.g., only the portion of the swipe contact at the end location). In some embodiments, a smoothing algorithm is applied to the intensities of the swipe contact prior to determining the characteristic intensity of the contact. For example, the smoothing algorithm optionally includes one or more of: an unweighted sliding-average smoothing algorithm, a triangular smoothing algorithm, a median filter smoothing algorithm, and/or an exponential smoothing algorithm. In some circumstances, these smoothing algorithms eliminate narrow spikes or dips in the intensities of the swipe contact for purposes of determining a characteristic intensity.

The intensity of a contact on the touch-sensitive surface is characterized relative to one or more intensity thresholds, such as a contact-detection intensity threshold, a light press intensity threshold, a deep press intensity threshold, and/or one or more other intensity thresholds. In some embodiments, the light press intensity threshold corresponds to an intensity at which the device will perform operations typically associated with clicking a button of a physical mouse or a trackpad. In some embodiments, the deep press intensity threshold corresponds to an intensity at which the device will perform operations that are different from operations typically associated with clicking a button of a physical mouse or a trackpad. In some embodiments, when a contact is detected with a characteristic intensity below the light press intensity threshold (e.g., and above a nominal contact-detection intensity threshold below which the contact is no longer detected), the device will move a focus selector in accordance with movement of the contact on the touch-sensitive surface without performing an operation associated with the light press intensity threshold or the deep press intensity threshold. Generally, unless otherwise stated, these intensity thresholds are consistent between different sets of user interface figures.

An increase of characteristic intensity of the contact from an intensity below the light press intensity threshold to an intensity between the light press intensity threshold and the deep press intensity threshold is sometimes referred to as a “light press” input. An increase of characteristic intensity of the contact from an intensity below the deep press intensity threshold to an intensity above the deep press intensity threshold is sometimes referred to as a “deep press” input. An increase of characteristic intensity of the contact from an intensity below the contact-detection intensity threshold to an intensity between the contact-detection intensity threshold and the light press intensity threshold is sometimes referred to as detecting the contact on the touch-surface. A decrease of characteristic intensity of the contact from an intensity above the contact-detection intensity threshold to an intensity below the contact-detection intensity threshold is sometimes referred to as detecting liftoff of the contact from the touch-surface. In some embodiments, the contact-detection intensity threshold is zero. In some embodiments, the contact-detection intensity threshold is greater than zero.

In some embodiments described herein, one or more operations are performed in response to detecting a gesture that includes a respective press input or in response to detecting the respective press input performed with a respective contact (or a plurality of contacts), where the respective press input is detected based at least in part on detecting an increase in intensity of the contact (or plurality of contacts) above a press-input intensity threshold. In some embodiments, the respective operation is performed in response to detecting the increase in intensity of the respective contact above the press-input intensity threshold (e.g., a “down stroke” of the respective press input). In some embodiments, the press input includes an increase in intensity of the respective contact above the press-input intensity threshold and a subsequent decrease in intensity of the contact below the press-input intensity threshold, and the respective operation is performed in response to detecting the subsequent decrease in intensity of the respective contact below the press-input threshold (e.g., an “up stroke” of the respective press input).

In some embodiments, the device employs intensity hysteresis to avoid accidental inputs sometimes termed “jitter,” where the device defines or selects a hysteresis intensity threshold with a predefined relationship to the press-input

intensity threshold (e.g., the hysteresis intensity threshold is X intensity units lower than the press-input intensity threshold or the hysteresis intensity threshold is 75%, 90%, or some reasonable proportion of the press-input intensity threshold). Thus, in some embodiments, the press input includes an increase in intensity of the respective contact above the press-input intensity threshold and a subsequent decrease in intensity of the contact below the hysteresis intensity threshold that corresponds to the press-input intensity threshold, and the respective operation is performed in response to detecting the subsequent decrease in intensity of the respective contact below the hysteresis intensity threshold (e.g., an “up stroke” of the respective press input). Similarly, in some embodiments, the press input is detected only when the device detects an increase in intensity of the contact from an intensity at or below the hysteresis intensity threshold to an intensity at or above the press-input intensity threshold and, optionally, a subsequent decrease in intensity of the contact to an intensity at or below the hysteresis intensity, and the respective operation is performed in response to detecting the press input (e.g., the increase in intensity of the contact or the decrease in intensity of the contact, depending on the circumstances).

For ease of explanation, the descriptions of operations performed in response to a press input associated with a press-input intensity threshold or in response to a gesture including the press input are, optionally, triggered in response to detecting either: an increase in intensity of a contact above the press-input intensity threshold, an increase in intensity of a contact from an intensity below the hysteresis intensity threshold to an intensity above the press-input intensity threshold, a decrease in intensity of the contact below the press-input intensity threshold, and/or a decrease in intensity of the contact below the hysteresis intensity threshold corresponding to the press-input intensity threshold. Additionally, in examples where an operation is described as being performed in response to detecting a decrease in intensity of a contact below the press-input intensity threshold, the operation is, optionally, performed in response to detecting a decrease in intensity of the contact below a hysteresis intensity threshold corresponding to, and lower than, the press-input intensity threshold.

3. Digital Assistant System

FIG. 7A illustrates a block diagram of digital assistant system 700 in accordance with various examples. In some examples, digital assistant system 700 is implemented on a standalone computer system. In some examples, digital assistant system 700 is distributed across multiple computers. In some examples, some of the modules and functions of the digital assistant are divided into a server portion and a client portion, where the client portion resides on one or more user devices (e.g., devices 104, 122, 200, 400, or 600) and communicates with the server portion (e.g., server system 108) through one or more networks, e.g., as shown in FIG. 1. In some examples, digital assistant system 700 is an implementation of server system 108 (and/or DA server 106) shown in FIG. 1. It should be noted that digital assistant system 700 is only one example of a digital assistant system, and that digital assistant system 700 can have more or fewer components than shown, can combine two or more components, or can have a different configuration or arrangement of the components. The various components shown in FIG. 7A are implemented in hardware, software instructions for execution by one or more processors, firmware, including one or more signal processing and/or application specific integrated circuits, or a combination thereof.

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Digital assistant system **700** includes memory **702**, one or more processors **704**, input/output (I/O) interface **706**, and network communications interface **708**. These components can communicate with one another over one or more communication buses or signal lines **710**.

In some examples, memory **702** includes a non-transitory computer-readable medium, such as high-speed random access memory and/or a non-volatile computer-readable storage medium (e.g., one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices).

In some examples, I/O interface **706** couples input/output devices **716** of digital assistant system **700**, such as displays, keyboards, touch screens, and microphones, to user interface module **722**. I/O interface **706**, in conjunction with user interface module **722**, receives user inputs (e.g., voice input, keyboard inputs, touch inputs, etc.) and processes them accordingly. In some examples, e.g., when the digital assistant is implemented on a standalone user device, digital assistant system **700** includes any of the components and I/O communication interfaces described with respect to devices **200**, **400**, or **600** in FIGS. 2A, 4, 6A-6B, respectively. In some examples, digital assistant system **700** represents the server portion of a digital assistant implementation, and can interact with the user through a client-side portion residing on a user device (e.g., devices **104**, **200**, **400**, or **600**).

In some examples, the network communications interface **708** includes wired communication port(s) **712** and/or wireless transmission and reception circuitry **714**. The wired communication port(s) receives and send communication signals via one or more wired interfaces, e.g., Ethernet, Universal Serial Bus (USB), FIREWIRE, etc. The wireless circuitry **714** receives and sends RF signals and/or optical signals from/to communications networks and other communications devices. The wireless communications use any of a plurality of communications standards, protocols, and technologies, such as GSM, EDGE, CDMA, TDMA, Bluetooth, Wi-Fi, VoIP, Wi-MAX, or any other suitable communication protocol. Network communications interface **708** enables communication between digital assistant system **700** with networks, such as the Internet, an intranet, and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN), and/or a metropolitan area network (MAN), and other devices.

In some examples, memory **702**, or the computer-readable storage media of memory **702**, stores programs, modules, instructions, and data structures including all or a subset of: operating system **718**, communications module **720**, user interface module **722**, one or more applications **724**, and digital assistant module **726**. In particular, memory **702**, or the computer-readable storage media of memory **702**, stores instructions for performing the processes described below. One or more processors **704** execute these programs, modules, and instructions, and reads/writes from/to the data structures.

Operating system **718** (e.g., Darwin, RTXC, LINUX, UNIX, iOS, OS X, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communications between various hardware, firmware, and software components.

Communications module **720** facilitates communications between digital assistant system **700** with other devices over network communications interface **708**. For example, communications module **720** communicates with RF circuitry

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208 of electronic devices such as devices **200**, **400**, and **600** shown in FIGS. 2A, 4, 6A-6B, respectively. Communications module **720** also includes various components for handling data received by wireless circuitry **714** and/or wired communications port **712**.

User interface module **722** receives commands and/or inputs from a user via I/O interface **706** (e.g., from a keyboard, touch screen, pointing device, controller, and/or microphone), and generate user interface objects on a display. User interface module **722** also prepares and delivers outputs (e.g., speech, sound, animation, text, icons, vibrations, haptic feedback, light, etc.) to the user via the I/O interface **706** (e.g., through displays, audio channels, speakers, touch-pads, etc.).

Applications **724** include programs and/or modules that are configured to be executed by one or more processors **704**. For example, if the digital assistant system is implemented on a standalone user device, applications **724** include user applications, such as games, a calendar application, a navigation application, or an email application. If digital assistant system **700** is implemented on a server, applications **724** include resource management applications, diagnostic applications, or scheduling applications, for example.

Memory **702** also stores digital assistant module **726** (or the server portion of a digital assistant). In some examples, digital assistant module **726** includes the following sub-modules, or a subset or superset thereof: input/output processing module **728**, speech-to-text (STT) processing module **730**, natural language processing module **732**, dialogue flow processing module **734**, task flow processing module **736**, service processing module **738**, and speech synthesis processing module **740**. Each of these modules has access to one or more of the following systems or data and models of the digital assistant module **726**, or a subset or superset thereof: ontology **760**, vocabulary index **744**, user data **748**, task flow models **754**, service models **756**, and ASR systems **758**.

In some examples, using the processing modules, data, and models implemented in digital assistant module **726**, the digital assistant can perform at least some of the following: converting speech input into text; identifying a user's intent expressed in a natural language input received from the user; actively eliciting and obtaining information needed to fully infer the user's intent (e.g., by disambiguating words, games, intentions, etc.); determining the task flow for fulfilling the inferred intent; and executing the task flow to fulfill the inferred intent.

In some examples, as shown in FIG. 7B, I/O processing module **728** interacts with the user through I/O devices **716** in FIG. 7A or with a user device (e.g., devices **104**, **200**, **400**, or **600**) through network communications interface **708** in FIG. 7A to obtain user input (e.g., a speech input) and to provide responses (e.g., as speech outputs) to the user input. I/O processing module **728** optionally obtains contextual information associated with the user input from the user device, along with or shortly after the receipt of the user input. The contextual information includes user-specific data, vocabulary, and/or preferences relevant to the user input. In some examples, the contextual information also includes software and hardware states of the user device at the time the user request is received, and/or information related to the surrounding environment of the user at the time that the user request was received. In some examples, I/O processing module **728** also sends follow-up questions to, and receive answers from, the user regarding the user request. When a user request is received by I/O processing

module 728 and the user request includes speech input, I/O processing module 728 forwards the speech input to STT processing module 730 (or speech recognizer) for speech-to-text conversions.

STT processing module 730 includes one or more ASR systems 758. The one or more ASR systems 758 can process the speech input that is received through I/O processing module 728 to produce a recognition result. Each ASR system 758 includes a front-end speech pre-processor. The front-end speech pre-processor extracts representative features from the speech input. For example, the front-end speech pre-processor performs a Fourier transform on the speech input to extract spectral features that characterize the speech input as a sequence of representative multi-dimensional vectors. Further, each ASR system 758 includes one or more speech recognition models (e.g., acoustic models and/or language models) and implements one or more speech recognition engines. Examples of speech recognition models include Hidden Markov Models, Gaussian-Mixture Models, Deep Neural Network Models, n-gram language models, and other statistical models. Examples of speech recognition engines include the dynamic time warping based engines and weighted finite-state transducers (WFST) based engines. The one or more speech recognition models and the one or more speech recognition engines are used to process the extracted representative features of the front-end speech pre-processor to produce intermediate recognitions results (e.g., phonemes, phonemic strings, and sub-words), and ultimately, text recognition results (e.g., words, word strings, or sequence of tokens). In some examples, the speech input is processed at least partially by a third-party service or on the user's device (e.g., device 104, 200, 400, or 600) to produce the recognition result. Once STT processing module 730 produces recognition results containing a text string (e.g., words, or sequence of words, or sequence of tokens), the recognition result is passed to natural language processing module 732 for intent deduction. In some examples, STT processing module 730 produces multiple candidate text representations of the speech input. Each candidate text representation is a sequence of words or tokens corresponding to the speech input. In some examples, each candidate text representation is associated with a speech recognition confidence score. Based on the speech recognition confidence scores, STT processing module 730 ranks the candidate text representations and provides the n-best (e.g., n highest ranked) candidate text representation(s) to natural language processing module 732 for intent deduction, where n is a predetermined integer greater than zero. For example, in one example, only the highest ranked (n=1) candidate text representation is passed to natural language processing module 732 for intent deduction. In another example, the five highest ranked (n=5) candidate text representations are passed to natural language processing module 732 for intent deduction.

More details on the speech-to-text processing are described in U.S. Utility application Ser. No. 13/236,942 for "Consolidating Speech Recognition Results," filed on Sep. 20, 2011, the entire disclosure of which is incorporated herein by reference.

In some examples, STT processing module 730 includes and/or accesses a vocabulary of recognizable words via phonetic alphabet conversion module 731. Each vocabulary word is associated with one or more candidate pronunciations of the word represented in a speech recognition phonetic alphabet. In particular, the vocabulary of recognizable words includes a word that is associated with a plurality of candidate pronunciations. For example, the vocabulary

includes the word "tomato" that is associated with the candidate pronunciations of /tə'meɪroʊ/ and /tə'matoʊ/. Further, vocabulary words are associated with custom candidate pronunciations that are based on previous speech inputs from the user. Such custom candidate pronunciations are stored in STT processing module 730 and are associated with a particular user via the user's profile on the device. In some examples, the candidate pronunciations for words are determined based on the spelling of the word and one or more linguistic and/or phonetic rules. In some examples, the candidate pronunciations are manually generated, e.g., based on known canonical pronunciations.

In some examples, the candidate pronunciations are ranked based on the commonness of the candidate pronunciation. For example, the candidate pronunciation /tə'meɪroʊ/ is ranked higher than /tə'matoʊ/, because the former is a more commonly used pronunciation (e.g., among all users, for users in a particular geographical region, or for any other appropriate subset of users). In some examples, candidate pronunciations are ranked based on whether the candidate pronunciation is a custom candidate pronunciation associated with the user. For example, custom candidate pronunciations are ranked higher than canonical candidate pronunciations. This can be useful for recognizing proper nouns having a unique pronunciation that deviates from canonical pronunciation. In some examples, candidate pronunciations are associated with one or more speech characteristics, such as geographic origin, nationality, or ethnicity. For example, the candidate pronunciation /tə'meɪroʊ/ is associated with the United States, whereas the candidate pronunciation /tə'matoʊ/ is associated with Great Britain. Further, the rank of the candidate pronunciation is based on one or more characteristics (e.g., geographic origin, nationality, ethnicity, etc.) of the user stored in the user's profile on the device. For example, it can be determined from the user's profile that the user is associated with the United States. Based on the user being associated with the United States, the candidate pronunciation /tə'meɪroʊ/ (associated with the United States) is ranked higher than the candidate pronunciation /tə'matoʊ/ (associated with Great Britain). In some examples, one of the ranked candidate pronunciations is selected as a predicted pronunciation (e.g., the most likely pronunciation).

When a speech input is received, STT processing module 730 is used to determine the phonemes corresponding to the speech input (e.g., using an acoustic model), and then attempt to determine words that match the phonemes (e.g., using a language model). For example, if STT processing module 730 first identifies the sequence of phonemes /tə'meɪroʊ/ corresponding to a portion of the speech input, it can then determine, based on vocabulary index 744, that this sequence corresponds to the word "tomato."

In some examples, STT processing module 730 uses approximate matching techniques to determine words in an utterance. Thus, for example, the STT processing module 730 determines that the sequence of phonemes /tə'meɪroʊ/ corresponds to the word "tomato," even if that particular sequence of phonemes is not one of the candidate sequence of phonemes for that word.

Natural language processing module 732 ("natural language processor") of the digital assistant takes the n-best candidate text representation(s) ("word sequence(s)" or "token sequence(s)") generated by STT processing module 730, and attempts to associate each of the candidate text representations with one or more "actionable intents" recognized by the digital assistant. An "actionable intent" (or "user intent") represents a task that can be performed by the

digital assistant, and can have an associated task flow implemented in task flow models 754. The associated task flow is a series of programmed actions and steps that the digital assistant takes in order to perform the task. The scope of a digital assistant's capabilities is dependent on the number and variety of task flows that have been implemented and stored in task flow models 754, or in other words, on the number and variety of "actionable intents" that the digital assistant recognizes. The effectiveness of the digital assistant, however, also depends on the assistant's ability to infer the correct "actionable intent(s)" from the user request expressed in natural language.

In some examples, in addition to the sequence of words or tokens obtained from STT processing module 730, natural language processing module 732 also receives contextual information associated with the user request, e.g., from I/O processing module 728. The natural language processing module 732 optionally uses the contextual information to clarify, supplement, and/or further define the information contained in the candidate text representations received from STT processing module 730. The contextual information includes, for example, user preferences, hardware, and/or software states of the user device, sensor information collected before, during, or shortly after the user request, prior interactions (e.g., dialogue) between the digital assistant and the user, and the like. As described herein, contextual information is, in some examples, dynamic, and changes with time, location, content of the dialogue, and other factors.

In some examples, the natural language processing is based on, e.g., ontology 760. Ontology 760 is a hierarchical structure containing many nodes, each node representing either an "actionable intent" or a "property" relevant to one or more of the "actionable intents" or other "properties." As noted above, an "actionable intent" represents a task that the digital assistant is capable of performing, i.e., it is "actionable" or can be acted on. A "property" represents a parameter associated with an actionable intent or a sub-aspect of another property. A linkage between an actionable intent node and a property node in ontology 760 defines how a parameter represented by the property node pertains to the task represented by the actionable intent node.

In some examples, ontology 760 is made up of actionable intent nodes and property nodes. Within ontology 760, each actionable intent node is linked to one or more property nodes either directly or through one or more intermediate property nodes. Similarly, each property node is linked to one or more actionable intent nodes either directly or through one or more intermediate property nodes. For example, as shown in FIG. 7C, ontology 760 includes a "restaurant reservation" node (i.e., an actionable intent node). Property nodes "restaurant," "date/time" (for the reservation), and "party size" are each directly linked to the actionable intent node (i.e., the "restaurant reservation" node).

In addition, property nodes "cuisine," "price range," "phone number," and "location" are sub-nodes of the property node "restaurant," and are each linked to the "restaurant reservation" node (i.e., the actionable intent node) through the intermediate property node "restaurant." For another example, as shown in FIG. 7C, ontology 760 also includes a "set reminder" node (i.e., another actionable intent node). Property nodes "date/time" (for setting the reminder) and "subject" (for the reminder) are each linked to the "set reminder" node. Since the property "date/time" is relevant to both the task of making a restaurant reservation and the task of setting a reminder, the property node "date/time" is linked

to both the "restaurant reservation" node and the "set reminder" node in ontology 760.

An actionable intent node, along with its linked property nodes, is described as a "domain." In the present discussion, each domain is associated with a respective actionable intent, and refers to the group of nodes (and the relationships there between) associated with the particular actionable intent. For example, ontology 760 shown in FIG. 7C includes an example of restaurant reservation domain 762 and an example of reminder domain 764 within ontology 760. The restaurant reservation domain includes the actionable intent node "restaurant reservation," property nodes "restaurant," "date/time," and "party size," and sub-property nodes "cuisine," "price range," "phone number," and "location." Reminder domain 764 includes the actionable intent node "set reminder," and property nodes "subject" and "date/time." In some examples, ontology 760 is made up of many domains. Each domain shares one or more property nodes with one or more other domains. For example, the "date/time" property node is associated with many different domains (e.g., a scheduling domain, a travel reservation domain, a movie ticket domain, etc.), in addition to restaurant reservation domain 762 and reminder domain 764.

While FIG. 7C illustrates two example domains within ontology 760, other domains include, for example, "find a movie," "initiate a phone call," "find directions," "schedule a meeting," "send a message," and "provide an answer to a question," "read a list," "providing navigation instructions," "provide instructions for a task" and so on. A "send a message" domain is associated with a "send a message" actionable intent node, and further includes property nodes such as "recipient(s)," "message type," and "message body." The property node "recipient" is further defined, for example, by the sub-property nodes such as "recipient name" and "message address."

In some examples, ontology 760 includes all the domains (and hence actionable intents) that the digital assistant is capable of understanding and acting upon. In some examples, ontology 760 is modified, such as by adding or removing entire domains or nodes, or by modifying relationships between the nodes within the ontology 760.

In some examples, nodes associated with multiple related actionable intents are clustered under a "super domain" in ontology 760. For example, a "travel" super-domain includes a cluster of property nodes and actionable intent nodes related to travel. The actionable intent nodes related to travel includes "airline reservation," "hotel reservation," "car rental," "get directions," "find points of interest," and so on. The actionable intent nodes under the same super domain (e.g., the "travel" super domain) have many property nodes in common. For example, the actionable intent nodes for "airline reservation," "hotel reservation," "car rental," "get directions," and "find points of interest" share one or more of the property nodes "start location," "destination," "departure date/time," "arrival date/time," and "party size."

In some examples, each node in ontology 760 is associated with a set of words and/or phrases that are relevant to the property or actionable intent represented by the node. The respective set of words and/or phrases associated with each node are the so-called "vocabulary" associated with the node. The respective set of words and/or phrases associated with each node are stored in vocabulary index 744 in association with the property or actionable intent represented by the node. For example, returning to FIG. 7B, the vocabulary associated with the node for the property of "restaurant" includes words such as "food," "drinks," "cui-

sine,” “hungry,” “eat,” “pizza,” “fast food,” “meal,” and so on. For another example, the vocabulary associated with the node for the actionable intent of “initiate a phone call” includes words and phrases such as “call,” “phone,” “dial,” “ring,” “call this number,” “make a call to,” and so on. The vocabulary index 744 optionally includes words and phrases in different languages.

Natural language processing module 732 receives the candidate text representations (e.g., text string(s) or token sequence(s)) from STT processing module 730, and for each candidate representation, determines what nodes are implicated by the words in the candidate text representation. In some examples, if a word or phrase in the candidate text representation is found to be associated with one or more nodes in ontology 760 (via vocabulary index 744), the word or phrase “triggers” or “activates” those nodes. Based on the quantity and/or relative importance of the activated nodes, natural language processing module 732 selects one of the actionable intents as the task that the user intended the digital assistant to perform. In some examples, the domain that has the most “triggered” nodes is selected. In some examples, the domain having the highest confidence value (e.g., based on the relative importance of its various triggered nodes) is selected. In some examples, the domain is selected based on a combination of the number and the importance of the triggered nodes. In some examples, additional factors are considered in selecting the node as well, such as whether the digital assistant has previously correctly interpreted a similar request from a user.

User data 748 includes user-specific information, such as user-specific vocabulary, user preferences, user address, user’s default and secondary languages, user’s contact list, and other short-term or long-term information for each user. In some examples, natural language processing module 732 uses the user-specific information to supplement the information contained in the user input to further define the user intent. For example, for a user request “invite my friends to my birthday party,” natural language processing module 732 is able to access user data 748 to determine who the “friends” are and when and where the “birthday party” would be held, rather than requiring the user to provide such information explicitly in his/her request.

It should be recognized that in some examples, natural language processing module 732 is implemented using one or more machine learning mechanisms (e.g., neural networks). In particular, the one or more machine learning mechanisms are configured to receive a candidate text representation and contextual information associated with the candidate text representation. Based on the candidate text representation and the associated contextual information, the one or more machine learning mechanisms are configured to determine intent confidence scores over a set of candidate actionable intents. Natural language processing module 732 can select one or more candidate actionable intents from the set of candidate actionable intents based on the determined intent confidence scores. In some examples, an ontology (e.g., ontology 760) is also used to select the one or more candidate actionable intents from the set of candidate actionable intents.

Other details of searching an ontology based on a token string are described in U.S. Utility application Ser. No. 12/341,743 for “Method and Apparatus for Searching Using An Active Ontology,” filed Dec. 22, 2008, the entire disclosure of which is incorporated herein by reference.

In some examples, once natural language processing module 732 identifies an actionable intent (or domain) based on the user request, natural language processing module 732

generates a structured query to represent the identified actionable intent. In some examples, the structured query includes parameters for one or more nodes within the domain for the actionable intent, and at least some of the parameters are populated with the specific information and requirements specified in the user request. For example, the user says “Make me a dinner reservation at a sushi place at 7.” In this case, natural language processing module 732 is able to correctly identify the actionable intent to be “restaurant reservation” based on the user input. According to the ontology, a structured query for a “restaurant reservation” domain includes parameters such as {Cuisine}, {Time}, {Date}, {Party Size}, and the like. In some examples, based on the speech input and the text derived from the speech input using STT processing module 730, natural language processing module 732 generates a partial structured query for the restaurant reservation domain, where the partial structured query includes the parameters {Cuisine=“Sushi”} and {Time=“7 pm”}. However, in this example, the user’s utterance contains insufficient information to complete the structured query associated with the domain. Therefore, other necessary parameters such as {Party Size} and {Date} are not specified in the structured query based on the information currently available. In some examples, natural language processing module 732 populates some parameters of the structured query with received contextual information. For example, in some examples, if the user requested a sushi restaurant “near me,” natural language processing module 732 populates a {location} parameter in the structured query with GPS coordinates from the user device.

In some examples, natural language processing module 732 identifies multiple candidate actionable intents for each candidate text representation received from STT processing module 730. Further, in some examples, a respective structured query (partial or complete) is generated for each identified candidate actionable intent. Natural language processing module 732 determines an intent confidence score for each candidate actionable intent and ranks the candidate actionable intents based on the intent confidence scores. In some examples, natural language processing module 732 passes the generated structured query (or queries), including any completed parameters, to task flow processing module 736 (“task flow processor”). In some examples, the structured query (or queries) for the m-best (e.g., m highest ranked) candidate actionable intents are provided to task flow processing module 736, where m is a predetermined integer greater than zero. In some examples, the structured query (or queries) for the m-best candidate actionable intents are provided to task flow processing module 736 with the corresponding candidate text representation(s).

Other details of inferring a user intent based on multiple candidate actionable intents determined from multiple candidate text representations of a speech input are described in U.S. Utility application Ser. No. 14/298,725 for “System and Method for Inferring User Intent From Speech Inputs,” filed Jun. 6, 2014, the entire disclosure of which is incorporated herein by reference.

Task flow processing module 736 is configured to receive the structured query (or queries) from natural language processing module 732, complete the structured query, if necessary, and perform the actions required to “complete” the user’s ultimate request. In some examples, the various procedures necessary to complete these tasks are provided in task flow models 754. In some examples, task flow models 754 include procedures for obtaining additional information from the user and task flows for performing actions associated with the actionable intent.

As described above, in order to complete a structured query, task flow processing module 736 needs to initiate additional dialogue with the user in order to obtain additional information, and/or disambiguate potentially ambiguous utterances. When such interactions are necessary, task flow processing module 736 invokes dialogue flow processing module 734 to engage in a dialogue with the user. In some examples, dialogue flow processing module 734 determines how (and/or when) to ask the user for the additional information and receives and processes the user responses. The questions are provided to and answers are received from the users through I/O processing module 728. In some examples, dialogue flow processing module 734 presents dialogue output to the user via audio and/or visual output, and receives input from the user via spoken or physical (e.g., clicking) responses. Continuing with the example above, when task flow processing module 736 invokes dialogue flow processing module 734 to determine the "party size" and "date" information for the structured query associated with the domain "restaurant reservation," dialogue flow processing module 734 generates questions such as "For how many people?" and "On which day?" to pass to the user. Once answers are received from the user, dialogue flow processing module 734 then populates the structured query with the missing information, or pass the information to task flow processing module 736 to complete the missing information from the structured query.

Once task flow processing module 736 has completed the structured query for an actionable intent, task flow processing module 736 proceeds to perform the ultimate task associated with the actionable intent. Accordingly, task flow processing module 736 executes the steps and instructions in the task flow model according to the specific parameters contained in the structured query. For example, the task flow model for the actionable intent of "restaurant reservation" includes steps and instructions for contacting a restaurant and actually requesting a reservation for a particular party size at a particular time. For example, using a structured query such as: {restaurant reservation, restaurant=ABC Café, date=3/12/2012, time=7 pm, party size=5}, task flow processing module 736 performs the steps of: (1) logging onto a server of the ABC Café or a restaurant reservation system such as OPENTABLE®, (2) entering the date, time, and party size information in a form on the website, (3) submitting the form, and (4) making a calendar entry for the reservation in the user's calendar.

In some examples, task flow processing module 736 employs the assistance of service processing module 738 ("service processing module") to complete a task requested in the user input or to provide an informational answer requested in the user input. For example, service processing module 738 acts on behalf of task flow processing module 736 to make a phone call, set a calendar entry, invoke a map search, invoke or interact with other user applications installed on the user device, and invoke or interact with third-party services (e.g., a restaurant reservation portal, a social networking website, a banking portal, etc.). In some examples, the protocols and application programming interfaces (API) required by each service are specified by a respective service model among service models 756. Service processing module 738 accesses the appropriate service model for a service and generates requests for the service in accordance with the protocols and APIs required by the service according to the service model.

For example, if a restaurant has enabled an online reservation service, the restaurant submits a service model specifying the necessary parameters for making a reservation and

the APIs for communicating the values of the necessary parameter to the online reservation service. When requested by task flow processing module 736, service processing module 738 establishes a network connection with the online reservation service using the web address stored in the service model, and sends the necessary parameters of the reservation (e.g., time, date, party size) to the online reservation interface in a format according to the API of the online reservation service.

In some examples, natural language processing module 732, dialogue flow processing module 734, and task flow processing module 736 are used collectively and iteratively to infer and define the user's intent, obtain information to further clarify and refine the user intent, and finally generate a response (i.e., an output to the user, or the completion of a task) to fulfill the user's intent. The generated response is a dialogue response to the speech input that at least partially fulfills the user's intent. Further, in some examples, the generated response is output as a speech output. In these examples, the generated response is sent to speech synthesis processing module 740 (e.g., speech synthesizer) where it can be processed to synthesize the dialogue response in speech form. In yet other examples, the generated response is data content relevant to satisfying a user request in the speech input.

In examples where task flow processing module 736 receives multiple structured queries from natural language processing module 732, task flow processing module 736 initially processes the first structured query of the received structured queries to attempt to complete the first structured query and/or execute one or more tasks or actions represented by the first structured query. In some examples, the first structured query corresponds to the highest ranked actionable intent. In other examples, the first structured query is selected from the received structured queries based on a combination of the corresponding speech recognition confidence scores and the corresponding intent confidence scores. In some examples, if task flow processing module 736 encounters an error during processing of the first structured query (e.g., due to an inability to determine a necessary parameter), the task flow processing module 736 can proceed to select and process a second structured query of the received structured queries that corresponds to a lower ranked actionable intent. The second structured query is selected, for example, based on the speech recognition confidence score of the corresponding candidate text representation, the intent confidence score of the corresponding candidate actionable intent, a missing necessary parameter in the first structured query, or any combination thereof.

Speech synthesis processing module 740 is configured to synthesize speech outputs for presentation to the user. Speech synthesis processing module 740 synthesizes speech outputs based on text provided by the digital assistant. For example, the generated dialogue response is in the form of a text string. Speech synthesis processing module 740 converts the text string to an audible speech output. Speech synthesis processing module 740 uses any appropriate speech synthesis technique in order to generate speech outputs from text, including, but not limited to, concatenative synthesis, unit selection synthesis, diphone synthesis, domain-specific synthesis, formant synthesis, articulatory synthesis, hidden Markov model (HMM) based synthesis, and sinewave synthesis. In some examples, speech synthesis processing module 740 is configured to synthesize individual words based on phonemic strings corresponding to the words. For example, a phonemic string is associated with a word in the generated dialogue response. The phonemic

string is stored in metadata associated with the word. Speech synthesis processing module **740** is configured to directly process the phonemic string in the metadata to synthesize the word in speech form.

In some examples, instead of (or in addition to) using speech synthesis processing module **740**, speech synthesis is performed on a remote device (e.g., the server system **108**), and the synthesized speech is sent to the user device for output to the user. For example, this can occur in some implementations where outputs for a digital assistant are generated at a server system. And because server systems generally have more processing power or resources than a user device, it is possible to obtain higher quality speech outputs than would be practical with client-side synthesis.

Additional details on digital assistants can be found in the U.S. Utility application Ser. No. 12/987,982, entitled "Intelligent Automated Assistant," filed Jan. 10, 2011, and U.S. Utility application Ser. No. 13/251,088, entitled "Generating and Processing Task Items That Represent Tasks to Perform," filed Sep. 30, 2011, the entire disclosures of which are incorporated herein by reference.

4. Integration of a Digital Assistant with Telephony

FIGS. **8-12** illustrate systems and processes for continuous dialog with a digital assistant. For example, as shown in FIG. **8A**, electronic device **800** may include any device described herein, including but not limited to devices **104**, **200**, **400**, and **600** (FIGS. **1**, **2A**, **4**, and **6A-6B**). Thus, it will be appreciated that the electronic devices depicted in FIGS. **8-12** may correspond to a user device of any type, such as a phone, laptop computer, desktop computer, tablet, wearable device (e.g., smart watch), home speaker, and the like. Furthermore, the processes described herein may be performed by a server with information delivered to and from the device, performed on the device, or a combination thereof.

With reference to FIG. **8A**, in general, a home screen of electronic device **800** is depicted. In particular, the user may be navigating to various applications on the device, interacting with the device to perform one or more specific tasks, and the like. In some examples, an incoming call from a caller is received at the electronic device. The incoming call may be received, and in response, a communication session is established between the caller and a digital assistant of the electronic device. Establishing a communication session between the caller and the digital assistant may include establishing the call between a device associated with the caller and the electronic device (e.g., device **800**), such that the caller initially interacts with the digital assistant and not with the user of the electronic device. In some examples, establishing the communication session may include initially establishing communication between the caller's electronic device and a digital assistant server.

In some examples, a digital assistant object **802** may be displayed along with an indication **804** (e.g., a banner at a top section of the display) including caller identification details. In this case, the caller may be unknown, such as a user not included within a contact database associated with electronic device **800** (e.g., a stored contact identifier). A calling number may be included within indication **804**. In some cases, a calling number may not be available. For example, the indication may be only "Unknown" or "Blocked." Displaying the digital assistant object **802** along with indication **804** may indicate to the user that the digital assistant has established communication with the caller associated with the incoming call. In some examples, only the digital assistant object **802** is displayed (i.e., without indication **804**) upon establishing communication with the

caller. In this case, in response to a user gaze directed at the digital assistant object **802**, indication **804** is displayed. For example, the user may initially not be looking at the digital assistant object or the device **800** when the call is initially established. Then, the user may look at digital assistant object **802**, which causes the caller details to be shown to the user via indication **804**.

In general, various trust levels may be associated with callers based on certain factors. For example, an incoming caller identity may be determined, such as an identity corresponding to a stored contact identifier. The trust level may be based on the stored contact identifier (or lack of an identified contact) and/or based on a group associated with the contact identifier. For example, any known contact may be assigned a "medium" trust level. Certain contacts may be assigned a "high" trust level, such as contacts belonging to specific groups (e.g., "family," "VIP," etc.). A high trust level may also be assigned based on a communication frequency value associated with the identification of the caller. In particular, any contact having a communication frequency value exceeding a communication frequency threshold (e.g., the user interacts with the caller at least once per day) may be assigned a "high" trust level. In contrast, a "low" trust level may be assigned to certain callers, such as unknown callers, callers associated with contacts added to a "block" list, and the like. The trust levels may be arranged in various ways, (e.g., trust levels between one and ten), general trust descriptions (e.g., "very high trust," "medium-high trust," etc.), and the like.

In some examples, voice identification and speech recognition is utilized in order to determine the identity of a caller. For instance, various voiceprints representing speech characteristics of specific persons may be stored on the electronic device. In particular, these voiceprints may correspond to speech profiles of persons such as family members or other close contacts of the user. The generating and storing of voiceprints may require prior approval from the person corresponding to the respective voiceprint in order for such voiceprints to be utilized. When an incoming call is received, incoming speech from the calling party may be compared to various voiceprints stored on the device in order to determine whether speech characteristics of the incoming speech matches speech characteristics of a stored voiceprint. In accordance with a determination that the incoming speech characteristics of the incoming speech matches (e.g., to a sufficient degree of confidence) speech characteristics of a stored voiceprint, an identity of the caller may be determined to correspond to a respective identity associated with the stored voiceprint.

In some examples, in accordance with a determination that the identification of the caller is unknown, a further determination is made whether the caller corresponds to an automated calling system. In general, automated calling systems may generally include systems that initiate communications to users regarding various subjects, such as political calls, advertising calls, polling inquiries, appointment reminders (e.g., medical, dental, etc.), and the like. Users may wish to obtain information from such calls, and/or may not wish to interact with such callers, such as those associated with unwanted spam calls or potentially fraudulent calls. In order to determine whether the caller corresponds to an automated system, the electronic device and digital assistant may identify audio attributes of the incoming call and compare the attributes to stored attributes. For example, incoming call audio originating from automated system calls may include unique characteristics distinct from those of a call originating from a non-automated

system. For instance, automated system calls may generally include synthesized (e.g., artificial, non-human) speech, which is distinguishable from natural human speech based on attributes such as pitch, speech rate, intonation, stress, rhythm, and other prosodic features.

Various prompt-response mechanisms may also be utilized to determine whether the caller corresponds to an automated calling system. For example, the digital assistant may provide, to the caller, a prompt including an interrogatory, such as "Let me ask you some questions. What is the opposite of Monday?" In this example, the interrogatory may include a nonsensical or otherwise illogical question, such that a human user would respond with a predictable response and/or response pattern, such as a pause followed by a clarifying question (e.g., "What do you mean?", "Excuse me?" etc.). In contrast, an automated calling system may have difficulty interpreting such an interrogatory, may provide a default response, and/or may simply provide a pre-configured default recording, such as "Hi, this is Senator Smith" In some examples, the identified auditory pattern of the incoming call is classified according to pattern recognition (e.g., using one or more data-trained pattern recognition models). For instance, in accordance with a determination that the identified auditory pattern of the incoming call corresponds to a stored auditory pattern corresponding to a human user, determination is made that the caller does not correspond to an automated system.

In some examples, in accordance with determination that the caller corresponds to an automated system, the communication session between the caller and digital assistant may be ended and/or the incoming call may be routed to a voicemail system associated with the electronic device. For example, digital assistant object **802** and indication **804** may be momentarily displayed while the digital assistant interacts with the automated system caller. Once the determination is made that the caller corresponds to an automated system, digital assistant object and indication **804** are removed from the display. In some examples, digital assistant object **802** and indication **804** are not displayed until a determination is made that an unknown caller does not correspond to an automated system. In such cases, digital assistant object **802** and indication **804** would not be displayed to the user to the extent the unknown caller is determined to be an automated system.

In some examples, in accordance with determination that the caller does not correspond to an automated system, a prompt including an interrogatory requesting a callee name is provided to the caller. In general, the prompt may request that the caller identify the name of the party the caller is trying to reach, such as "Hi, I'm the personal assistant for this phone number. Who are you trying to reach?" An input responsive to the prompt may be compared to a stored callee name. Based on a determination whether the received input corresponds to the stored callee name, a specific trust level is assigned to the caller. For example, the stored callee name may correspond to "Dan," and the input responsive to the prompt may include "Daniel" (a variant of the stored callee name). Given the sufficient similarity between "Dan" and "Daniel," (e.g., based on a set of acceptably similar names), determination is made that the received input corresponds to the stored callee name. Accordingly, the caller may be assigned a "high" trust level. In contrast, in accordance with a determination that the received input does not correspond to the stored callee name, the caller may be assigned a lesser trust level (e.g., "medium" or "low"). In some cases, the caller may be assigned significantly lesser trust level, such as "low," if no response is provided to the interrogatory.

In accordance with a determination that the identification of the caller is known, a response to the caller is provided by the digital assistant. For example, the unknown caller with number **(555) 123-4567** may correspond to a receptionist at a doctor's office calling to remind the user of their upcoming medical appointment. The response provided to the caller may depend upon the trust level associated with the caller. In general, a caller associated with a higher trust level may be provided with information regarding the user's availability or other information about reaching the user, such as additional contact addresses or other user preferences. For example, a schedule of the user may be obtained including information indicating that the user is currently in a meeting until a time of 11:00 AM. Accordingly, a caller associated with a "high" trust level may be provided a response such as "Dan is in the weekly team meeting until 11 AM. Would you like to leave a message?" In contrast, a caller associated with a "medium" trust level may be provide a response such as "Dan is unavailable until 11 AM. Would you like to leave a message?" Furthermore, a user with a "low" trust level may be provided a response such as "Dan is busy. Would you like to leave a message?"

In some examples, a known caller may correspond to an automated system. For instance, the user may have previously stored a contact identifier corresponding to the user's doctor's office. In accordance with a determination that an incoming call from the doctor's office is associated with a known caller, and a determination that the caller further corresponds to an automated system, a specific response to the automated system may be provided. For example, upon establishing communication with the automated system, the automated system may provide an output "Hello, this is a reminder regarding John's annual appointment on Wednesday at 1 PM. Please press one to confirm the appointment, or press two to cancel the appointment." In response to the output, the digital assistant may provide a notification to the user including corresponding options in order to respond to the automated system, such as a notification with selectable affordances. For example, a notification may include the prompt "Would you like to confirm your annual appointment on Wednesday at 1 PM?," including selectable options "Yes," "No," or "Cancel Appointment." Based on the user selection, the digital assistant may provide the appropriate response to the automated system (e.g., responding to the automated system with an input of "one" if the user selects "Yes").

Referring to FIG. **8B**, in some examples, an output including information corresponding to the communication session is provided at the electronic device. In general, a transcription **808** of the communication session may be provided to the user of the electronic device via the display. Transcription **808** may be accessed, for example, based on a user pressing an affordance (e.g., a "listen-in" button) included within the incoming call indication **804** depicted on FIG. **8A**. The transcription may be provided to the user in real-time or in substantially real-time as the conversation takes place between the digital assistant and the caller. From the call window including transcription **808**, the user may further end the call or route the call to voicemail using various displayed affordances. In some examples, as discussed in more detail with respect to FIGS. **12A-12D**, during the communication session between the caller and a digital assistant, the user may provide an input (e.g., a typed input), and a corresponding output is provided to the caller by the digital assistant (e.g., a text-to-speech output). In some examples, the user may decide to answer the call, and accordingly, may provide an input such as pressing a

“pickup” affordance. In response to receiving the input, the user is included as a participant in the communication session. The digital assistant may remain a participant of the communication, as discussed in more detail with respect to FIGS. 11A-11B. Alternatively, the digital assistant may be removed from the communication session.

In some examples, a summary of the communication session may be provided. For example, the user may opt to not view the entire transcription of the communication session. Here, once the digital assistant obtains information from the caller, the information may be provided and/or summarized within indication 804 or elsewhere on the display. For example, the digital assistant may provide a prompt to the caller with the interrogatory “What is the call about?” The caller may provide input such as “This is a reminder about Dan’s 1 PM appointment today.” Accordingly, the digital assistant may recognize the input using natural language processing and in particular may obtain a paraphrase of the input based on recognition of a reminder type input, for example. In particular, the digital assistant may provide a summary within indication 804 including the text “The caller would like to remind you of your 1 PM appointment.” In some examples, the digital assistant may provide the same input that is provided by the caller without paraphrasing.

Referring to FIG. 9, the output may include select output types based on the trust level of the caller. In particular, a caller associated with a contact named “Jim Snow” may be identified as a high trust level caller. For example, “Jim Snow” may correspond to a contact identifier within the user’s contact database, and may be further included within a “favorites” group. Certain levels of trust may be associated with special options. For example, high trust level callers may be given an option to request an interruption of the callee. Accordingly, the digital assistant may provide a prompt such as “Dan is in a currently meeting until 11 AM. Would you like to interrupt Dan?” In response to an input indicating that the caller would like to interrupt the callee, an output is provided within indication 902, such as “Jim Snow would like to interrupt you,” including options for the callee to act upon (e.g., answer call, drop call, send to voicemail, etc.).

In general, the digital assistant may provide proactive assistance prior to a voice communication being established and/or during an established voice communication. For instance, various outputs may be provided before or while a call is being initiated (e.g., a reminder, a notification, or other information related to the voice communication). In some examples, a voice communication is established between the electronic device and a second electronic device in response to an incoming call from the second electronic device. While the incoming call is being received (e.g., displayed as an incoming call) and prior to the user of the electronic device accepting the call, a notification including a reminder may be provided to the user regarding a second user of the second electronic device, such as “Remember that tomorrow is Jim’s birthday.” In particular, metadata associated with the caller identification may be retrieved in order to provide relevant information to the user regarding the calling user, such as information related to significant events, recent conversations, social media posts, and the like. In some examples, the reminder may be provided prior to or after the user initiates a call to the second user (e.g., shown concurrently with a “call” affordance and/or shown immediately after the user initiates the call).

In general, a voice communication may be established between the electronic device and a second electronic

device. While the voice communication is established, an input may be received from the user, such as the user pressing an affordance on an active call screen. For example, an affordance may be displayed including an option for a digital assistant to take the place of the user in the established voice communication (e.g., a “swap” affordance). In response to receiving the input, a digital assistant session is initiated to include a digital assistant as a participant of the voice communication. In some examples, the user of the electronic device is removed as a participant from the voice communication. Generally, the voice communication may remain established between the electronic device and the second electronic device. After the digital assistant is added as a participant to the voice communication, the user of the second electronic device may interact with the digital assistant within the voice communication. Removal of the user as a participant of the voice communication may include muting the user, or may include ceasing to provide incoming call audio to the electronic device. As discussed further herein, an output including an ongoing transcription of the voice communication between the digital assistant and the second user may be provided on the display of the electronic device.

Referring to FIG. 10A, various outputs may be provided during the voice communication with the second electronic device. In particular, once the digital assistant is included as a participant in the voice communication, the user may be removed as an active participant in the voice communication such that the second user associated with the second electronic device interacts with the digital assistant of the electronic device. While the second user interacts with the digital assistant, an output 1002 is provided to the user of the electronic device, the output 1002 including a transcription of the voice communication. In general, audio corresponding to the voice communication may be processed using speech recognition in order to produce a textual representation of the second user’s speech. During the voice communication, the user of the electronic device may send messages to the second user in various ways. For example, the user may provide text input 1004 in order to provide a message to the second user (e.g., via a virtual keyboard on the display). In response to receiving text input 1004, the digital assistant may provide an audible output within the voice communication to the second user. In particular, text-to-speech processing may be performed on text input 1004 in order to obtain an audible output corresponding to the entered text (e.g., synthesized speech). In some examples, the user may provide a message to the second user using speech input. For example, the user may trigger audio sampling and natural language processing (e.g., by selecting an affordance or uttering a trigger phrase) of speech input. The captured speech input may be provided directly to the second user in the voice communication (e.g., the second user heard the user’s actual voice). In some examples, the digital assistant provides the speech input in the voice of the digital assistant. In particular, a textual representation of the user’s speech input may be obtained (e.g., using speech recognition), such that the textual representation is converted to an audible output which is then provided to the second user via the digital assistant. The digital assistant may also paraphrase or otherwise modify the user’s speech input. For example, the user may utter “Hey Siri, tell Joe that I’m running late but, um, should be ready to chat in, like, a few minutes.” The utterance may then be processed using natural language processing in order to obtain a corresponding user intent and a paraphrase of the input, such as “Dan

is running late but will be available soon.” The paraphrase may then be provided as an audible output within the voice communication.

In general, the second user of the second electronic device may interact with the digital assistant within the voice communication in various ways. For example, the second user may ask questions to the digital assistant, such as “When will Dan be available?” As discussed with respect to FIGS. 8-9, the second user may be provided access to various information regarding the user depending on an established trust level. To the extent the second user is associated with a high trust level, information regarding the user, such as scheduling information, may be provided to the second user. For example, the digital assistant may respond within the voice communication with “Dan is in a meeting until 11 AM, and then free until 1 PM.” The second user may also ask the digital assistant other questions unrelated to the user, such as “What’s the weather in San Francisco right now?” The digital assistant may then respond with appropriate information such as “It’s currently 50 degrees and cloudy in San Francisco.” The user of the electronic device may also interact with the digital assistant without the interaction being noticeable by the second user. For example, the user may provide various requests to the digital assistant, such as “Hey Siri, set a reminder at 5 PM to call Joe back.” In this case, the user’s speech input and the digital assistant’s corresponding response are not provided within the voice communication.

In some examples, the user may resume active participation in the voice communication. For instance, the user may provide an input indicative of an intent to join the voice communication. For the example, the user may press an affordance on the display or may provide a speech input (e.g., “Hey Siri, patch me into the call.”) In response to the input, the user may be connected to the voice communication such that the electronic device receives audio from the second electronic device and provides audio to the second electronic device. The digital assistant may also remain a participant of the voice communication, such that the user and/or second user may interact with the digital assistant. Digital assistant interactions among multiple parties of a voice communication are also discussed in more detail with respect to FIGS. 11A-11B and 12A-12D. The digital assistant may also be removed as a participant of the voice communication. For example, the user may press a displayed affordance or provide a speech input such as “Hey Siri, you’re dismissed.” In response to the input, the digital assistant may be removed as a participant. The digital assistant may also be removed automatically once the user resumes active participation in the voice communication (e.g., based on a user preference or other setting).

Other types of assistance may be provided during the voice communication. For example, audio quality may be monitored during the voice communication. If the monitored audio quality drops below a threshold audio quality, then the digital assistant may provide a respective output (e.g., “There appears to be noise coming from the number ending in 6217.”). Various methods may be used to measure audio quality such as those based on Mean Opinion Score (MOS) or Perceptual Evaluation of Speech Quality (PESQ). In some examples, notifications or other reminders may be provided within the voice communication (or only to one or more select parties within the voice communication) based on content of the voice communication. For example, a participant of the voice communication may utter “Does anyone know how late The Pizza Place is open?” The digital assistant may recognize parameters, such as a business name

(e.g., “The Pizza Place”), and may further recognize a user intent of the corresponding utterance, such as a request to obtain hours of operation. Accordingly, the digital assistant may obtain results corresponding to the intent and parameters, such as “The Pizza Place closes at 10 PM.”

The digital assistant may also provide enhanced functionality for handling automated call systems. In particular, a menu structure corresponding to an automated call system may be obtained. Automated systems may generally be associated with a menu hierarchy for navigating to different services offered by the automated system. For example, when contacting a bank associated with an automated system, an initial prompt may be provided to the user including options to enter a specific input (e.g., via pressing a numerical affordance, speaking a number, uttering a predetermined word or phrase, etc.). In particular, an initial level of the menu structure for the bank may include “Press one for hours of operation, press two for account details, press three to speak to an operator.” If the user enters an input corresponding to “two,” an additional level may correspond to “Press one for savings accounts, press two for checking accounts, press three for credit card accounts.” Thus, the obtained menu structure may include all the respective levels associated with navigating the bank’s telephony-based services. Accordingly, the user may provide a speech input such as “call the bank and check my savings account balance.” Using the obtained menu structure, the digital assistant may identify a routing instruction based on the speech input. In particular, given the user intent corresponding to identifying a savings account balance, the routing instruction may include calling the bank and entering the sequential inputs “two” and “two.” Once the appropriate results are obtained (e.g., the user’s savings account balance), the digital assistant may provide an audible and/or displayed output with the results.

In some examples, the user may wish to bypass an automated call system and speak with a live operator. For instance, an obtained menu structure may indicate that in order to reach a live operator, the sequential inputs “one,” “three,” “five,” and “zero” must be entered once the call is connected. Accordingly, the user may provide a speech input such as “call the ticket office and route me to a customer service agent.” Using the obtained menu structure, the digital assistant may identify a routing instruction based on the speech input. In particular, given the user intent corresponding to a desire to speak to a live agent, the routing instruction may include calling the respective ticket office and entering the sequential inputs “one,” “three,” “five,” and “zero.” While the digital assistant navigates the automated call system, the user may engage in other activities on the device (e.g., interacting with one or more other applications), while the navigation occurs in the background. Once the digital assistant connects with the live operator, a short exchange may occur, such as the digital assistant providing an output to the live operator (e.g., “Hello, I’m the caller’s digital assistant. Please hold while the caller joins this call.”). In some examples, the user may be immediately connected to the live operator without the digital assistant providing the output. The digital assistant may then remain on the call between the user and the live operator for further assistance, or may alternatively be removed from the call.

In general, the digital assistant may be used to initiate conference calls between a plurality of users. In some examples, the user may provide a speech input including a reference to a plurality of contact identifiers. For instance, the user may utter “Hey Siri, start a call with Joe, Wendy, and Mike.” In some examples, the speech input may include

a single reference associated with a plurality of contact identifiers. In particular, the user may have previously configured a plurality of contact identifiers to be associated with a single group name. For example, the user may have associated contacts “Sarah,” “John,” “Jim,” and “Laura” within a group titled “Ski Group.” Accordingly, the user may utter “Hey Siri, start a call with the Ski Group.” The speech input may also include a request to schedule the call at a later time and/or date. In some examples, the speech input may be directed to establishing a video call with a plurality of contacts.

For each contact identifier referenced by the first speech input, a respective communication may be initiated between the digital assistant and each respective party referenced by the speech input. In particular, in response to the input “Hey Siri, start a call with Joe, Wendy, and Mike,” a first call may be initiated to a first contact named “Joe,” a second call may be initiated to a second contact named “Wendy,” and the third call may be initiated to a third contact named “Mike.” While the calls are being initiated, an output including connection status information may be provided to the calling user of the electronic device. In particular, the connection status information may include information indicating whether communication has been successfully established to each respective contact (e.g., “Mike (did not answer)”). The digital assistant may also interact with callees. For example, a respective callee, such as “Joe” may answer the call from the digital assistant, which may cause the digital assistant to provide an output such as “Hi Joe. I’m Dan’s personal assistant. Please hold while we set up a call.” For successfully established communications, the information provided to the calling user may also include an elapsed time since the communication was established (e.g., “Joe (answered, connected for 0:45)”). The information may also include a transcription of an interaction between the digital assistant and a user of a successfully established communication. For example, the callee may inquire about call details. Using caller and caller identifiers, Siri may respond to the callee question and a transcription is provided accordingly, such as “Dan said ‘Who is calling again.’ Siri responded ‘The caller is Dan.’”

In some examples, the calling user may input a request for information from the respective callees. For example, the request for information may include a speech input or a typed input with one or more specific questions (e.g., “Please submit your preferred date for our annual meeting.”) The digital assistant may then provide the question to the callee (e.g., via an audible output) upon successfully establishing the communication with the respective callee. Any answers provided by the callee may then be provided to the calling user (e.g., via the provided transcription, by a separate notification or message, etc.)

In general, accordance with a determination that connection completion criteria has been satisfied, at least one communication of the plurality of initiated communications is identified based on the connection completion criteria. Particularly, connection completion criteria may require that at least one successful communication be established. Determination is thus made whether at least one initiated communication between the digital assistant and a respective party resulted in a successfully established communication. For instance, the digital assistant may have successfully established communication with contacts “Joe” and “Wendy,” but may not have established communication with contact “Mike.” Accordingly, the connection completion criteria is determined to be satisfied. If the connection completion criteria is not satisfied, then no communication

session is established (e.g., the user is provided a message or notification that no callee party answered).

In general, when at least one successfully established communication is identified, a communication session is established by combining all successfully established communications. For example, each successful call leg is merged into a single conference call between the caller and each callee that was successfully reached by the digital assistant. During the communication session, various parties may initiate requests directed to the digital assistant. With reference to FIG. 11A, an exemplary call interface is displayed corresponding to an established communication session and digital assistant interaction. Digital assistant requests may be associated with an intent to communicate either publicly or privately with the digital assistant. Specifically, an intent to communicate publicly with the digital assistant may result in the user’s digital assistant request and/or the digital assistant response being provided with in the communication session to be received by all participating parties. In contrast, an intent to communicate privately with the digital assistant may result in the user’s digital assistant request and/or digital assistant response being provided only to the requesting user. The intent to communicate either privately or publicly may be conveyed by utilizing a specific input (e.g., pressing a physical button on the device or a specific displayed affordance) or by providing a specific speech input (e.g., “Hey Siri, respond to the group with the nearest Starbucks location). The intent may also be determined by a user preference, such as a default preference that any digital assistant request provided during a communication session be private unless indicated otherwise. With reference to FIG. 11A, during the establishing communication session, the calling user may provide a request 1102 to the digital assistant, such as “Hey Siri, where is the nearest Starbucks?” The digital assistant may provide a response 1104 to the user (and not to the remaining participants of the call). The response may include a displayed response, and may further include an audible response based on the displayed response (e.g., the audible response may be output while reducing the volume of the corresponding communication session). In some examples, an audible response is provided to the user, while the outgoing audio data (e.g., the audio data from the user’s device) is silenced such that the remaining participants of the call do not hear the audio data.

With reference to FIGS. 12A-12D, the calling user may further interact within the context of the communication session by text. Such interaction may be initiated and maintained based on various device settings, such as one or more accessibility settings. In particular, the user may enter text input which is then provided audibly by the digital assistant to the other users within the communication session. Speech input from the other users may also be transcribed and provided to the calling user. The user may also communicate with the digital assistant via text input. In some examples, the user may initiate the communication session by typing an input to the digital assistant, such as typing “Start a call with Joe, Wendy, and Mike.” In some examples, an initial output 1202 is provided by the digital assistant within the communication session indicating that the calling user is communicating by text. The initial output may be provided audibly to the callee users, and may be provided textually to the calling user. A speech input may be received within the communication session from contact “Joe,” such that the speech input is converted to textual form and provided as a text output 1204 to the calling user. In response the calling user may type various responses which

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are then converted to audible form and provided, by the digital assistant, within the communication session.

In some examples, the calling user may provide a typed input **1206** directed to the digital assistant. The typed input may be determined to be directed to the digital assistant based on a specific trigger word or phrase, for example, such as a sentence beginning with “Siri.” With reference to FIG. **12C**, the digital assistant may provide response **1208**. Response **1208** may be provided textually to the calling user. In some examples, to the extent input **1206** is associated with an intent to communicate with the digital assistant publicly (e.g., based on a corresponding user preference), response **1208** is also provided audibly within the communication session to the other callee parties. The callee parties may also interact with the digital assistant. Such interaction may first require permission from the calling user, for example, based on a user setting allowing such interactions. A digital assistant response from a callee party may be provided publicly within the communication session or may be provided privately to the callee party in accordance with a respective intent of the callee party to communicate either publicly or privately.

During the communication session, the calling user may provide a second speech input with an additional reference to another contact identifier not included in the initial set of contact identifiers. For example, the user may utter “Hey Siri, add Laura to this call.” Accordingly, the digital assistant may initiate a communication to the contact “Laura.” If communication is successfully established, the communication between the digital assistant and contact “Laura” may be merged with the established communication session between the other users. The user may also remove various participants from the established communication session. For example, the calling user may have accidentally included a participant in the initial set of contact identifiers. Accordingly, the user may utter “Hey Siri, remove Mike from this call.” The digital assistant may then proceed to identify the appropriate contact associated with the removal request (e.g., an “end call” request associated with the specific contact). The calling user request and any corresponding digital assistant response may be provided publicly or privately as discussed herein.

FIGS. **13-15** illustrate systems and processes for integrating a digital assistant with telephony. For example, the electronic device may include any device described herein, including but not limited to devices **104**, **200**, **400**, and **600** (FIGS. **1**, **2A**, **4**, and **6A-6B**). Thus, it will be appreciated that electronic devices associated with FIGS. **13-15** may correspond to a user device of any type, such as a phone, laptop computer, desktop computer, tablet, wearable device (e.g., smart watch), home speaker, and the like. Furthermore, the processes described herein may be performed by a server with information delivered to and from the device, performed on the device, or a combination thereof.

FIG. **13** illustrates process **1300** for integrating a digital assistant with telephony according to various examples. Process **1300** is performed, for example, using one or more electronic devices implementing a digital assistant. In some examples, process **1300** is performed using a client-server system (e.g., system **100**), and the blocks of process **1300** are divided up in any manner between the server (e.g., DA server **106**) and a client device. In other examples, the blocks of process **1300** are divided up between the server and multiple client devices (e.g., a mobile phone and a smart watch). Thus, while portions of process **1300** are described herein as being performed by particular devices of a client-server system, it will be appreciated that process **1300** is not

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so limited. In other examples, process **1300** is performed using only a client device (e.g., user device **104**) or only multiple client devices. In process **1300**, some blocks are, optionally, combined, the order of some blocks is, optionally, changed, and some blocks are, optionally, omitted. In some examples, additional steps may be performed in combination with the process **1300**.

With reference to FIG. **13**, at block **1302**, incoming call from a caller is received at the electronic device. At block **1304**, a communication session between the caller and the digital assistant of the electronic device is established. In some examples, in response to establishing the communication session between the caller and a digital assistant of the electronic device, an output is provided indicating the digital assistant has answered the incoming call. In some examples, in response to establishing a communication session between the caller and a digital assistant of the electronic device, a digital assistant object is displayed, a user gaze directed at the digital assistant object is detected, and in response to detecting the user gaze directed at the digital assistant object, an output is provided indicating the digital assistant has answered the incoming call. By utilizing leveraging signals such as user gaze, the system enhances display aspects by providing relevant information when necessary. Only providing information when necessary makes the device more efficient by avoiding providing unwanted information. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

At block **1306**, a determination is made whether the identification of the caller is unknown. In some examples, a determination is made that the identification of the caller is known based on identification of a stored contact identifier corresponding to the caller. In some examples, in accordance with the stored contact identifier corresponding to a first contact type, a first trust level is assigned to the caller, and in accordance with the stored contact identifier corresponding to a second contact type, a second trust level is assigned to the caller, wherein the second trust level is lower than the first trust level. In some examples, a determination is made that the identification of the caller is known based on at least one of an identification of a stored contact identifier corresponding to the caller and an identity recognized based on a comparison of a stored voiceprint and speech input received from the caller, and a communication frequency value associated with the identification of the caller is identified. In some examples, in accordance with the communication frequency value exceeding a communication frequency threshold value, a first trust level is assigned to the caller, and in accordance with the communication frequency value not exceeding the communication frequency threshold value, a second trust level is assigned to the caller, wherein the second trust level is lower than the first trust level. By considering trust levels associated with callers, the system improves device functionality by customizing device outputs. Customizing device outputs makes the device more efficient by only providing information if appropriate. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

At block **1308**, in accordance with a determination that the identification of the caller is unknown, a determination is made whether the caller corresponds to an automated calling system. In some examples, determining whether the caller corresponds to an automated system includes identifying audio attributes of the incoming call, comparing the identified audio attributes to stored audio attributes, and in

accordance with a determination that the identified audio attributes correspond to the stored audio attributes, determining that the caller does not correspond to an automated system. In some examples, determining whether the caller corresponds to an automated system includes providing, to the caller, a prompt including an interrogatory, receiving, from the caller, a response to the prompt, identifying an auditory pattern associated with the response, classifying the identified auditory based on pattern recognition, and determining, based on the classification, whether the caller corresponds to an automated system. In some examples, in accordance with determination that the caller corresponds to an automated system, the communication session between the caller and the digital assistant is ended. In some examples, in accordance with determination that the caller corresponds to an automated system, an instruction is transmitted to route the caller to a voicemail system. In some examples, in accordance with a determination that the caller does not correspond to an automated system, a prompt including an interrogatory requesting a callee name is provided to the caller, an input responsive to the prompt is received from the caller, the received input is compared to a stored callee name, and in accordance with a determination that the received input corresponds to the stored callee name, the identification of the caller is determined to be known. In some examples, in accordance with a determination that the received input corresponds to the stored callee name, a first trust level is assigned to the caller, and in accordance with a determination that the received input does not correspond to the stored callee name, a second trust level is assigned to the caller, wherein the second trust level is lower than the first trust level. By considering trust levels associated with callers, the system improves device functionality by customizing device outputs. Customizing device outputs makes the device more efficient by only providing information if appropriate. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

At block **1310**, in accordance with a determination that the identification of the caller is known, a response to the caller is provided by the digital assistant. In some examples, providing, by the digital assistant, a response to the caller includes, in accordance with a determination that the caller is associated with a first trust level, providing a first response to the caller, and, in accordance with a determination that the caller is associated with a second trust level lower than the first trust level, providing a second response to the caller, wherein the second response is different than the first response. In some examples, providing the first response to the caller includes providing a first set of information to the caller, and providing the second response to the caller includes providing a second set of information to the caller, wherein the first set of information includes the second set of information and additional information not included in the second set of information. In some examples, providing, by the digital assistant, a response to the caller includes obtaining information corresponding to a schedule of a user of the electronic device, and providing, to the caller, at least a portion of information included in the schedule. In some examples, providing, by the digital assistant, a response to the caller includes obtaining information corresponding to a schedule of a user of the electronic device, and providing, based on the schedule, a date and time corresponding to future availability of the user. By providing callee information to respective callers, the system improves device functionality by providing information on behalf the user. Providing information on behalf of the user makes the device

more efficient by eliminating the need for the user to provide the information in, for example, a separate communication. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

At block **1312**, an output including information corresponding to the communication session is provided at the electronic device. In some examples, during the communication session between the caller and a digital assistant of the electronic device, an input is received from a user of the electronic device, and an output is provided to the caller by the digital assistant, wherein the output is based on the input from the user. In some examples, providing, at the electronic device, an output including information corresponding to the communication session includes, while providing the response to the caller, providing a transcription of the response at the electronic device, and while receiving audio input from the caller to the digital assistant, providing a transcription of the audio input at the electronic device. In some examples, an input is received from a user of the electronic device, and in response to receiving the input, the user is included as a participant in the communication session. In some examples, speech input is received from the user, and the speech input is transmitted to the caller as part of the communication session. By providing a transcription to the callee, the system improves device functionality by providing multi-modal outputs. Providing a transcription of the communication session to the user makes the device more efficient by removing the need to provide an audible output of the session to the user. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

The operations described above with reference to FIG. **13** are optionally implemented by components depicted in FIGS. **1-4**, **6A-6B**, and **7A-7C**. For example, the operations of process **1300** may be implemented by one or more of operating system **718**, applications module **724**, I/O processing module **728**, STT processing module **730**, natural language processing module **732**, vocabulary index **744**, task flow processing module **736**, service processing module **738**, media service(s) **120-1**, or processor(s) **220**, **410**, and **704**. It would be clear to a person having ordinary skill in the art how other processes are implemented based on the components depicted in FIGS. **1-4**, **6A-6B**, and **7A-7C**.

FIG. **14** illustrates process **1400** for integrating a digital assistant with telephony according to various examples. Process **1400** is performed, for example, using one or more electronic devices implementing a digital assistant. In some examples, process **1400** is performed using a client-server system (e.g., system **100**), and the blocks of process **1400** are divided up in any manner between the server (e.g., DA server **106**) and a client device. In other examples, the blocks of process **1400** are divided up between the server and multiple client devices (e.g., a mobile phone and a smart watch). Thus, while portions of process **1400** are described herein as being performed by particular devices of a client-server system, it will be appreciated that process **1400** is not so limited. In other examples, process **1400** is performed using only a client device (e.g., user device **104**) or only multiple client devices. In process **1400**, some blocks are, optionally, combined, the order of some blocks is, optionally, changed, and some blocks are, optionally, omitted. In some examples, additional steps may be performed in combination with the process **1400**.

With reference to FIG. **14**, at block **1402**, a voice communication is established between the electronic device and

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a second electronic device. In some examples, the voice communication is established in response to a request from the second electronic device, and while the voice communication is established, a reminder is provided to the user regarding a second user of the second electronic device. In some examples, prior to establishing the voice communication, a user input associated with a contact identifier is received, and in response to receiving the user input, a reminder is provided to the user regarding a second user associated with the contact identifier. In some examples, the second electronic device of the established voice communication is associated with the contact identifier. By providing notifications and reminders, the system enhances device functionality by providing contextually relevant reminders. Providing contextually relevant reminders makes the device more efficient by increasing the chances of the user acting on the reminders and avoiding searching for such information later. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

At block **1404**, an input is received from a user of the electronic device. In some examples, the input from the user of the electronic device is received during the established voice communication. At block **1406**, in response to receiving the input, a digital assistant session is initiated to include a digital assistant as a participant of the voice communication. In some examples, initiating a digital assistant session to include a digital assistant as a participant of the voice communication includes removing the user of the electronic device as a participant of the established voice communication, and replacing the removed user with the digital assistant as a participant of the established voice communication. In some examples, accordance with a determination that an audio quality associated with the established voice communication is below a threshold quality level, the digital assistant is caused to provide, within the voice communication, an output associated with the audio quality. By providing notifications regarding audio quality, the system enhances device functionality by avoiding poor call quality and avoiding causing the users to repeat speech. Providing call quality information thus makes the device more efficient by quickly identifying when a call connection is poor. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

At block **1408**, speech input is received from the second electronic device, wherein the speech input is received as part of the voice communication. In some examples, at least one parameter is identified from the speech input received from the second electronic device, and in accordance with a determination that a result is applicable to the at least one parameter, an output including the result is provided to the user. In some examples, a second input including a text message is received from the user, and an audible output is provided, by the digital assistant, within the voice communication, wherein the output includes a text-to-speech representation of the text message. In some examples, a second input including a second speech input is received from the user, and a textual representation of the second speech input is obtained. In some examples, an audible output is provided, by the digital assistant, within the voice communication, wherein the output includes a text-to-speech representation of the second speech input. By providing notifications within an active call, the system enhances device functionality by providing contextually relevant information. Providing contextually relevant information makes the device more efficient by avoiding having call

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participants search for such information during the call. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

At block **1410**, the electronic device causes the digital assistant to provide, within the voice communication, a response to the speech input. In some examples, a menu structure corresponding to an automated call system is obtained. In some examples, a routing instruction is identified using the obtained menu structure and based on the received speech input from the second electronic device. In some examples, the digital assistant is caused to provide the response to the received speech input based on the routing instruction. In some examples, a second input including a second speech input is received from the user, and an audible output is provided, by the digital assistant, within the voice communication, wherein the output is based on the second speech input. In some examples, the audible output includes a paraphrase of the second speech input. In some examples, during the established voice communication, a second speech input is received from the user, the second speech input including a request directed to the digital assistant. In some examples, an output responsive to the second speech input is provided to the user. In some examples, providing, to the user, an output responsive to the second speech input includes forgoing providing the second speech input within the established voice session, and providing the response to the user without providing the response within the established voice session. By providing responses to user queries, the system enhances device functionality by providing contextually relevant information in response to user requests. Providing contextually relevant information makes the device more efficient by avoiding having call participants search for such information during the call. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

At block **1412**, while the voice communication occurs, an output is provided including an ongoing transcription of the voice communication. In some examples, providing an output including an ongoing transcription of the voice communication includes in response to receiving, from the second electronic device, speech input as part of the voice communication, obtaining a textual representation of the speech input, providing the textual representation of the speech input to the user, and forgoing providing the speech input to the user. In some examples, providing an output including an ongoing transcription of the voice communication includes causing the digital assistant to provide, within the voice communication, an audible response to the speech input, and in response causing the digital assistant to provide, within the voice communication, the audible response to the speech input, obtaining a textual representation of the audible response, providing the textual representation of the audible response to the user; and forgoing providing the audible response to the user. In some examples, a second input is received from the user of the electronic device, and in response to receiving the second input, the user is included as a participant in the voice communication. In some examples, speech input is received from the user, and the speech input is transmitted as part of the voice communication. By allowing text input to be conveyed by the digital assistant during a communication, the system enhances device functionality by improving accessibility for users with limited capabilities. Improving accessibility makes the device more efficient by enhancing input modalities for a wide range of users. Thus, these

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features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

The operations described above with reference to FIG. 14 are optionally implemented by components depicted in FIGS. 1-4, 6A-6B, and 7A-7C. For example, the operations of process 1300 may be implemented by one or more of operating system 718, applications module 724, I/O processing module 728, STT processing module 730, natural language processing module 732, vocabulary index 744, task flow processing module 736, service processing module 738, media service(s) 120-1, or processor(s) 220, 410, and 704. It would be clear to a person having ordinary skill in the art how other processes are implemented based on the components depicted in FIGS. 1-4, 6A-6B, and 7A-7C.

FIGS. 15A-15B illustrates process 1500 for integrating a digital assistant with telephony according to various examples. Process 1500 is performed, for example, using one or more electronic devices implementing a digital assistant. In some examples, process 1500 is performed using a client-server system (e.g., system 100), and the blocks of process 1500 are divided up in any manner between the server (e.g., DA server 106) and a client device. In other examples, the blocks of process 1500 are divided up between the server and multiple client devices (e.g., a mobile phone and a smart watch). Thus, while portions of process 1500 are described herein as being performed by particular devices of a client-server system, it will be appreciated that process 1500 is not so limited. In other examples, process 1500 is performed using only a client device (e.g., user device 104) or only multiple client devices. In process 1500, some blocks are, optionally, combined, the order of some blocks is, optionally, changed, and some blocks are, optionally, omitted. In some examples, additional steps may be performed in combination with the process 1500.

With reference to FIG. 15A, at block 1502, a first speech input directed to a digital assistant is received from a user of the electronic device, wherein the first speech input includes a reference to a plurality of contact identifiers. In some examples, the reference to a plurality of contact identifiers includes a single reference associated with the plurality of contact identifiers. At block 1504, for each contact identifier referenced by the first speech input, a respective communication is initiated between the digital assistant and a respective party associated with a respective contact identifier of the plurality of contact identifiers. In some examples, an output including connection status information is provided to the user based on the plurality of initiated communications. In some examples, the connection status information comprises, for each contact identifier referenced by the first speech input, an output indicating whether a respective initiated communication has resulted in a successfully established communication between the digital assistant and a respective party associated with a respective contact identifier. In some examples, the connection status information includes a parameter indicating an elapsed time since communication was successfully established between the digital assistant and a respective party associated with a respective contact identifier. In some examples, the connection status information includes a transcription of at least one interaction between the digital assistant and a respective party associated with a respective contact identifier. By accepting speech input to establish a call among multiple users, the system improves device functionality by consolidating required inputs into a smaller amount. Consolidating required inputs makes the device more efficient by enhancing input modalities for a wide range of users. Thus, these

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features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

At block 1506, a determination is made whether connection completion criteria has been satisfied. In some examples, the determination that connection completion criteria has been satisfied includes determining whether at least one initiated communication between the digital assistant and a respective party resulted in a successfully established communication between the digital assistant and the respective party. In some examples, prior to establishing a communication session, a communication is established between the digital assistant and a respective party associated with a respective contact identifier of the plurality of contact identifiers, a respective speech input directed to the digital assistant is received from the respective party, and a response based on the respective speech input is provided to the respective party. By providing responses to user queries, the system enhances device functionality by providing contextually relevant information in response to user requests. Providing contextually relevant information makes the device more efficient by avoiding having call participants search for such information during the call. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

With reference to FIG. 15B, at block 1508, in accordance with a determination that the connection completion criteria has been satisfied, at least one communication of the plurality of initiated communications is identified based on the connection completion criteria. In some examples, a request for information is received from the user, and while initiating a respective communication between the digital assistant and a respective party associated with a respective contact identifier of the plurality of contact identifiers, the request for information is provided to the respective party. In some examples, prior to establishing the communication session, an input responsive to the request for information is received, and the input responsive to the request for information is provided to the user. At block 1510, in accordance with a determination that the connection completion criteria has not been satisfied, establishing a communication session based on the first speech input is foregone. By providing responses to user queries, the system enhances device functionality by providing contextually relevant information in response to user requests. Providing contextually relevant information makes the device more efficient by avoiding having call participants search for such information during the call. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

At block 1512, a communication session is established, by combining the at least one identified communication, between the electronic device and at least one party associated with a contact identifier of the plurality of contact identifiers. In some examples, during the established communication session, a second speech input including a reference to a second contact identifier not included in the plurality of contact identifiers is received from the user. In some examples, a communication is established between the digital assistant and a respective party associated with the second contact identifier. In some examples, the established communication between the digital assistant and the respective party associated with the second contact identifier is combined with the established communication session. In some examples, during the established communication session, a third speech input is received from the user, the third

speech input including a reference to a respective contact identifier of the plurality of contact identifiers, wherein the third speech input includes a removal request. In some examples, in accordance with a determination that a respective party associated with the respective contact identifier is a participant in the established communication session, the respective party associated with the respective contact identifier is removed from the established communication session. By allowing the user to add or remove call participants using the speech, device functionality is improved by allowing the user to use the device for other non-speech based inputs. Allowing the user to use the device for non-speech base inputs makes the device more efficient by increasing the functions available to the user. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

In some examples, during the established communication session, a fourth input directed to the digital assistant is received from the user, and an output responsive to the fourth input is provided to the user. In some examples, providing, to the user, an output responsive to the fourth input includes, in accordance with a determination that the fourth input is associated with an input indicative of an intent to communicate privately, forgoing providing the fourth input within the established communication session, and providing the response to the user without providing the response within the established communication session. In some examples, providing, to the user, an output responsive to the fourth input includes, in accordance with a determination that the fourth input is not associated with an input indicative of an intent to communicate privately, providing the fourth input within the established communication session, and providing, within the established communication session, the response to the user. In some examples, the fourth input includes one of a textual input and a speech input. In some examples, during the established communication session, a fifth speech input directed to the digital assistant is received from a party of the established communication session, wherein the party is not the user. In some examples, the fifth speech input is provided within the communication session, and a response to the fifth input is provided, from the digital assistant, within the established communication session. By allowing text input to be conveyed by the digital assistant during a communication, the system enhances device functionality by improving accessibility for users with limited capabilities. Improving accessibility makes the device more efficient by enhancing input modalities for a wide range of users. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

The operations described above with reference to FIGS. 15A-15B are optionally implemented by components depicted in FIGS. 1-4, 6A-6B, and 7A-7C. For example, the operations of process 1300 may be implemented by one or more of operating system 718, applications module 724, I/O processing module 728, STT processing module 730, natural language processing module 732, vocabulary index 744, task flow processing module 736, service processing module 738, media service(s) 120-1, or processor(s) 220, 410, and 704. It would be clear to a person having ordinary skill in the art how other processes are implemented based on the components depicted in FIGS. 1-4, 6A-6B, and 7A-7C.

In accordance with some implementations, a computer-readable storage medium (e.g., a non-transitory computer readable storage medium) is provided, the computer-read-

able storage medium storing one or more programs for execution by one or more processors of an electronic device, the one or more programs including instructions for performing any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises means for performing any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises a processing unit configured to perform any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises one or more processors and memory storing one or more programs for execution by the one or more processors, the one or more programs including instructions for performing any of the methods or processes described herein.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in the art are thereby enabled to best utilize the techniques and various embodiments with various modifications as are suited to the particular use contemplated.

Although the disclosure and examples have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims.

As described above, one aspect of the present technology is the gathering and use of data available from various sources to improve integrating digital assistants with telephony. The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, twitter IDs, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other identifying or personal information.

The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, personal information such as a user's contact database may be used to improve digital assistant interactions within the context of an ongoing communication session. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, user interaction history (e.g., phone or text interactions) may be leveraged to improve integrating digital assistants with telephony.

The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are

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generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to data such as a contacts database. In yet another example, users can select to limit detail provided with respect to contact information. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user’s privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data at a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the

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present technology are not rendered inoperable due to the lack of all or a portion of such personal information data. For example, digital assistant integration with telephony can be facilitated by inferring preferences based on non-personal information data or a bare minimum amount of personal information, such as anonymized contact information, other non-personal information available to the system, or publicly available information.

What is claimed is:

1. An electronic device, comprising:
 - one or more processors; and
 - a memory storing one or more programs configured to be executed by the one or more processors, the one or more programs including instructions for:
 - establishing a voice communication between the electronic device and a second electronic device;
 - receiving a first input from a user of the electronic device;
 - in response to receiving the first input, initiating a digital assistant session to include a digital assistant as a participant of the voice communication;
 - causing a prompt to be provided, by the digital assistant, to the second electronic device;
 - receiving, from the second electronic device, a first speech input as part of the voice communication, wherein the first speech input is responsive to the prompt;
 - in accordance with a determination that the first speech input corresponds to a predetermined type, providing a paraphrase of the first speech input to the user of the electronic device;
 - causing the digital assistant to provide, within the voice communication, a response to the first speech input;
 - receiving a second input from the user, wherein the second input includes a second speech input;
 - causing the digital assistant to provide, within the voice communication, a paraphrase of the second speech input; and
 - while the voice communication occurs, providing an output including an ongoing transcription of the voice communication.
2. The electronic device of claim 1, the one or more programs including instructions for:
 - receiving a user input associated with a contact identifier;
 - in response to receiving the user input:
 - providing a reminder to the user regarding a second user associated with the contact identifier; and
 - establishing the voice communication between the electronic device and the second electronic device, wherein the second electronic device is associated with the contact identifier.
3. The electronic device of claim 1, wherein initiating a digital assistant session to include a digital assistant as a participant of the voice communication comprises:
 - removing the user of the electronic device as a participant of the established voice communication; and
 - replacing the removed user with the digital assistant as a participant of the established voice communication.
4. The electronic device of claim 1, the one or more programs including instructions for:
 - receiving, from the user, a third input including a text message; and
 - providing, by the digital assistant, an audible output within the voice communication, wherein the output includes a text-to-speech representation of the text message.
5. The electronic device of claim 1, the one or more programs including instructions for:

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during the established voice communication:
 receiving, from the user, a third speech input including
 a request directed to the digital assistant; and
 providing, to the user, an output responsive to the third
 speech input. 5

6. The electronic device of claim 1, the one or more
 programs including instructions for:
 obtaining a menu structure corresponding to an automated
 call system;
 identifying, based on the first speech input from the 10
 second electronic device, a routing instruction using the
 obtained menu structure; and
 causing the digital assistant to provide the response to the
 first speech input based on 15
 the routing instruction.

7. The electronic device of claim 1, the one or more
 programs including instructions for:
 in accordance with a determination that an audio quality
 associated with the established voice communication is 20
 below a threshold quality level:
 causing the digital assistant to provide, within the voice
 communication, an output associated with the audio
 quality.

8. The electronic device of claim 1, the one or more 25
 programs including instructions for:
 identifying, from the first speech input received from the
 second electronic device, at least one parameter; and
 in accordance with a determination that a reminder is
 applicable to the at least one parameter: 30
 providing, to the user, an output including the reminder.

9. The electronic device of claim 1, wherein providing an
 output including an ongoing transcription of the voice
 communication comprises:
 in response to receiving, from the second electronic 35
 device, the first speech input as part of the voice
 communication:
 obtaining a textual representation of the first speech
 input;
 providing the textual representation of the first speech 40
 input to the user; and
 forgoing providing the first speech input to the user.

10. The electronic device of claim 1, wherein providing an
 output including an ongoing transcription of the voice
 communication comprises: 45
 causing the digital assistant to provide, within the voice
 communication, an audible response to the first speech
 input;
 in response causing the digital assistant to provide, within
 the voice communication, the audible response to the 50
 first speech input:
 obtaining a textual representation of the audible
 response;
 providing the textual representation of the audible
 response to the user; and 55
 forgoing providing the audible response to the user.

11. The electronic device of claim 1, the one or more
 programs including instructions for:
 receiving a third input from the user of the electronic
 device; 60
 in response to receiving the third input;
 including the user as a participant in the voice commu-
 nication;
 removing the digital assistant as a participant from the
 voice communication;
 receiving a third input including a third speech input from 65
 the user; and

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transmitting, as part of the voice communication, the third
 speech input.

12. A computer-implemented method, comprising:
 at an electronic device with one or more processors and
 memory:
 establishing a voice communication between the elec-
 tronic device and a second electronic device;
 receiving a first input from a user of the electronic
 device;
 in response to receiving the first input, initiating a
 digital assistant session to include a digital assistant
 as a participant of the voice communication;
 causing a prompt to be provided, by the digital assis-
 tant, to the second electronic device;
 receiving, from the second electronic device, a first
 speech input as part of the voice communication,
 wherein the first speech input is responsive to the
 prompt;
 in accordance with a determination that the first speech
 input corresponds to a predetermined type, providing
 a paraphrase of the first speech input to the user of
 the electronic device;
 causing the digital assistant to provide, within the voice
 communication, a response to the first speech input;
 receiving a second input from the user, wherein the
 second input includes a second speech input;
 causing the digital assistant to provide, within the voice
 communication, a paraphrase of the second speech
 input; and
 while the voice communication occurs, providing an
 output including an ongoing transcription of the
 voice communication.

13. The method of claim 12, comprising:
 receiving a user input associated with a contact identifier;
 in response to receiving the user input:
 providing a reminder to the user regarding a second
 user associated with the contact identifier; and
 establishing the voice communication between the
 electronic device and the second electronic device,
 wherein the second electronic device is associated
 with the contact identifier.

14. The method of claim 12, wherein initiating a digital
 assistant session to include a digital assistant as a participant
 of the voice communication comprises:
 removing the user of the electronic device as a participant
 of the established voice communication; and
 replacing the removed user with the digital assistant as a
 participant of the established voice communication.

15. The method of claim 12, comprising:
 receiving, from the user, a third input including a text
 message; and
 providing, by the digital assistant, an audible output
 within the voice communication, wherein the output
 includes a text-to-speech representation of the text
 message.

16. The method of claim 12, comprising:
 during the established voice communication:
 receiving, from the user, a third speech input including
 a request directed to the digital assistant; and
 providing, to the user, an output responsive to the third
 speech input.

17. The method of claim 12, comprising:
 obtaining a menu structure corresponding to an automated
 call system;
 identifying, based on the first speech input from the
 second electronic device, a routing instruction using the
 obtained menu structure; and

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causing the digital assistant to provide the response to the first speech input based on the routing instruction.

18. The method of claim 12, comprising:

in accordance with a determination that an audio quality associated with the established voice communication is below a threshold quality level:

causing the digital assistant to provide, within the voice communication, an output associated with the audio quality.

19. The method of claim 12, comprising:

identifying, from the first speech input received from the second electronic device, at least one parameter; and in accordance with a determination that a reminder is applicable to the at least one parameter:

providing, to the user, an output including the reminder.

20. The method of claim 12, wherein providing an output including an ongoing transcription of the voice communication comprises:

in response to receiving, from the second electronic device, the first speech input as part of the voice communication:

obtaining a textual representation of the first speech input;

providing the textual representation of the first speech input to the user; and

forgoing providing the first speech input to the user.

21. The method of claim 12, wherein providing an output including an ongoing transcription of the voice communication comprises:

causing the digital assistant to provide, within the voice communication, an audible response to the first speech input;

in response causing the digital assistant to provide, within the voice communication, the audible response to the first speech input:

obtaining a textual representation of the audible response;

providing the textual representation of the audible response to the user; and

forgoing providing the audible response to the user.

22. The method of claim 12, comprising:

receiving a third input from the user of the electronic device;

in response to receiving the third input: including the user as a participant in the voice communication;

removing the digital assistant as a participant from the voice communication;

receiving a third input including a third speech input from the user; and

transmitting, as part of the voice communication, the third speech input.

23. A non-transitory computer-readable storage medium storing one or more programs

configured to be executed by one or more processors of an electronic device, the one or more programs including instructions for:

establishing a voice communication between the electronic device and a second electronic device;

receiving a first input from a user of the electronic device;

in response to receiving the first input, initiating a digital assistant session to include a digital assistant as a participant of the voice communication;

causing a prompt to be provided, by the digital assistant, to the second electronic device;

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receiving, from the second electronic device, a first speech input as part of the voice communication, wherein the first speech input is responsive to the prompt;

in accordance with a determination that the first speech input corresponds to a predetermined type, providing a paraphrase of the first speech input to the user of the electronic device;

causing the digital assistant to provide, within the voice communication, a response to the first speech input; receiving a second input from the user, wherein the second input includes a second speech input;

causing the digital assistant to provide, within the voice communication, a paraphrase of the second speech input; and

while the voice communication occurs, providing an output including an ongoing transcription of the voice communication.

24. The computer-readable storage medium of claim 23, the one or more programs including instructions for: receiving a user input associated with a contact identifier; in response to receiving the user input:

providing a reminder to the user regarding a second user associated with the contact identifier; and

establishing the voice communication between the electronic device and the second electronic device, wherein the second electronic device is associated with the contact identifier.

25. The computer-readable storage medium of claim 23, wherein initiating a digital assistant session to include a digital assistant as a participant of the voice communication comprises:

removing the user of the electronic device as a participant of the established voice communication; and

replacing the removed user with the digital assistant as a participant of the established voice communication.

26. The computer-readable storage medium of claim 23, the one or more programs including instructions for: receiving, from the user, a third input including a text message; and

providing, by the digital assistant, an audible output within the voice communication, wherein the output includes a text-to-speech representation of the text message.

27. The computer-readable storage medium of claim 23, the one or more programs including instructions for: during the established voice communication:

receiving, from the user, a third speech input including a request directed to the digital assistant; and

providing, to the user, an output responsive to the third speech input.

28. The computer-readable storage medium of claim 23, the one or more programs including instructions for: obtaining a menu structure corresponding to an automated call system;

identifying, based on the first speech input from the second electronic device, a routing instruction using the obtained menu structure; and

causing the digital assistant to provide the response to the first speech input based on the routing instruction.

29. The computer-readable storage medium of claim 23, the one or more programs including instructions for: in accordance with a determination that an audio quality associated with the established voice communication is below a threshold quality level:

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causing the digital assistant to provide, within the voice communication, an output associated with the audio quality.

30. The computer-readable storage medium of claim 23, the one or more programs including instructions for: 5
 identifying, from the first speech input received from the second electronic device, at least one parameter; and
 in accordance with a determination that a reminder is applicable to the at least one parameter:
 providing, to the user, an output including the reminder. 10

31. The computer-readable storage medium of claim 23, wherein providing an output including an ongoing transcription of the voice communication comprises:

in response to receiving, from the second electronic device, the first speech input as part of the voice communication: 15

obtaining a textual representation of the first speech input;

providing the textual representation of the first speech input to the user; and 20

forgoing providing the first speech input to the user.

32. The computer-readable storage medium of claim 23, wherein providing an output including an ongoing transcription of the voice communication comprises:

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causing the digital assistant to provide, within the voice communication, an audible response to the first speech input;

in response causing the digital assistant to provide, within the voice communication, the audible response to the first speech input:

obtaining a textual representation of the audible response;

providing the textual representation of the audible response to the user; and

forgoing providing the audible response to the user.

33. The computer-readable storage medium of claim 23, the one or more programs including instructions for: receiving a third input from the user of the electronic device;

in response to receiving the third input:

including the user as a participant in the voice communication;

removing the digital assistant as a participant from the voice communication;

receiving a third input including a third speech input from the user; and

transmitting, as part of the voice communication, the third speech input.

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